

# Sid Mathur's Thematic ETF Screener – Daily Report

2026-05-08

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## Executive Summary

**Run date:** 2026-05-08    **Universe:** 728 ETFs across 27 clusters

### Headline state

- **Full sample PC1:** LONG since 2026-04-09 (20 days, 4.6%)
- **Clusters in long state:** 16 of 27 (59%)
- **ETFs in long state:** 436 of 728 (60%)

### Today's events

- Buy crossovers: 8
- Sell crossovers: 21
- Imminent buys (P > 50% within 5 days): 669
- Imminent sells (P > 50% within 5 days): 296

## Full Sample PC1

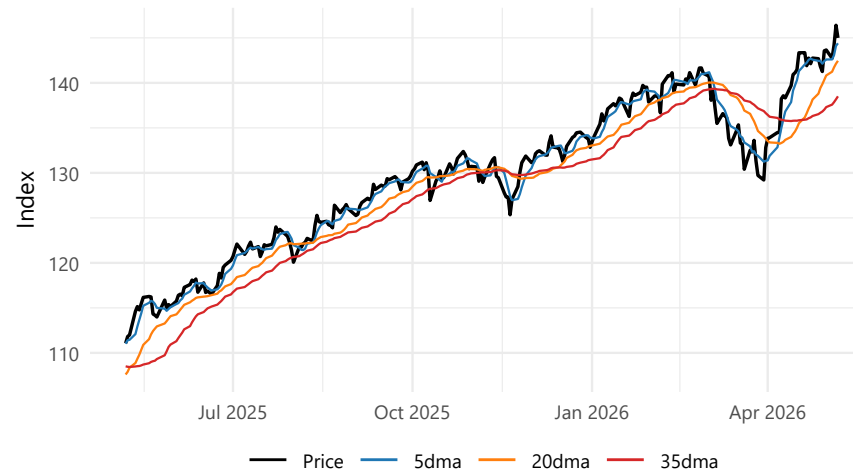


Figure 1: Full sample PC1 portfolio price with 5d / 20d / 35d MAs (last 12 months).

**Current state:** LONG since 2026-04-09 (20 days, 4.6%)

5dma: 144.399    20dma: 142.454    35dma: 138.506

Spread vs 35dma: +5.893    Spread vs 20dma: +1.944

P(buy crossover within 5 days): 100.0%    P(sell): 6.0%

## Top 10 PC1 weights

|     | ticker | weight | name                                       | category         |
|-----|--------|--------|--------------------------------------------|------------------|
| 108 | ACWI   | 0.0019 | iShares MSCI ACWI ETF                      | Global Equity    |
| 119 | AOA    | 0.0019 | iShares Core 80/20 Aggressive Allocatio... | Asset Allocation |
| 122 | AOR    | 0.0019 | iShares Core 60/40 Balanced Allocation ... | Asset Allocation |
| 126 | URTH   | 0.0019 | iShares MSCI World ETF                     | Global Equity    |
| 128 | CRBN   | 0.0019 | iShares Low Carbon Optimized MSCI ACWI ... | Global Equity    |
| 139 | SPGM   | 0.0019 | State Street SPDR Portfolio MSCI Global... | Global Equity    |
| 143 | VT     | 0.0019 | Vanguard Total World Stock Index ETF       | Global Equity    |

|     |      |        |                                          |                          |
|-----|------|--------|------------------------------------------|--------------------------|
| 186 | RSP  | 0.0019 | Invesco S&P 500 Equal Weight ETF         | U.S. Equal Weight Equity |
| 288 | FEX  | 0.0019 | First Trust Large Cap Core AlphaDEX Fund | Large Cap Blend Equities |
| 299 | JHMM | 0.0019 | John Hancock Multifactor Mid Cap ETF     | U.S. Mid Cap Equity      |

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## Cluster Overview

|    | id         | name                             | category                 | basket                    | n   | var_explained | signal | entry_date | days | cum_return | nss    |
|----|------------|----------------------------------|--------------------------|---------------------------|-----|---------------|--------|------------|------|------------|--------|
| 1  | cluster_1  | Mixed: IYY, SCHX, ILCB           | Mixed                    | SCHX+XNTK+XSD (cor=1.00)  | 112 | 0.873         | LONG   | 2026-04-10 | 19   | 10.8%      | 1.000  |
| 23 | cluster_23 | Mixed: SRLN, BKLN, FTSL          | Mixed                    | SRLN+LQDH+FLOT (cor=0.97) | 9   | 0.560         | LONG   | 2026-04-02 | 24   | 1.2%       | 1.000  |
| 24 | cluster_24 | Cash / Ultra-Short Fixed Inco... | Cash / Ultra-Short Fi... | GSY+NEAR+VNLA (cor=0.95)  | 9   | 0.456         | FLAT   |            |      |            | 1.000  |
| 25 | cluster_25 | Energy (5/8)                     | Energy                   | QCLN+REMX+ICLN (cor=0.99) | 8   | 0.774         | LONG   | 2026-04-10 | 19   | 10.5%      | 1.000  |
| 7  | cluster_7  | Emerging Markets Equity (19/2... | Emerging Markets Equi... | IEMG+DEM+FEMS (cor=1.00)  | 26  | 0.879         | LONG   | 2026-04-10 | 19   | 8.9%       | 0.996  |
| 6  | cluster_6  | Mixed: IWP, VOT, IMCG            | Mixed                    | IWP+BLOK+ARKQ (cor=0.99)  | 35  | 0.781         | LONG   | 2026-04-10 | 19   | 9.6%       | 0.989  |
| 20 | cluster_20 | Broad Commodities (6/10)         | Broad Commodities        | GSG+UCO+DBC (cor=1.00)    | 10  | 0.912         | FLAT   |            |      |            | 0.987  |
| 5  | cluster_5  | Mixed: VTV, SCHV, IWD            | Mixed                    | VTV+SDY+SPYV (cor=1.00)   | 49  | 0.899         | LONG   | 2026-04-10 | 19   | 2.2%       | 0.981  |
| 21 | cluster_21 | Inflation-Protected Fixed Inc... | Inflation-Protected F... | TIP+STPZ+SPIP (cor=1.00)  | 10  | 0.908         | LONG   | 2026-04-09 | 20   | 0.5%       | 0.920  |
| 19 | cluster_19 | Consumer Staples (6/11)          | Consumer Staples         | VDC+FUTY+PSCC (cor=1.00)  | 11  | 0.748         | LONG   | 2026-04-27 | 8    | -0.3%      | 0.886  |
| 17 | cluster_17 | Real Estate (14/14)              | Real Estate              | SCHH+RWR+INDS (cor=1.00)  | 14  | 0.962         | LONG   | 2026-04-10 | 19   | 3.8%       | 0.837  |
| 2  | cluster_2  | Mixed: IJH, MDY, IVOO            | Mixed                    | IJH+SPSM+IWR (cor=1.00)   | 107 | 0.889         | LONG   | 2026-04-09 | 20   | 3.8%       | 0.801  |
| 15 | cluster_15 | China Equity (13/15)             | China Equity             | CXSE+CNYA+MCHI (cor=0.99) | 15  | 0.804         | LONG   | 2026-04-13 | 18   | 2.9%       | 0.781  |
| 26 | cluster_26 | Currency (5/7)                   | Currency                 | UDN+BWX+FXF (cor=0.99)    | 7   | 0.763         | LONG   | 2026-04-13 | 18   | 0.6%       | 0.766  |
| 16 | cluster_16 | Energy (10/14)                   | Energy                   | FENY+AMLP+XES (cor=1.00)  | 14  | 0.861         | LONG   | 2026-04-29 | 6    | -4.4%      | 0.648  |
| 22 | cluster_22 | Japan Equity (7/9)               | Japan Equity             | EWJ+DXJ+VPL (cor=1.00)    | 9   | 0.888         | FLAT   |            |      |            | 0.597  |
| 8  | cluster_8  | High Yield Bond (10) + Prefer... | High Yield Bond          | JNK+PFF+PCY (cor=0.99)    | 22  | 0.772         | LONG   | 2026-04-09 | 20   | 0.8%       | 0.437  |
| 3  | cluster_3  | Mixed: VEA, SPDW, SCHF           | Mixed                    | VEA+VGK+VYMI (cor=1.00)   | 82  | 0.839         | FLAT   |            |      |            | 0.366  |
| 18 | cluster_18 | Municipal Bond (13/13)           | Municipal Bond           | VTEB+HYD+CMF (cor=0.99)   | 13  | 0.819         | FLAT   |            |      |            | 0.194  |
| 10 | cluster_10 | Healthcare (16/17)               | Healthcare               | FXH+XBI+VHT (cor=0.99)    | 17  | 0.764         | FLAT   |            |      |            | 0.180  |
| 11 | cluster_11 | Materials (13/17)                | Materials                | GNR+IYM+PICK (cor=0.99)   | 17  | 0.756         | FLAT   |            |      |            | 0.165  |
| 12 | cluster_12 | Financials (13/15)               | Financials               | VFH+KBE+KCE (cor=1.00)    | 15  | 0.883         | FLAT   |            |      |            | 0.121  |
| 27 | cluster_27 | Mixed: UUP, EUO, TBT             | Mixed                    | DUST+TBT+DZZ (cor=0.98)   | 6   | 0.459         | LONG   | 2026-05-01 | 4    | -4.6%      | -0.008 |
| 14 | cluster_14 | Mixed: INDA, INDY, EPI           | Mixed                    | INDA+ILF+EIDO (cor=0.97)  | 15  | 0.386         | FLAT   |            |      |            | -0.030 |
| 4  | cluster_4  | Mixed: BND, AGG, SCHZ            | Mixed                    | BND+BLV+BIV (cor=1.00)    | 59  | 0.833         | LONG   | 2026-05-07 | 0    | 0.0%       | -0.320 |
| 13 | cluster_13 | Mixed: DBP, GLTR, NUGT           | Mixed                    | GLTR+GDJX+AGQ (cor=1.00)  | 15  | 0.831         | FLAT   |            |      |            | -0.583 |
| 9  | cluster_9  | Leveraged / Inverse (16/22)      | Leveraged / Inverse      | SDS+UVXY+TZA (cor=1.00)   | 22  | 0.722         | FLAT   |            |      |            | -0.973 |

**Cluster Details****Cluster 1 – Mixed: IYY, SCHX, ILCB****Members:** 112   **Var explained:** 0.873   **Signal:** LONG since 2026-04-10 (19 days, 10.8%)**Category breakdown**

| category                     | n_members | abs_weight |
|------------------------------|-----------|------------|
| Technology                   | 23        | 0.2010     |
| U.S. Large Cap Growth        | 11        | 0.1027     |
| U.S. Large Cap Equity        | 9         | 0.0852     |
| U.S. Total Market Equity     | 7         | 0.0664     |
| Global Equity                | 7         | 0.0651     |
| Consumer Discretionary       | 6         | 0.0499     |
| Leveraged / Inverse          | 5         | 0.0471     |
| U.S. Equity                  | 4         | 0.0373     |
| Asset Allocation             | 4         | 0.0341     |
| Communication Services       | 4         | 0.0322     |
| U.S. Mega Cap Equity         | 3         | 0.0282     |
| U.S. Quality / Factor Equity | 3         | 0.0274     |
| U.S. Multifactor Equity      | 3         | 0.0273     |
| Unknown                      | 3         | 0.0259     |
| U.S. ESG Equity              | 2         | 0.0189     |
| Growth Equity                | 2         | 0.0184     |
| Commodity                    | 1         | 0.0094     |
| Large Cap Blend Equities     | 1         | 0.0094     |
| Large Cap Value Equities     | 1         | 0.0093     |
| U.S. Mega Cap Growth         | 1         | 0.0093     |
| U.S. Equal Weight Equity     | 1         | 0.0092     |
| Dividend Equity              | 1         | 0.0091     |
| U.S. Dividend Equity         | 1         | 0.0088     |
| Options Income               | 1         | 0.0085     |
| U.S. Small Cap Technology    | 1         | 0.0083     |
| Alternatives                 | 1         | 0.0082     |
| Trend-Following Equity       | 1         | 0.0080     |
| International IPO Equity     | 1         | 0.0074     |
| Volatility                   | 1         | 0.0072     |
| U.S. Trend-Following Equity  | 1         | 0.0072     |
| Israel Equity                | 1         | 0.0070     |
| Singapore Equity             | 1         | 0.0065     |

**Replication basket** – SCHX+XNTK+XSD (cor=1.00)   basket lead at lag 1: -0.007

| ticker | weight_ols | name                                     | category              | etf_signal |
|--------|------------|------------------------------------------|-----------------------|------------|
| SCHX   | 0.820      | Schwab U.S. Large-Cap ETF                | U.S. Large Cap Equity | LONG       |
| XNTK   | 0.175      | State Street SPDR NYSE Technology ETF    | Technology            | LONG       |
| XSD    | 0.052      | State Street SPDR S&P Semiconductor E... | Technology            | LONG       |

## Members

|     | ticker | weight | name                                       | category                 | etf_signal |
|-----|--------|--------|--------------------------------------------|--------------------------|------------|
| 10  | SPY    | 0.0095 | State Street SPDR S&P 500 ETF Trust        | U.S. Large Cap Equity    | LONG       |
| 70  | IWV    | 0.0095 | iShares Russell 3000 Fund                  | U.S. Total Market Equity | LONG       |
| 71  | IYY    | 0.0095 | iShares Dow Jones U.S. ETF                 | U.S. Total Market Equity | LONG       |
| 74  | SPYX   | 0.0095 | State Street SPDR S&P 500 Fossil Fuel R... | U.S. Equity              | LONG       |
| 76  | ESGU   | 0.0095 | iShares ESG Aware MSCI USA ETF             | U.S. ESG Equity          | LONG       |
| 77  | IWB    | 0.0095 | iShares Russell 1000 ETF                   | U.S. Large Cap Equity    | LONG       |
| 78  | IWL    | 0.0095 | iShares Russell Top 200 ETF                | U.S. Equity              | LONG       |
| 82  | VTHR   | 0.0095 | Vanguard Russell 3000 ETF                  | U.S. Total Market Equity | LONG       |
| 85  | VONE   | 0.0095 | Vanguard Russell 1000 ETF                  | U.S. Large Cap Equity    | LONG       |
| 86  | MGC    | 0.0095 | Vanguard Mega Cap ETF                      | U.S. Mega Cap Equity     | LONG       |
| 87  | VV     | 0.0095 | Vanguard Large-Cap ETF                     | U.S. Large Cap Equity    | LONG       |
| 88  | ILCB   | 0.0095 | iShares Morningstar Large-Cap ETF          | U.S. Large Cap Equity    | LONG       |
| 89  | ITOT   | 0.0095 | iShares Core S&P Total U.S. Stock Marke... | U.S. Total Market Equity | LONG       |
| 90  | IVV    | 0.0095 | iShares Core S&P 500 ETF                   | U.S. Large Cap Equity    | LONG       |
| 91  | SCHB   | 0.0095 | Schwab U.S. Broad Market ETF               | U.S. Total Market Equity | LONG       |
| 92  | SCHX   | 0.0095 | Schwab U.S. Large-Cap ETF                  | U.S. Large Cap Equity    | LONG       |
| 93  | SPLG   | 0.0095 |                                            |                          | LONG       |
| 94  | SPTM   | 0.0095 | State Street SPDR Portfolio S&P 1500 Co... | U.S. Total Market Equity | LONG       |
| 95  | VOO    | 0.0095 | Vanguard S&P 500 ETF                       | U.S. Large Cap Equity    | LONG       |
| 96  | VTI    | 0.0095 | Vanguard Total Stock Market ETF            | U.S. Total Market Equity | LONG       |
| 99  | SSO    | 0.0095 | ProShares Ultra S&P500                     | Leveraged / Inverse      | LONG       |
| 102 | UPRO   | 0.0095 | ProShares UltraPro S&P 500                 | Leveraged / Inverse      | LONG       |
| 103 | SPXL   | 0.0095 | Direxion Daily S&P 500 Bull 3X ETF         | Leveraged / Inverse      | LONG       |
| 14  | ACWI   | 0.0094 | iShares MSCI ACWI ETF                      | Global Equity            | LONG       |
| 19  | URTH   | 0.0094 | iShares MSCI World ETF                     | Global Equity            | LONG       |
| 42  | ONEQ   | 0.0094 | Fidelity Nasdaq Composite Index ETF        | U.S. Equity              | LONG       |
| 44  | IWF    | 0.0094 | iShares Russell 1000 Growth Fund           | U.S. Large Cap Growth    | LONG       |
| 53  | VONG   | 0.0094 | Vanguard Russell 1000 Growth ETF           | U.S. Large Cap Growth    | LONG       |
| 55  | ILCG   | 0.0094 | iShares Morningstar Large-Cap Growth ETF   | U.S. Large Cap Growth    | LONG       |
| 56  | IUSG   | 0.0094 | iShares Core S&P U.S. Growth ETF           | U.S. Large Cap Growth    | LONG       |
| 68  | DSI    | 0.0094 | iShares ESG MSCI KLD 400 ETF               | Large Cap Blend Equities | LONG       |
| 69  | SUSA   | 0.0094 | iShares ESG Optimized MSCI USA ETF         | U.S. ESG Equity          | LONG       |
| 72  | LRGF   | 0.0094 | iShares U.S. Equity Factor ETF             | U.S. Multifactor Equity  | LONG       |
| 73  | OEF    | 0.0094 | iShares S&P 100 Fund                       | U.S. Mega Cap Equity     | LONG       |

|     |      |        |                                            |                             |      |
|-----|------|--------|--------------------------------------------|-----------------------------|------|
| 83  | GSLC | 0.0094 | Goldman Sachs ActiveBeta U.S. Large Cap... | Commodity                   | LONG |
| 11  | QQQ  | 0.0093 | Invesco QQQ Trust, Series 1                | U.S. Large Cap Growth       | LONG |
| 20  | CRBN | 0.0093 | iShares Low Carbon Optimized MSCI ACWI ... | Global Equity               | LONG |
| 21  | SPGM | 0.0093 | State Street SPDR Portfolio MSCI Global... | Global Equity               | LONG |
| 22  | VT   | 0.0093 | Vanguard Total World Stock Index ETF       | Global Equity               | LONG |
| 41  | NULG | 0.0093 | Nuveen ESG Large-Cap Growth ETF            | U.S. Large Cap Growth       | LONG |
| 43  | IWY  | 0.0093 | iShares Russell Top 200 Growth ETF         | Growth Equity               | LONG |
| 45  | IVW  | 0.0093 | iShares S&P 500 Growth ETF                 | U.S. Large Cap Growth       | LONG |
| 48  | VOOG | 0.0093 | Vanguard S&P 500 Growth ETF                | U.S. Large Cap Growth       | LONG |
| 54  | MGK  | 0.0093 | Vanguard Mega Cap Growth ETF               | U.S. Mega Cap Growth        | LONG |
| 57  | SCHG | 0.0093 | Schwab U.S. Large-Cap Growth ETF           | U.S. Large Cap Growth       | LONG |
| 58  | SPYG | 0.0093 | State Street SPDR Portfolio S&P 500 Gro... | U.S. Large Cap Growth       | LONG |
| 59  | VUG  | 0.0093 | Vanguard Growth ETF                        | U.S. Large Cap Growth       | LONG |
| 60  | TILT | 0.0093 | FlexShares Mornigstar US Market Factors... | U.S. Multifactor Equity     | LONG |
| 66  | JHML | 0.0093 | John Hancock Multifactor Large Cap ETF     | U.S. Large Cap Equity       | LONG |
| 75  | XLG  | 0.0093 | Invesco S&P 500 Top 50 ETF                 | U.S. Mega Cap Equity        | LONG |
| 79  | QUAL | 0.0093 | iShares MSCI USA Quality Factor ETF        | U.S. Quality / Factor Eq... | LONG |
| 84  | EPS  | 0.0093 | WisdomTree U.S. LargeCap Fund              | Large Cap Value Equities    | LONG |
| 97  | TQQQ | 0.0093 | ProShares UltraPro QQQ                     | Leveraged / Inverse         | LONG |
| 106 | QLD  | 0.0093 | ProShares Ultra QQQ                        | Leveraged / Inverse         | LONG |
| 15  | AOA  | 0.0092 | iShares Core 80/20 Aggressive Allocatio... | Asset Allocation            | LONG |
| 25  | QQEW | 0.0092 | First Trust Nasdaq-100 Select Equal Wei... | U.S. Equal Weight Equity    | LONG |
| 37  | IGM  | 0.0092 | iShares Expanded Tech Sector ETF           | Technology                  | LONG |
| 40  | IYW  | 0.0092 | iShares U.S. Technology ETF                | Technology                  | LONG |
| 47  | VGT  | 0.0092 | Vanguard Information Tech ETF              | Technology                  | LONG |
| 52  | FTEC | 0.0092 | Fidelity MSCI Information Technology In... | Technology                  | LONG |
| 63  | IOO  | 0.0092 | iShares Global 100 ETF                     | Global Equity               | LONG |
| 65  | CATH | 0.0092 | Global X S&P 500 Catholic Values ETF       | Global Equity               | LONG |
| 1   | XLK  | 0.0091 | State Street Technology Select Sector S... | Technology                  | LONG |
| 29  | XT   | 0.0091 | iShares Future Exponential Technologies... | Technology                  | LONG |
| 36  | PWB  | 0.0091 | Invesco Large Cap Growth ETF               | Growth Equity               | LONG |
| 38  | IXN  | 0.0091 | iShares Global Tech ETF                    | Technology                  | LONG |
| 64  | QDF  | 0.0091 | FlexShares Quality Dividend Index Fund     | Dividend Equity             | LONG |
| 8   | AIQ  | 0.0090 | Global X Artificial Intelligence & Tech... | Technology                  | LONG |
| 18  | AOR  | 0.0090 | iShares Core 60/40 Balanced Allocation ... | Asset Allocation            | LONG |
| 26  | QTEC | 0.0090 | First Trust NASDAQ-100-Technology Secto... | Technology                  | LONG |
| 31  | FXL  | 0.0090 | First Trust Technology AlphaDEX            | Technology                  | LONG |
| 32  | XNTK | 0.0090 | State Street SPDR NYSE Technology ETF      | Technology                  | LONG |
| 61  | SPHB | 0.0090 | Invesco S&P 500 High Beta ETF              | U.S. Equity                 | LONG |
| 80  | QUS  | 0.0090 | State Street SPDR MSCI USA StrategicFac... | U.S. Quality / Factor Eq... | LONG |
| 81  | SPHQ | 0.0090 | Invesco S&P 500 Quality ETF                | U.S. Quality / Factor Eq... | LONG |
| 28  | TDIV | 0.0089 | First Trust NASDAQ Technology Dividend ... | Technology                  | LONG |
| 7   | ROBO | 0.0088 | ROBO Global Robotics and Automation Ind... | Technology                  | LONG |
| 39  | IYC  | 0.0088 | iShares U.S. Consumer Discretionary ETF    | Consumer Discretionary      | FLAT |
| 67  | DGRW | 0.0088 | WisdomTree U.S. Quality Dividend Growth... | U.S. Dividend Equity        | LONG |

|     |      |        |                                            |                             |      |
|-----|------|--------|--------------------------------------------|-----------------------------|------|
| 46  | VCR  | 0.0087 | Vanguard Consumer Discretion ETF           | Consumer Discretionary      | LONG |
| 51  | FDIS | 0.0087 | Fidelity MSCI Consumer Discretionary In... | Consumer Discretionary      | LONG |
| 107 | FIVG | 0.0087 |                                            |                             | LONG |
| 108 | OMFL | 0.0087 | Invesco Russell 1000 Dynamic Multifacto... | U.S. Multifactor Equity     | LONG |
| 2   | XLY  | 0.0086 | State Street Consumer Discretionary Sel... | Consumer Discretionary      | LONG |
| 6   | BOTZ | 0.0086 | Global X Robotics & Artificial Intellig... | Technology                  | LONG |
| 23  | NXTG | 0.0086 | First Trust Indxx NextG ETF                | Technology                  | LONG |
| 35  | QYLD | 0.0085 | Global X NASDAQ 100 Covered Call ETF       | Options Income              | LONG |
| 111 | QTUM | 0.0085 | Defiance Quantum ETF                       | Technology                  | LONG |
| 4   | SOXX | 0.0084 | iShares PHLX SOX Semiconductor Sector I... | Technology                  | LONG |
| 5   | SMH  | 0.0084 | VanEck Semiconductor ETF                   | Technology                  | LONG |
| 49  | VOX  | 0.0084 | Vanguard Communication Services ETF        | Communication Services      | LONG |
| 50  | FCOM | 0.0084 | Fidelity MSCI Communication Services In... | Communication Services      | LONG |
| 17  | AOM  | 0.0083 | iShares Core 40/60 Moderate Allocation ... | Asset Allocation            | LONG |
| 30  | XSD  | 0.0083 | State Street SPDR S&P Semiconductor ETF    | Technology                  | LONG |
| 101 | PSCT | 0.0083 | Invesco S&P SmallCap Information Techno... | U.S. Small Cap Technology   | LONG |
| 12  | QAI  | 0.0082 | NYLI Hedge Multi-Strategy Tracker ETF      | Alternatives                | LONG |
| 27  | PSI  | 0.0082 | Invesco Semiconductors ETF                 | Technology                  | LONG |
| 105 | FTXL | 0.0082 | First Trust Nasdaq Semiconductor ETF       | Technology                  | LONG |
| 3   | XLC  | 0.0081 | State Street Communication Services Sel... | Communication Services      | FLAT |
| 110 | PTF  | 0.0081 | Invesco Dorsey Wright Technology Moment... | Technology                  | LONG |
| 34  | PTNQ | 0.0080 | Pacer Trendpilot 100 ETF                   | Trend-Following Equity      | LONG |
| 24  | PRNT | 0.0078 | 3D Printing (The) ETF                      | Technology                  | LONG |
| 104 | RTH  | 0.0078 | VanEck Retail ETF                          | Consumer Discretionary      | LONG |
| 109 | IGN  | 0.0077 |                                            |                             | LONG |
| 16  | AOK  | 0.0076 | iShares Core 30/70 Conservative Allocat... | Asset Allocation            | LONG |
| 9   | ESPO | 0.0074 | VanEck Video Gaming and eSports ETF        | Consumer Discretionary      | FLAT |
| 33  | FPXI | 0.0074 | First Trust International Equity Opport... | International IPO Equity    | LONG |
| 98  | IYZ  | 0.0073 | iShares U.S. Telecommunications ETF        | Communication Services      | FLAT |
| 62  | PTLC | 0.0072 | Pacer Trendpilot US Large Cap ETF          | U.S. Trend-Following Equ... | LONG |
| 112 | SVXY | 0.0072 | ProShares Short VIX Short Term Futures ... | Volatility                  | LONG |
| 100 | EIS  | 0.0070 | iShares MSCI Israel ETF                    | Israel Equity               | LONG |
| 13  | EWS  | 0.0065 | iShares MSCI Singapore ETF                 | Singapore Equity            | FLAT |

## Cluster 2 – Mixed: IJH, MDY, IVOO

Members: 107 Var explained: 0.889 Signal: LONG since 2026-04-09 (20 days, 3.8%)

## Category breakdown

| category                              | n_members | abs_weight |
|---------------------------------------|-----------|------------|
| Industrials                           | 11        | 0.0983     |
| U.S. Small Cap Equity                 | 10        | 0.0967     |
| U.S. Mid Cap Equity                   | 9         | 0.0876     |
| U.S. Small Cap Value                  | 6         | 0.0575     |
| U.S. Small Cap Growth                 | 6         | 0.0567     |
| Consumer Discretionary                | 6         | 0.0512     |
| U.S. Mid Cap Value                    | 5         | 0.0484     |
| Leveraged / Inverse                   | 5         | 0.0476     |
| U.S. Mid Cap Growth                   | 3         | 0.0287     |
| U.S. Equity                           | 3         | 0.0282     |
| U.S. Value Factor Equity              | 3         | 0.0278     |
| U.S. Multifactor Equity               | 3         | 0.0277     |
| Utilities                             | 3         | 0.0261     |
| U.S. Equal Weight Equity              | 2         | 0.0193     |
| Dividend Equity                       | 2         | 0.0186     |
| U.S. Dividend Equity                  | 2         | 0.0182     |
| U.S. Small Cap Low Volatility         | 2         | 0.0178     |
| Unknown                               | 1         | 0.0098     |
| U.S. Small Cap Fundamental Equity     | 1         | 0.0098     |
| U.S. Mid/Small Cap Equity             | 1         | 0.0098     |
| U.S. Small Cap ESG Equity             | 1         | 0.0098     |
| Mid Cap Value Equities                | 1         | 0.0098     |
| U.S. Small/Mid Cap Fundamental Equity | 1         | 0.0097     |
| U.S. Small Cap Multifactor Equity     | 1         | 0.0097     |
| U.S. Mid Cap Dividend Equity          | 1         | 0.0096     |
| U.S. ESG Equity                       | 1         | 0.0096     |
| U.S. Size Factor Equity               | 1         | 0.0095     |
| U.S. Extended Market Equity           | 1         | 0.0095     |
| U.S. Small/Mid Cap Equity             | 1         | 0.0095     |
| Large Cap Blend Equities              | 1         | 0.0095     |
| Small Cap Value Equities              | 1         | 0.0095     |
| U.S. Mid Cap Quality Equity           | 1         | 0.0095     |
| Growth Equity                         | 1         | 0.0094     |
| Value Equity                          | 1         | 0.0094     |
| U.S. Small Cap Dividend Equity        | 1         | 0.0094     |
| U.S. Buyback Equity                   | 1         | 0.0093     |
| U.S. Small Cap Industrials            | 1         | 0.0092     |
| U.S. Small Cap Momentum Equity        | 1         | 0.0092     |
| U.S. Micro Cap Equity                 | 1         | 0.0090     |
| U.S. Moat Equity                      | 1         | 0.0089     |
| U.S. Small Cap Consumer Discretionary | 1         | 0.0088     |
| U.S. Mid Cap Low Volatility           | 1         | 0.0087     |
| Real Estate                           | 1         | 0.0076     |

**Replication basket** – IJH+SPSM+IWR (cor=1.00) basket lead at lag 1: -0.008

| ticker | weight_ols | name                                     | category              | etf_signal |
|--------|------------|------------------------------------------|-----------------------|------------|
| IJH    | 0.323      | iShares Core S&P Mid-Cap ETF             | U.S. Mid Cap Equity   | LONG       |
| SPSM   | 0.401      | State Street SPDR Portfolio S&P 600 S... | U.S. Small Cap Equity | LONG       |
| IWR    | 0.293      | iShares Russell Mid-Cap ETF              | U.S. Mid Cap Equity   | LONG       |

## Members

|     | ticker | weight | name                                       | category                    | etf_signal |
|-----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 16  | JHMM   | 0.0098 | John Hancock Multifactor Mid Cap ETF       | U.S. Mid Cap Equity         | LONG       |
| 30  | SCHM   | 0.0098 | Schwab U.S. Mid Cap ETF                    | U.S. Mid Cap Equity         | LONG       |
| 32  | FNX    | 0.0098 | First Trust Mid Cap Core AlphaDEX Fund     | U.S. Mid/Small Cap Equity   | LONG       |
| 35  | NUSC   | 0.0098 | Nuveen ESG Small-Cap ETF                   | U.S. Small Cap ESG Equity   | LONG       |
| 39  | MDY    | 0.0098 | State Street SPDR S&P MIDCAP 400 ETF Tr... | U.S. Mid Cap Equity         | LONG       |
| 47  | IVOO   | 0.0098 | Vanguard S&P Mid-Cap 400 ETF               |                             | LONG       |
| 54  | IJH    | 0.0098 | iShares Core S&P Mid-Cap ETF               | U.S. Mid Cap Equity         | LONG       |
| 55  | SPMD   | 0.0098 | State Street SPDR Portfolio S&P 400 Mid... | U.S. Mid Cap Equity         | LONG       |
| 57  | VB     | 0.0098 | Vanguard Small-Cap ETF                     | U.S. Small Cap Equity       | LONG       |
| 69  | EZM    | 0.0098 | WisdomTree U.S. MidCap Fund                | Mid Cap Value Equities      | LONG       |
| 73  | FNDA   | 0.0098 | Schwab Fundamental U.S. Small Company E... | U.S. Small Cap Fundament... | LONG       |
| 86  | VBR    | 0.0098 | Vanguard Small-Cap Value ETF               | U.S. Small Cap Value        | LONG       |
| 105 | MVV    | 0.0098 | ProShares Ultra MidCap400                  | Leveraged / Inverse         | LONG       |
| 21  | EQAL   | 0.0097 | Invesco Russell 1000 Equal Weight ETF      | U.S. Equal Weight Equity    | LONG       |
| 23  | IWR    | 0.0097 | iShares Russell Mid-Cap ETF                | U.S. Mid Cap Equity         | LONG       |
| 33  | FYX    | 0.0097 | First Trust Small Cap Core AlphaDEX Fund   | U.S. Small Cap Equity       | LONG       |
| 36  | PRFZ   | 0.0097 | Invesco RAFI US 1500 Small-Mid ETF         | U.S. Small/Mid Cap Funda... | LONG       |
| 37  | SMLF   | 0.0097 | iShares U.S. Small-Cap Equity Factor ETF   | U.S. Small Cap Multifact... | LONG       |
| 40  | IJT    | 0.0097 | iShares S&P SmallCap 600 Growth ETF        | U.S. Small Cap Growth       | LONG       |
| 42  | VIOG   | 0.0097 | Vanguard S&P Small-Cap 600 Growth ETF      | U.S. Small Cap Growth       | LONG       |
| 45  | SLYG   | 0.0097 | State Street SPDR S&P 600 Small Cap Gro... | U.S. Small Cap Growth       | LONG       |
| 48  | VIOO   | 0.0097 | Vanguard S&P Small-Cap 600 ETF             | U.S. Small Cap Equity       | LONG       |
| 51  | IJR    | 0.0097 | iShares Core S&P Small-Cap ETF             | U.S. Small Cap Equity       | LONG       |
| 56  | SPSM   | 0.0097 | State Street SPDR Portfolio S&P 600 Sma... | U.S. Small Cap Equity       | LONG       |
| 58  | SCHA   | 0.0097 | Schwab U.S. Small-Cap ETF                  | U.S. Small Cap Equity       | LONG       |
| 77  | IWS    | 0.0097 | iShares Russell Mid-Cap Value ETF          | U.S. Mid Cap Value          | LONG       |
| 79  | IJJ    | 0.0097 | iShares S&P Mid-Cap 400 Value ETF          | U.S. Mid Cap Value          | LONG       |
| 81  | IVOV   | 0.0097 | Vanguard S&P Mid-Cap 400 Value ETF         | U.S. Mid Cap Value          | LONG       |
| 82  | MDYV   | 0.0097 | State Street SPDR S&P 400 Mid Cap Value... | U.S. Mid Cap Value          | LONG       |
| 98  | RWK    | 0.0097 | Invesco S&P MidCap 400 Revenue ETF         | U.S. Mid Cap Equity         | LONG       |
| 5   | IWM    | 0.0096 | iShares Russell 2000 Index Fund            | U.S. Small Cap Equity       | LONG       |
| 6   | RSP    | 0.0096 | Invesco S&P 500 Equal Weight ETF           | U.S. Equal Weight Equity    | LONG       |

|     |      |        |                                            |                             |      |
|-----|------|--------|--------------------------------------------|-----------------------------|------|
| 7   | FXR  | 0.0096 | First Trust Industrials AlphaDEX           | Industrials                 | FLAT |
| 29  | IMCB | 0.0096 | iShares Morningstar Mid-Cap ETF            | U.S. Mid Cap Equity         | LONG |
| 41  | IJK  | 0.0096 | iShares S&P Mid-Cap 400 Growth ETF         | U.S. Mid Cap Growth         | LONG |
| 44  | MDYG | 0.0096 | State Street SPDR S&P 400 Mid Cap Growt... | U.S. Mid Cap Growth         | LONG |
| 49  | VTWO | 0.0096 | Vanguard Russell 2000 ETF                  | U.S. Small Cap Equity       | LONG |
| 67  | DON  | 0.0096 | WisdomTree U.S. MidCap Dividend Fund       | U.S. Mid Cap Dividend Eq... | LONG |
| 76  | IWN  | 0.0096 | iShares Russell 2000 Value ETF             | U.S. Small Cap Value        | LONG |
| 85  | VTWV | 0.0096 | Vanguard Russell 2000 Value ETF            | U.S. Small Cap Value        | LONG |
| 88  | UWM  | 0.0096 | ProShares Ultra Russell2000                | Leveraged / Inverse         | LONG |
| 89  | ETHO | 0.0096 | Amplify Etho Climate Leadership U.S. ETF   | U.S. ESG Equity             | LONG |
| 95  | TNA  | 0.0096 | Direxion Small Cap Bull 3X ETF             | Leveraged / Inverse         | LONG |
| 97  | JPME | 0.0096 | JPMorgan Diversified Return U.S. Mid Ca... | U.S. Equity                 | LONG |
| 100 | JVAL | 0.0096 | JPMorgan U.S. Value Factor ETF             | U.S. Value Factor Equity    | LONG |
| 106 | SDVY | 0.0096 | First Trust SMID Cap Rising Dividend Ac... | Dividend Equity             | LONG |
| 9   | FEX  | 0.0095 | First Trust Large Cap Core AlphaDEX Fund   | Large Cap Blend Equities    | LONG |
| 25  | SIZE | 0.0095 | iShares MSCI USA Size Factor ETF           | U.S. Size Factor Equity     | LONG |
| 31  | VO   | 0.0095 | Vanguard Mid-Cap ETF                       | U.S. Mid Cap Equity         | LONG |
| 43  | IVOG | 0.0095 | Vanguard S&P Mid-Cap 400 Growth ETF        | U.S. Mid Cap Growth         | LONG |
| 53  | VXF  | 0.0095 | Vanguard Extended Market ETF               | U.S. Extended Market Equ... | LONG |
| 68  | EES  | 0.0095 | WisdomTree U.S. SmallCap Fund              | Small Cap Value Equities    | LONG |
| 70  | PSC  | 0.0095 | Principal U.S. Small-Cap ETF               | U.S. Small Cap Equity       | LONG |
| 80  | IJS  | 0.0095 | iShares S&P SmallCap 600 Value ETF         | U.S. Small Cap Value        | LONG |
| 83  | SLYV | 0.0095 | State Street SPDR S&P 600 Small Cap Val... | U.S. Small Cap Value        | LONG |
| 84  | VIOV | 0.0095 | Vanguard S&P Small-Cap 600 Value ETF       | U.S. Small Cap Value        | LONG |
| 90  | RWJ  | 0.0095 | Invesco S&P SmallCap 600 Revenue ETF       | U.S. Small Cap Equity       | LONG |
| 93  | XMHQ | 0.0095 | Invesco S&P MidCap Quality ETF             | U.S. Mid Cap Quality Equ... | FLAT |
| 107 | ISMD | 0.0095 | Inspire Small/Mid Cap ETF                  | U.S. Small/Mid Cap Equity   | LONG |
| 13  | RDVY | 0.0094 | First Trust Rising Dividend Achievers E... | U.S. Dividend Equity        | LONG |
| 19  | CFA  | 0.0094 | VictoryShares US 500 Volatility Wtd ETF    | U.S. Multifactor Equity     | LONG |
| 24  | JPUS | 0.0094 | JPMorgan Diversified Return U.S. Equity... | U.S. Multifactor Equity     | LONG |
| 52  | ISCG | 0.0094 | iShares Morningstar Small-Cap Growth ETF   | Growth Equity               | LONG |
| 66  | DES  | 0.0094 | WisdomTree U.S. SmallCap Dividend Fund     | U.S. Small Cap Dividend ... | LONG |
| 87  | VOE  | 0.0094 | Vanguard Mid-Cap Value ETF                 | U.S. Mid Cap Value          | LONG |
| 96  | RFV  | 0.0094 | Invesco S&P MidCap 400 Pure Value ETF      | Leveraged / Inverse         | LONG |
| 104 | FYT  | 0.0094 | First Trust Small Cap Value AlphaDEX Fu... | Value Equity                | LONG |
| 8   | PKW  | 0.0093 | Invesco BuyBack Achievers ETF              | U.S. Buyback Equity         | FLAT |
| 15  | IYJ  | 0.0093 | iShares U.S. Industrials ETF               | Industrials                 | LONG |
| 22  | ONEV | 0.0093 | State Street SPDR Russell 1000 Low Vola... | U.S. Equity                 | FLAT |
| 27  | VIS  | 0.0093 | Vanguard Industrials ETF                   | Industrials                 | LONG |
| 28  | FIDU | 0.0093 | Fidelity MSCI Industrials Index ETF        | Industrials                 | LONG |
| 38  | IWO  | 0.0093 | iShares Russell 2000 Growth Fund           | U.S. Small Cap Growth       | LONG |
| 99  | OMFS | 0.0093 | Invesco Russell 2000 Dynamic Multifacto... | U.S. Equity                 | LONG |
| 26  | VLUE | 0.0092 | iShares MSCI USA Value Factor ETF          | U.S. Value Factor Equity    | LONG |
| 46  | VTWG | 0.0092 | Vanguard Russell 2000 Growth ETF           | U.S. Small Cap Growth       | LONG |
| 50  | VBK  | 0.0092 | Vanguard Small-Cap Growth ETF              | U.S. Small Cap Growth       | LONG |

|     |      |        |                                            |                             |      |
|-----|------|--------|--------------------------------------------|-----------------------------|------|
| 59  | FXD  | 0.0092 | First Trust Cons. Discret. AlphaDEX        | Consumer Discretionary      | FLAT |
| 91  | PSCI | 0.0092 | Invesco S&P SmallCap Industrials ETF       | U.S. Small Cap Industria... | LONG |
| 92  | PAVE | 0.0092 | Global X U.S. Infrastructure Developmen... | Industrials                 | LONG |
| 103 | XSMO | 0.0092 | Invesco S&P SmallCap Momentum ETF          | U.S. Small Cap Momentum ... | LONG |
| 18  | AIVL | 0.0091 | WisdomTree U.S. AI Enhanced Value Fund     | U.S. Value Factor Equity    | LONG |
| 64  | RSPD | 0.0091 | Invesco S&P 500 Equal Weight Consumer D... | Consumer Discretionary      | FLAT |
| 78  | SMMV | 0.0091 | iShares MSCI USA Small-Cap Min Vol Fact... | U.S. Small Cap Low Volat... | FLAT |
| 101 | RZV  | 0.0091 | Invesco S&P SmallCap 600 Pure Value ETF    | Leveraged / Inverse         | LONG |
| 1   | XLI  | 0.0090 | State Street Industrial Select Sector S... | Industrials                 | LONG |
| 12  | FIW  | 0.0090 | First Trust Water ETF                      | Utilities                   | FLAT |
| 17  | REGL | 0.0090 | ProShares S&P MidCap 400 Dividend Arist... | Dividend Equity             | FLAT |
| 34  | IWC  | 0.0090 | iShares Microcap ETF                       | U.S. Micro Cap Equity       | LONG |
| 14  | MOAT | 0.0089 | VanEck Morningstar Wide Moat ETF           | U.S. Moat Equity            | FLAT |
| 20  | CFO  | 0.0089 | VictoryShares US 500 Enhanced Volatilit... | U.S. Multifactor Equity     | LONG |
| 61  | FTXR | 0.0089 | First Trust Nasdaq Transportation ETF      | Industrials                 | LONG |
| 10  | PHO  | 0.0088 | Invesco Water Resources ETF                | Utilities                   | LONG |
| 65  | SMDV | 0.0088 | ProShares Russell 2000 Dividend Growers... | U.S. Dividend Equity        | FLAT |
| 102 | PSCD | 0.0088 | Invesco S&P SmallCap Consumer Discretio... | U.S. Small Cap Consumer ... | FLAT |
| 72  | XTN  | 0.0087 | State Street SPDR S&P Transportation ETF   | Industrials                 | FLAT |
| 74  | XMLV | 0.0087 | Invesco S&P MidCap Low Volatility ETF      | U.S. Mid Cap Low Volatil... | FLAT |
| 75  | XSLV | 0.0087 | Invesco S&P SmallCap Low Volatility ETF    | U.S. Small Cap Low Volat... | LONG |
| 94  | PKB  | 0.0087 | Invesco Building & Construction ETF        | Industrials                 | LONG |
| 2   | XRT  | 0.0085 | State Street SPDR S&P Retail ETF           | Consumer Discretionary      | FLAT |
| 63  | IYT  | 0.0085 | iShares U.S. Transportation ETF            | Industrials                 | FLAT |
| 60  | PEJ  | 0.0084 | Invesco Leisure and Entertainment ETF      | Consumer Discretionary      | FLAT |
| 71  | XHB  | 0.0084 | State Street SPDR S&P Homebuilders ETF     | Consumer Discretionary      | FLAT |
| 11  | CGW  | 0.0082 | Invesco S&P Global Water Index ETF         | Utilities                   | FLAT |
| 3   | JETS | 0.0076 | U.S. Global Jets ETF                       | Industrials                 | LONG |
| 4   | REM  | 0.0076 | iShares Mortgage Real Estate ETF           | Real Estate                 | LONG |
| 62  | ITB  | 0.0076 | iShares U.S. Home Construction ETF         | Consumer Discretionary      | FLAT |

**Cluster 3 – Mixed: VEA, SPDW, SCHF****Members:** 82 **Var explained:** 0.839 **Signal:** FLAT**Category breakdown**

| category               | n_members | abs_weight |
|------------------------|-----------|------------|
| International Equity   | 21        | 0.2686     |
| Developed ex-US Equity | 9         | 0.1179     |

|                               |   |        |
|-------------------------------|---|--------|
| International Dividend Equity | 6 | 0.0748 |
| Global ex-US Equity           | 5 | 0.0656 |
| Europe Equity                 | 4 | 0.0518 |
| Alternatives                  | 4 | 0.0464 |
| Eurozone Equity               | 3 | 0.0372 |
| Asia-Pacific Equity           | 2 | 0.0242 |
| Real Estate                   | 2 | 0.0225 |
| Global Equity                 | 2 | 0.0222 |
| Emerging Markets Fixed Income | 2 | 0.0193 |
| Currency                      | 2 | 0.0181 |
| Commodity                     | 1 | 0.0132 |
| Unknown                       | 1 | 0.0130 |
| Global ex-US Small Cap Equity | 1 | 0.0129 |
| Germany Equity                | 1 | 0.0124 |
| France Equity                 | 1 | 0.0124 |
| United Kingdom Equity         | 1 | 0.0123 |
| Italy Equity                  | 1 | 0.0122 |
| Europe Financials             | 1 | 0.0120 |
| Sweden Equity                 | 1 | 0.0120 |
| Netherlands Equity            | 1 | 0.0120 |
| Australia Equity              | 1 | 0.0119 |
| Spain Equity                  | 1 | 0.0116 |
| Canada Equity                 | 1 | 0.0114 |
| Switzerland Equity            | 1 | 0.0114 |
| Belgium Equity                | 1 | 0.0113 |
| Austria Equity                | 1 | 0.0112 |
| Poland Equity                 | 1 | 0.0104 |
| South Africa Equity           | 1 | 0.0099 |
| New Zealand Equity            | 1 | 0.0091 |
| Mexico Equity                 | 1 | 0.0089 |

**Replication basket** – VEA+VGK+VYMI (cor=1.00) basket lead at lag 1: 0.000

| ticker | weight_ols | name                                     | category                 | etf_signal |
|--------|------------|------------------------------------------|--------------------------|------------|
| VEA    | 0.479      | Vanguard FTSE Developed Markets ETF      | Developed ex-US Equity   | FLAT       |
| VGK    | 0.240      | Vanguard FTSEEuropean ETF                | Europe Equity            | FLAT       |
| VYMI   | 0.223      | Vanguard International High Dividend ... | International Dividen... | FLAT       |

## Members

| ticker | weight | name | category | etf_signal |
|--------|--------|------|----------|------------|
|--------|--------|------|----------|------------|

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 2  | EFA  | 0.0133 | iShares MSCI EAFE ETF                      | Developed ex-US Equity      | FLAT |
| 3  | VEA  | 0.0133 | Vanguard FTSE Developed Markets ETF        | Developed ex-US Equity      | FLAT |
| 63 | IEFA | 0.0133 | iShares Core MSCI EAFE ETF                 | Developed ex-US Equity      | FLAT |
| 64 | SCHF | 0.0133 | Schwab International Equity ETF            | Developed ex-US Equity      | FLAT |
| 65 | SPDW | 0.0133 | State Street SPDR Portfolio Developed W... | Developed ex-US Equity      | FLAT |
| 21 | DWM  | 0.0132 | WisdomTree International Equity Fund       | International Dividend E... | FLAT |
| 34 | TLTD | 0.0132 | FlexShares Morningstar Developed Market... | International Equity        | FLAT |
| 49 | GSIE | 0.0132 | Goldman Sachs ActiveBeta International ... | Commodity                   | FLAT |
| 52 | ESGD | 0.0132 | iShares ESG Aware MSCI EAFE ETF            | International Equity        | FLAT |
| 28 | PXF  | 0.0131 | Invesco RAFI Developed Markets ex-U.S. ... | International Equity        | LONG |
| 38 | ACWX | 0.0131 | iShares MSCI ACWI ex U.S. ETF              | Global ex-US Equity         | LONG |
| 39 | CWI  | 0.0131 | State Street SPDR MSCI ACWI ex-US ETF      | Global ex-US Equity         | LONG |
| 41 | INTF | 0.0131 | iShares International Equity Factor ETF    | International Equity        | FLAT |
| 42 | IQLT | 0.0131 | iShares MSCI Intl Quality Factor ETF       | International Equity        | FLAT |
| 44 | QEFA | 0.0131 | State Street SPDR MSCI EAFE StrategicFa... | International Equity        | FLAT |
| 58 | IEUR | 0.0131 | iShares Core MSCI Europe ETF               | Europe Equity               | FLAT |
| 59 | IXUS | 0.0131 | iShares Core MSCI Total International S... | Global ex-US Equity         | LONG |
| 60 | VEU  | 0.0131 | Vanguard FTSE All World Ex US ETF          | Global ex-US Equity         | LONG |
| 61 | VGK  | 0.0131 | Vanguard FTSEEurope ETF                    | Europe Equity               | FLAT |
| 62 | VXUS | 0.0131 | Vanguard Total International Stock ETF     | Global ex-US Equity         | LONG |
| 6  | IEV  | 0.0130 | iShares Europe ETF                         | Europe Equity               | FLAT |
| 32 | EFV  | 0.0130 | iShares MSCI EAFE Value ETF                | Developed ex-US Equity      | FLAT |
| 35 | JPIN | 0.0130 | JPMorgan Diversified Return Internation... | International Equity        | FLAT |
| 46 | RODM | 0.0130 | Hartford Multifactor Developed Markets ... | International Equity        | FLAT |
| 47 | VYMI | 0.0130 | Vanguard International High Dividend Yi... | International Dividend E... | FLAT |
| 48 | FNDF | 0.0130 | Schwab Fundamental International Equity... | International Equity        | FLAT |
| 56 | SCHC | 0.0130 | Schwab International Small-Cap Equity E... | Developed ex-US Equity      | FLAT |
| 75 | FEU  | 0.0130 |                                            |                             | FLAT |
| 24 | IQDF | 0.0129 | FlexShares International Quality Divide... | International Equity        | FLAT |
| 31 | SCZ  | 0.0129 | iShares MSCI EAFE Small-Cap ETF            | Developed ex-US Equity      | FLAT |
| 33 | FNDC | 0.0129 | Schwab Fundamental International Small ... | International Equity        | LONG |
| 57 | VSS  | 0.0129 | Vanguard FTSE All-Wld ex-US SmCp Idx ETF   | Global ex-US Small Cap E... | LONG |
| 8  | DLS  | 0.0128 | WisdomTree International SmallCap Fund     | International Equity        | FLAT |
| 19 | EZU  | 0.0128 | iShares MSCI Eurozone ETF                  | Eurozone Equity             | FLAT |
| 43 | IVLU | 0.0128 | iShares MSCI Intl Value Factor ETF         | International Equity        | FLAT |
| 72 | HFXI | 0.0128 | NYLI FTSE International Equity Currency... | International Equity        | FLAT |
| 29 | EFG  | 0.0127 | iShares MSCI EAFE Growth ETF               | Developed ex-US Equity      | FLAT |
| 45 | FEZ  | 0.0127 | State Street SPDR EURO STOXX 50 ETF        | Eurozone Equity             | FLAT |
| 54 | VIGI | 0.0127 | Vanguard International Dividend Appreci... | International Dividend E... | FLAT |
| 4  | FEP  | 0.0126 | First Trust Europe AlphaDEX Fund           | Europe Equity               | FLAT |
| 30 | GWX  | 0.0126 | State Street SPDR S&P International Sma... | International Equity        | LONG |
| 40 | IMTM | 0.0126 | iShares MSCI Intl Momentum Factor ETF      | International Equity        | FLAT |
| 1  | EWG  | 0.0124 | iShares MSCI Germany Index Fund            | Germany Equity              | FLAT |
| 5  | VIDI | 0.0124 | Vident International Equity Strategy ETF   | International Equity        | LONG |
| 16 | EWQ  | 0.0124 | iShares MSCI France Index Fund             | France Equity               | FLAT |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 20 | IDV  | 0.0124 | iShares International Select Dividend E... | International Dividend E... | FLAT |
| 53 | HDEF | 0.0124 | Xtrackers MSCI EAFE High Dividend Yield... | International Equity        | FLAT |
| 17 | EWU  | 0.0123 | iShares MSCI United Kingdom ETF            | United Kingdom Equity       | FLAT |
| 50 | IDLV | 0.0123 | Invesco S&P International Developed Low... | International Equity        | FLAT |
| 79 | IPKW | 0.0123 | Invesco International BuyBack Achievers... | International Equity        | FLAT |
| 82 | EWI  | 0.0122 | iShares MSCI Italy ETF                     | Italy Equity                | FLAT |
| 22 | EPP  | 0.0121 | iShares MSCI Pacific Ex-Japan Index Fund   | Asia-Pacific Equity         | FLAT |
| 51 | EFAV | 0.0121 | iShares MSCI EAFE Min Vol Factor ETF       | International Equity        | FLAT |
| 23 | EUFN | 0.0120 | iShares MSCI Europe Financials ETF         | Europe Financials           | FLAT |
| 71 | EWD  | 0.0120 | iShares MSCI Sweden ETF                    | Sweden Equity               | FLAT |
| 77 | EWN  | 0.0120 | iShares MSCI Netherlands Index Fund        | Netherlands Equity          | FLAT |
| 81 | BBAX | 0.0120 | JPMorgan BetaBuilders Developed Asia Pa... | Asia-Pacific Equity         | FLAT |
| 13 | EWA  | 0.0119 | iShares MSCI Australia Index Fund          | Australia Equity            | FLAT |
| 27 | DWX  | 0.0119 | State Street SPDR S&P International Div... | International Dividend E... | FLAT |
| 12 | HEZU | 0.0118 | iShares Currency Hedged MSCI Eurozone E... | Eurozone Equity             | FLAT |
| 36 | DBEF | 0.0118 | Xtrackers MSCI EAFE Hedged Equity ETF      | Alternatives                | FLAT |
| 37 | HEFA | 0.0118 | iShares Currency Hedged MSCI EAFE ETF      | Alternatives                | FLAT |
| 11 | PID  | 0.0117 | Invesco International Dividend Achiever... | International Dividend E... | FLAT |
| 15 | EWP  | 0.0116 | iShares MSCI Spain ETF                     | Spain Equity                | FLAT |
| 9  | HEDJ | 0.0114 | WisdomTree Europe Hedged Equity Fund       | Alternatives                | FLAT |
| 10 | IHDG | 0.0114 | WisdomTree International Hedged Quality... | Alternatives                | FLAT |
| 14 | EWL  | 0.0114 | iShares MSCI Switzerland ETF               | Switzerland Equity          | FLAT |
| 18 | EWC  | 0.0114 | iShares MSCI Canada Index Fund             | Canada Equity               | LONG |
| 25 | NFRA | 0.0114 | FlexShares STOXX Global Broad Infrastru... | Global Equity               | LONG |
| 55 | VNQI | 0.0114 | Vanguard Global ex-U.S. Real Estate ETF    | Real Estate                 | FLAT |
| 80 | EWK  | 0.0113 | iShares MSCI Belgium ETF                   | Belgium Equity              | FLAT |
| 76 | EWO  | 0.0112 | iShares MSCI Austria ETF                   | Austria Equity              | FLAT |
| 7  | RWX  | 0.0111 | State Street SPDR Dow Jones Internation... | Real Estate                 | FLAT |
| 26 | IGF  | 0.0108 | iShares Global Infrastructure ETF          | Global Equity               | LONG |
| 70 | EPOL | 0.0104 | iShares MSCI Poland ETF                    | Poland Equity               | FLAT |
| 74 | EZA  | 0.0099 | iShares MSCI South Africa Index Fund       | South Africa Equity         | LONG |
| 69 | FXA  | 0.0097 | Invesco CurrencyShares Australian Dolla... | Currency                    | LONG |
| 66 | EBND | 0.0096 | State Street SPDR Bloomberg Emerging Ma... | Emerging Markets Fixed I... | FLAT |
| 67 | EMLC | 0.0096 | VanEck J. P. Morgan EM Local Currency B... | Emerging Markets Fixed I... | FLAT |
| 73 | ENZL | 0.0091 | iShares MSCI New Zealand ETF               | New Zealand Equity          | LONG |
| 68 | EWX  | 0.0089 | iShares MSCI Mexico ETF                    | Mexico Equity               | FLAT |
| 78 | FXC  | 0.0084 | Invesco CurrencyShares Canadian Dollar ... | Currency                    | LONG |

**Cluster 4 – Mixed: BND, AGG, SCHZ****Members:** 59 **Var explained:** 0.833 **Signal:** LONG since 2026-05-07 (0 days, 0.0%)

## Category breakdown

| category                         | n_members | abs_weight |
|----------------------------------|-----------|------------|
| Fixed Income                     | 19        | 0.3226     |
| Investment Grade Corporate Bond  | 13        | 0.2258     |
| Aggregate Fixed Income           | 6         | 0.1100     |
| Mortgage-Backed Fixed Income     | 6         | 0.0965     |
| Active Fixed Income              | 5         | 0.0874     |
| Cash / Ultra-Short Fixed Income  | 3         | 0.0462     |
| International Fixed Income       | 2         | 0.0308     |
| Target Maturity Bond             | 2         | 0.0304     |
| Leveraged / Inverse Fixed Income | 1         | 0.0178     |
| Agency Fixed Income              | 1         | 0.0163     |
| Taxable Municipal Bond           | 1         | 0.0162     |

**Replication basket** – BND+BLV+BIV (cor=1.00) basket lead at lag 1: 0.003

| ticker | weight_ols | name                                | category               | etf_signal |
|--------|------------|-------------------------------------|------------------------|------------|
| BND    | 0.508      | Vanguard Total Bond Market ETF      | Aggregate Fixed Income | LONG       |
| BLV    | 0.132      | Vanguard Long-Term Bond ETF         | Fixed Income           | FLAT       |
| BIV    | 0.245      | Vanguard Intermediate-Term Bond ETF | Fixed Income           | FLAT       |

## Members

|    | ticker | weight | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 26 | AGG    | 0.0185 | iShares Core U.S. Aggregate Bond ETF       | Aggregate Fixed Income      | LONG       |
| 28 | BND    | 0.0185 | Vanguard Total Bond Market ETF             | Aggregate Fixed Income      | LONG       |
| 21 | BIV    | 0.0184 | Vanguard Intermediate-Term Bond ETF        | Fixed Income                | FLAT       |
| 30 | SCHZ   | 0.0184 | Schwab US Aggregate Bond ETF               | Aggregate Fixed Income      | LONG       |
| 31 | SPAB   | 0.0184 | State Street SPDR Portfolio Aggregate B... | Aggregate Fixed Income      | FLAT       |
| 20 | IUSB   | 0.0183 | iShares Core Universal USD Bond ETF        | Aggregate Fixed Income      | LONG       |
| 2  | IEF    | 0.0182 | iShares 7-10 Year Treasury Bond ETF        | Fixed Income                | FLAT       |
| 17 | IGIB   | 0.0181 | iShares 5-10 Year Investment Grade Corp... | Investment Grade Corpora... | FLAT       |
| 7  | FBND   | 0.0180 | Fidelity Total Bond ETF                    | Active Fixed Income         | LONG       |
| 24 | VCIT   | 0.0180 | Vanguard Intermediate-Term Corporate Bo... | Investment Grade Corpora... | FLAT       |
| 43 | SPTI   | 0.0180 | State Street SPDR Portfolio Intermediat... | Fixed Income                | FLAT       |
| 47 | VGIT   | 0.0180 | Vanguard Intermediate-Term Treasury ETF    | Fixed Income                | FLAT       |
| 13 | AGGY   | 0.0179 | WisdomTree Yield Enhanced U.S. Aggregat... | Aggregate Fixed Income      | LONG       |
| 46 | SCHR   | 0.0179 | Schwab Intermediate-Term U.S. Treasury ... | Fixed Income                | FLAT       |
| 6  | GTO    | 0.0178 | Invesco Total Return Bond ETF              | Active Fixed Income         | FLAT       |

|    |      |        |                                             |                             |      |
|----|------|--------|---------------------------------------------|-----------------------------|------|
| 11 | GVI  | 0.0178 | iShares Intermediate Government/Credit ...  | Fixed Income                | LONG |
| 15 | SPIB | 0.0178 | State Street SPDR Portfolio Intermediat...  | Investment Grade Corpora... | FLAT |
| 40 | IEI  | 0.0178 | iShares 3-7 Year Treasury Bond ETF          | Fixed Income                | FLAT |
| 59 | UST  | 0.0178 | ProShares Ultra 7-10 Year Treasury          | Leveraged / Inverse Fixe... | FLAT |
| 27 | USIG | 0.0177 | iShares Broad USD Investment Grade Corp...  | Investment Grade Corpora... | FLAT |
| 50 | MBB  | 0.0177 | iShares MBS ETF                             | Mortgage-Backed Fixed In... | FLAT |
| 51 | VMBS | 0.0177 | Vanguard Mortgage-Backed Securities ETF     | Mortgage-Backed Fixed In... | LONG |
| 5  | BOND | 0.0176 | PIMCO Active Bond Exchange-Traded Fund ...  | Active Fixed Income         | LONG |
| 12 | QLTA | 0.0176 | iShares Aaa A Rated Corporate Bond ETF      | Investment Grade Corpora... | LONG |
| 52 | SPMB | 0.0176 | State Street SPDR Portfolio Mortgage Ba...  | Mortgage-Backed Fixed In... | FLAT |
| 3  | LQD  | 0.0175 | iShares iBoxx \$ Investment Grade Corpor... | Investment Grade Corpora... | LONG |
| 19 | ISTB | 0.0175 | iShares Core 1-5 Year USD Bond ETF          | Fixed Income                | FLAT |
| 10 | CORP | 0.0174 | PIMCO Investment Grade Corporate Bond I...  | Investment Grade Corpora... | FLAT |
| 8  | FLCO | 0.0173 | Franklin Investment Grade Corporate ETF     | Active Fixed Income         | FLAT |
| 22 | BLV  | 0.0173 | Vanguard Long-Term Bond ETF                 | Fixed Income                | FLAT |
| 41 | TLH  | 0.0173 | iShares 10-20 Year Treasury Bond ETF        | Fixed Income                | FLAT |
| 45 | GOVT | 0.0173 | iShares U.S. Treasury Bond ETF              | Fixed Income                | FLAT |
| 35 | IGSB | 0.0172 | iShares 1-5 Year Investment Grade Corpo...  | Investment Grade Corpora... | FLAT |
| 37 | VCSH | 0.0172 | Vanguard Short-Term Corporate Bond ETF      | Investment Grade Corpora... | FLAT |
| 23 | BSV  | 0.0171 | Vanguard Short-Term Bond ETF                | Fixed Income                | FLAT |
| 49 | GNMA | 0.0170 | iShares GNMA Bond ETF                       | Mortgage-Backed Fixed In... | LONG |
| 18 | IGLB | 0.0169 | iShares 10 Year Investment Grade Corpor...  | Investment Grade Corpora... | LONG |
| 44 | SPTL | 0.0169 | State Street SPDR Portfolio Long Term T...  | Fixed Income                | FLAT |
| 16 | SPLB | 0.0168 | State Street SPDR Portfolio Long Term C...  | Investment Grade Corpora... | LONG |
| 25 | VCLT | 0.0168 | Vanguard Long-Term Corporate Bond ETF       | Investment Grade Corpora... | LONG |
| 36 | SLQD | 0.0168 | iShares 0-5 Year Investment Grade Corpo...  | Investment Grade Corpora... | FLAT |
| 48 | VGLT | 0.0168 | Vanguard Long-Term Treasury ETF             | Fixed Income                | FLAT |
| 29 | TOTL | 0.0167 | State Street DoubleLine Total Return Ta...  | Active Fixed Income         | LONG |
| 1  | TLT  | 0.0166 | iShares 20+ Year Treasury Bond ETF          | Fixed Income                | FLAT |
| 39 | AGZ  | 0.0163 | iShares Agency Bond ETF                     | Agency Fixed Income         | LONG |
| 9  | BAB  | 0.0162 | Invesco Taxable Municipal Bond ETF          | Taxable Municipal Bond      | LONG |
| 42 | EDV  | 0.0161 | Vanguard Extended Duration Treasury ETF     | Fixed Income                | FLAT |
| 34 | SPSB | 0.0159 | State Street SPDR Portfolio Short Term ...  | Fixed Income                | FLAT |
| 53 | SHY  | 0.0159 | iShares 1-3 Year Treasury Bond ETF          | Fixed Income                | FLAT |
| 56 | VGSH | 0.0156 | Vanguard Short-Term Treasury ETF            | Cash / Ultra-Short Fixed... | FLAT |
| 57 | BNDX | 0.0155 | Vanguard Total International Bond ETF       | International Fixed Inco... | LONG |
| 33 | BSCQ | 0.0154 | Invesco BulletShares 2026 Corporate Bon...  | Target Maturity Bond        | LONG |
| 54 | SPTS | 0.0153 | State Street SPDR Portfolio Short Term ...  | Cash / Ultra-Short Fixed... | FLAT |
| 55 | SCHO | 0.0153 | Schwab Short-Term U.S. Treasury ETF         | Cash / Ultra-Short Fixed... | FLAT |
| 58 | IAGG | 0.0153 | iShares International Aggregate Bond Fu...  | International Fixed Inco... | LONG |
| 14 | IBDR | 0.0149 | iShares iBonds Dec 2026 Term Corporate ...  | Target Maturity Bond        | FLAT |
| 38 | CMBS | 0.0133 | iShares CMBS Bond ETF                       | Mortgage-Backed Fixed In... | FLAT |
| 4  | LMBS | 0.0130 | First Trust Low Duration Opportunities ...  | Mortgage-Backed Fixed In... | FLAT |
| 32 | LDUR | 0.0118 | PIMCO Enhanced Low Duration Active Exch...  | Fixed Income                | LONG |

**Cluster 5 – Mixed: VTV, SCHV, IWD****Members:** 49   **Var explained:** 0.899   **Signal:** LONG since 2026-04-10 (19 days, 2.2%)**Category breakdown**

| category                     | n_members | abs_weight |
|------------------------------|-----------|------------|
| U.S. Dividend Equity         | 21        | 0.4272     |
| U.S. Large Cap Value         | 10        | 0.2113     |
| Value Equity                 | 4         | 0.0820     |
| U.S. Low Volatility Equity   | 4         | 0.0775     |
| U.S. Fundamental Equity      | 2         | 0.0419     |
| Dividend Equity              | 2         | 0.0396     |
| U.S. Mega Cap Value          | 1         | 0.0211     |
| U.S. Revenue Weighted Equity | 1         | 0.0207     |
| U.S. Large Cap Equity        | 1         | 0.0201     |
| U.S. Quality / Factor Equity | 1         | 0.0200     |
| U.S. Defensive Equity        | 1         | 0.0198     |
| Global Equity                | 1         | 0.0188     |

**Replication basket** – VTV+SDY+SPYV (cor=1.00)   basket lead at lag 1: 0.010

| ticker | weight_ols | name                                     | category             | etf_signal |
|--------|------------|------------------------------------------|----------------------|------------|
| VTV    | 0.465      | Vanguard Value ETF                       | U.S. Large Cap Value | LONG       |
| SDY    | 0.270      | State Street SPDR S&P Dividend ETF       | U.S. Dividend Equity | LONG       |
| SPYV   | 0.243      | State Street SPDR Portfolio S&P 500 V... | U.S. Large Cap Value | LONG       |

**Members**

|    | ticker | weight | name                              | category             | etf_signal |
|----|--------|--------|-----------------------------------|----------------------|------------|
| 28 | VTV    | 0.0213 | Vanguard Value ETF                | U.S. Large Cap Value | LONG       |
| 16 | IWD    | 0.0212 | iShares Russell 1000 Value ETF    | U.S. Large Cap Value | LONG       |
| 20 | VONV   | 0.0212 | Vanguard Russell 1000 Value ETF   | U.S. Large Cap Value | LONG       |
| 26 | SCHV   | 0.0212 | Schwab U.S. Large-Cap Value ETF   | U.S. Large Cap Value | LONG       |
| 10 | NULV   | 0.0211 | Nuveen ESG Large-Cap Value ETF    | U.S. Large Cap Value | LONG       |
| 15 | IWX    | 0.0211 | iShares Russell Top 200 Value ETF | Value Equity         | LONG       |
| 17 | IVE    | 0.0211 | iShares S&P 500 Value ETF         | U.S. Large Cap Value | LONG       |
| 18 | VOOV   | 0.0211 | Vanguard S&P 500 Value ETF        | U.S. Large Cap Value | LONG       |
| 21 | MGV    | 0.0211 | Vanguard Mega Cap Value ETF       | U.S. Mega Cap Value  | LONG       |
| 23 | VYM    | 0.0211 | Vanguard High Dividend Yield ETF  | U.S. Dividend Equity | LONG       |

|    |      |        |                                             |                             |      |
|----|------|--------|---------------------------------------------|-----------------------------|------|
| 24 | ILCV | 0.0211 | iShares Morningstar Large-Cap Value ETF     | U.S. Large Cap Value        | LONG |
| 25 | IUSV | 0.0211 | iShares Core S&P U.S. Value ETF             | U.S. Large Cap Value        | LONG |
| 43 | DTD  | 0.0211 | WisdomTree U.S. Total Dividend Fund         | U.S. Dividend Equity        | LONG |
| 48 | DGRO | 0.0211 | iShares Core Dividend Growth ETF            | U.S. Dividend Equity        | LONG |
| 5  | PRF  | 0.0210 | Invesco RAFI US 1000 ETF                    | U.S. Fundamental Equity     | LONG |
| 27 | SPYV | 0.0210 | State Street SPDR Portfolio S&P 500 Val...  | U.S. Large Cap Value        | LONG |
| 42 | DLN  | 0.0210 | WisdomTree U.S. LargeCap Dividend Fund      | U.S. Dividend Equity        | LONG |
| 14 | FNDX | 0.0209 | Schwab Fundamental U.S. Large Company E...  | U.S. Fundamental Equity     | LONG |
| 29 | FTA  | 0.0208 | First Trust Large Cap Value AlphaDEX Fu...  | Value Equity                | FLAT |
| 40 | PFM  | 0.0208 | Invesco Dividend Achievers ETF              | U.S. Dividend Equity        | LONG |
| 6  | RWL  | 0.0207 | Invesco S&P 500 Revenue ETF                 | U.S. Revenue Weighted Eq... | LONG |
| 11 | SDY  | 0.0207 | State Street SPDR S&P Dividend ETF          | U.S. Dividend Equity        | LONG |
| 22 | SCHD | 0.0207 | Schwab US Dividend Equity ETF               | U.S. Dividend Equity        | LONG |
| 2  | FVD  | 0.0206 | First Trust VL Dividend                     | U.S. Dividend Equity        | FLAT |
| 9  | NOBL | 0.0206 | ProShares S&P 500 Dividend Aristocrats ...  | U.S. Dividend Equity        | FLAT |
| 33 | DVY  | 0.0206 | iShares Select Dividend ETF                 | U.S. Dividend Equity        | LONG |
| 37 | ONEY | 0.0206 | State Street SPDR Russell 1000 Yield Fo...  | U.S. Dividend Equity        | LONG |
| 3  | PWV  | 0.0205 | Invesco Large Cap Value ETF                 | Value Equity                | LONG |
| 32 | SDOG | 0.0205 | ALPS Sector Dividend Dogs ETF               | U.S. Dividend Equity        | LONG |
| 12 | FDVV | 0.0204 | Fidelity High Dividend ETF                  | U.S. Dividend Equity        | LONG |
| 49 | VIG  | 0.0204 | Vanguard Div Appreciation ETF               | U.S. Dividend Equity        | LONG |
| 38 | SPYD | 0.0202 | State Street SPDR Portfolio S&P 500 Hig...  | U.S. Dividend Equity        | LONG |
| 47 | LGLV | 0.0202 | State Street SPDR US Large Cap Low Volat... | U.S. Low Volatility Equi... | FLAT |
| 7  | DHS  | 0.0201 | WisdomTree U.S. High Dividend Fund          | U.S. Dividend Equity        | LONG |
| 45 | DIA  | 0.0201 | State Street SPDR Dow Jones Industrial ...  | U.S. Large Cap Equity       | LONG |
| 39 | FTCS | 0.0200 | First Trust Capital Strength ETF            | U.S. Quality / Factor Eq... | LONG |
| 41 | OUSA | 0.0200 | ALPS O'Shares U.S. Quality Dividend ETF     | Dividend Equity             | LONG |
| 8  | CDC  | 0.0198 | VictoryShares US EQ Income Enhanced Vol...  | U.S. Defensive Equity       | LONG |
| 35 | RPV  | 0.0197 | Invesco S&P 500 Pure Value ETF              | Value Equity                | LONG |
| 4  | FDL  | 0.0196 | First Trust Morningstar ETF                 | U.S. Dividend Equity        | LONG |
| 36 | SPHD | 0.0196 | Invesco S&P 500 High Dividend Low Volat...  | Dividend Equity             | FLAT |
| 46 | USMV | 0.0196 | iShares MSCI USA Min Vol Factor ETF         | U.S. Low Volatility Equi... | LONG |
| 34 | RDIV | 0.0194 | Invesco S&P Ultra Dividend Revenue ETF      | U.S. Dividend Equity        | LONG |
| 19 | HDV  | 0.0193 | iShares Core High Dividend ETF              | U.S. Dividend Equity        | LONG |
| 13 | LVHD | 0.0192 | Franklin U.S. Low Volatility High Divid...  | U.S. Low Volatility Equi... | LONG |
| 30 | PEY  | 0.0192 | Invesco High Yield Equity Dividend Achi...  | U.S. Dividend Equity        | LONG |
| 31 | DIV  | 0.0190 | Global X Super Dividend ETF                 | U.S. Dividend Equity        | LONG |
| 1  | ACWV | 0.0188 | iShares MSCI Global Min Vol Factor ETF      | Global Equity               | LONG |
| 44 | SPLV | 0.0186 | Invesco S&P 500 Low Volatility ETF          | U.S. Low Volatility Equi... | FLAT |

**Cluster 6 – Mixed: IWP, VOT, IMCG****Members:** 35   **Var explained:** 0.781   **Signal:** LONG since 2026-04-10 (19 days, 9.6%)**Category breakdown**

| category                       | n_members | abs_weight |
|--------------------------------|-----------|------------|
| Technology                     | 12        | 0.3384     |
| Industrials                    | 4         | 0.1017     |
| U.S. Mid Cap Growth            | 3         | 0.0944     |
| U.S. Momentum Equity           | 3         | 0.0855     |
| IPO Equity                     | 2         | 0.0599     |
| Convertible Bond               | 2         | 0.0598     |
| Momentum Equity                | 2         | 0.0584     |
| Financials                     | 2         | 0.0578     |
| Growth Equity                  | 1         | 0.0304     |
| U.S. Mid Cap Momentum Equity   | 1         | 0.0285     |
| Consumer Discretionary         | 1         | 0.0285     |
| U.S. Small Cap Momentum Equity | 1         | 0.0284     |
| Unknown                        | 1         | 0.0284     |

**Replication basket** – IWP+BLOK+ARKQ (cor=0.99)   basket lead at lag 1: 0.019

| ticker | weight_ols | name                                     | category            | etf_signal |
|--------|------------|------------------------------------------|---------------------|------------|
| IWP    | 0.684      | iShares Russell Midcap Growth ETF        | U.S. Mid Cap Growth | LONG       |
| BLOK   | 0.089      | Amplify Blockchain Technology ETF        | Technology          | LONG       |
| ARKQ   | 0.150      | ARK Autonomous Technology & Robotics ... | Technology          | LONG       |

**Members**

|    | ticker | weight | name                                       | category             | etf_signal |
|----|--------|--------|--------------------------------------------|----------------------|------------|
| 22 | IWP    | 0.0316 | iShares Russell Midcap Growth ETF          | U.S. Mid Cap Growth  | LONG       |
| 25 | VOT    | 0.0315 | Vanguard Mid-Cap Growth ETF                | U.S. Mid Cap Growth  | LONG       |
| 26 | IMCG   | 0.0314 | iShares Morningstar Mid-Cap Growth ETF     | U.S. Mid Cap Growth  | LONG       |
| 18 | FPX    | 0.0308 | First Trust US Equity Opportunities ETF    | IPO Equity           | LONG       |
| 16 | FTC    | 0.0306 | First Trust Large Cap Growth AlphaDEX F... | Technology           | LONG       |
| 20 | RPG    | 0.0304 | Invesco S&P 500 Pure Growth ETF            | Growth Equity        | LONG       |
| 17 | PDP    | 0.0302 | Invesco Dorsey Wright Momentum ETF         | Momentum Equity      | LONG       |
| 19 | CWB    | 0.0302 | State Street SPDR Bloomberg Convertible... | Convertible Bond     | LONG       |
| 32 | VFMO   | 0.0298 | Vanguard U.S. Momentum Factor ETF          | U.S. Momentum Equity | LONG       |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 23 | ICVT | 0.0296 | iShares Convertible Bond ETF               | Convertible Bond            | LONG |
| 7  | ARKQ | 0.0295 | ARK Autonomous Technology & Robotics ETF   | Technology                  | LONG |
| 15 | FINX | 0.0295 | Global X FinTech ETF                       | Financials                  | LONG |
| 13 | FDN  | 0.0293 | First Trust DJ Internet Index Fund         | Technology                  | LONG |
| 2  | SKYY | 0.0292 | First Trust Cloud Computing ETF            | Technology                  | LONG |
| 11 | IPO  | 0.0291 | Renaissance IPO ETF                        | IPO Equity                  | LONG |
| 12 | PNQI | 0.0289 | Invesco Nasdaq Internet ETF                | Technology                  | LONG |
| 6  | ARKW | 0.0287 | ARK Next Generation Internet ETF           | Technology                  | LONG |
| 5  | ARKK | 0.0285 | ARK Innovation ETF                         | Technology                  | LONG |
| 9  | IBUY | 0.0285 | Amplify Online Retail ETF                  | Consumer Discretionary      | FLAT |
| 21 | XMMO | 0.0285 | Invesco S&P MidCap Momentum ETF            | U.S. Mid Cap Momentum Eq... | LONG |
| 24 | MTUM | 0.0285 | iShares MSCI USA Momentum Factor ETF       | U.S. Momentum Equity        | LONG |
| 1  | IGV  | 0.0284 | iShares Expanded Tech-Software Sector E... | Technology                  | LONG |
| 14 | IPAY | 0.0284 | Amplify Digital Payments ETF               | Financials                  | LONG |
| 27 | DWAS | 0.0284 | Invesco Dorsey Wright SmallCap Momentum... | U.S. Small Cap Momentum ... | LONG |
| 35 | PBS  | 0.0284 |                                            |                             | LONG |
| 8  | FV   | 0.0282 | First Trust Dorsey Wright Focus 5 ETF      | Momentum Equity             | LONG |
| 10 | CIBR | 0.0278 | First Trust NASDAQ Cybersecurity ETF       | Technology                  | LONG |
| 4  | HACK | 0.0276 | Amplify Cybersecurity ETF                  | Technology                  | LONG |
| 34 | PRN  | 0.0275 | Invesco Dorsey Wright Industrials Momen... | Industrials                 | LONG |
| 31 | QMOM | 0.0272 | Alpha Architect U.S. Quantitative Momen... | U.S. Momentum Equity        | LONG |
| 3  | BLOK | 0.0264 | Amplify Blockchain Technology ETF          | Technology                  | LONG |
| 30 | XAR  | 0.0260 | State Street SPDR S&P Aerospace & Defen... | Industrials                 | FLAT |
| 28 | PPA  | 0.0249 | Invesco Aerospace & Defense ETF            | Industrials                 | FLAT |
| 33 | BLCN | 0.0235 | Siren NexGen Economy ETF                   | Technology                  | LONG |
| 29 | ITA  | 0.0234 | iShares U.S. Aerospace & Defense ETF       | Industrials                 | FLAT |

**Cluster 7 – Emerging Markets Equity (19/26)****Members:** 26 **Var explained:** 0.879 **Signal:** LONG since 2026-04-10 (19 days, 8.9%)**Category breakdown**

| category                | n_members | abs_weight |
|-------------------------|-----------|------------|
| Emerging Markets Equity | 19        | 0.7408     |
| Asia Equity             | 2         | 0.0780     |
| Leveraged / Inverse     | 1         | 0.0407     |
| Asia ex-Japan Equity    | 1         | 0.0401     |
| Taiwan Equity           | 1         | 0.0342     |
| Global Dividend Equity  | 1         | 0.0334     |

South Korea Equity 1 0.0328

**Replication basket** – IEMG+DEM+FEMS (cor=1.00) basket lead at lag 1: -0.022

| ticker | weight_ols | name                                     | category                 | etf_signal |
|--------|------------|------------------------------------------|--------------------------|------------|
| IEMG   | 0.850      | iShares Core MSCI Emerging Markets ETF   | Emerging Markets Equi... | LONG       |
| DEM    | 0.161      | WisdomTree Emerging Markets High Divi... | Emerging Markets Equi... | LONG       |
| FEMS   | 0.062      | First Trust Emerging Markets Small Ca... | Emerging Markets Equi... | FLAT       |

## Members

|    | ticker | weight | name                                       | category                | etf_signal |
|----|--------|--------|--------------------------------------------|-------------------------|------------|
| 21 | IEMG   | 0.0408 | iShares Core MSCI Emerging Markets ETF     | Emerging Markets Equity | LONG       |
| 2  | EEM    | 0.0407 | iShares MSCI Emerging Index Fund           | Emerging Markets Equity | LONG       |
| 25 | EDC    | 0.0407 | Direxion Emerging Markets Bull 3X ETF      | Leveraged / Inverse     | LONG       |
| 16 | GEM    | 0.0406 | Goldman Sachs ActiveBeta Emerging Marke... | Emerging Markets Equity | LONG       |
| 23 | SPEM   | 0.0406 | State Street SPDR Portfolio Emerging Ma... | Emerging Markets Equity | LONG       |
| 3  | VWO    | 0.0405 | Vanguard FTSE Emerging Markets ETF         | Emerging Markets Equity | LONG       |
| 20 | ESGE   | 0.0405 | iShares ESG Aware MSCI EM ETF              | Emerging Markets Equity | LONG       |
| 22 | SCHE   | 0.0404 | Schwab Emerging Markets Equity ETF         | Emerging Markets Equity | LONG       |
| 18 | XSOE   | 0.0403 | WisdomTree Emerging Markets Ex-State Ow... | Emerging Markets Equity | LONG       |
| 15 | EMGF   | 0.0402 | iShares Emerging Markets Equity Factor ... | Emerging Markets Equity | LONG       |
| 5  | AAXJ   | 0.0401 | iShares MSCI All Country Asia ex Japan ... | Asia ex-Japan Equity    | LONG       |
| 11 | EEMA   | 0.0396 | iShares MSCI Emerging Markets Asia ETF     | Emerging Markets Equity | LONG       |
| 14 | GMF    | 0.0395 | State Street SPDR S&P Emerging Asia Pac... | Asia Equity             | LONG       |
| 12 | PXH    | 0.0393 | Invesco RAFI Emerging Markets ETF          | Emerging Markets Equity | LONG       |
| 17 | FNDE   | 0.0393 | Schwab Fundamental Emerging Markets Equ... | Emerging Markets Equity | LONG       |
| 19 | EEMV   | 0.0387 | iShares MSCI Emerging Markets Min Vol F... | Emerging Markets Equity | LONG       |
| 8  | DGS    | 0.0385 | WisdomTree Emerging Market SmallCap Fund   | Emerging Markets Equity | FLAT       |
| 10 | AIA    | 0.0385 | iShares Asia 50 ETF                        | Asia Equity             | LONG       |
| 7  | DEM    | 0.0380 | WisdomTree Emerging Markets High Divide... | Emerging Markets Equity | LONG       |
| 6  | EWX    | 0.0374 | State Street SPDR S&P Emerging Markets ... | Emerging Markets Equity | LONG       |
| 4  | FEM    | 0.0369 | First Trust Emerging Markets AlphaDEX F... | Emerging Markets Equity | LONG       |
| 13 | DVYE   | 0.0353 | iShares Emerging Markets Dividend Index... | Emerging Markets Equity | FLAT       |
| 9  | EWT    | 0.0342 | iShares MSCI Taiwan ETF                    | Taiwan Equity           | LONG       |
| 24 | SDIV   | 0.0334 | Global X SuperDividend ETF                 | Global Dividend Equity  | LONG       |
| 26 | FEMS   | 0.0332 | First Trust Emerging Markets Small Cap ... | Emerging Markets Equity | FLAT       |
| 1  | EWY    | 0.0328 | iShares MSCI South Korea ETF               | South Korea Equity      | LONG       |

**Cluster 8 – High Yield Bond (10) + Preferred Stock (7)****Members:** 22   **Var explained:** 0.772   **Signal:** LONG since 2026-04-09 (20 days, 0.8%)**Category breakdown**

| category                      | n_members | abs_weight |
|-------------------------------|-----------|------------|
| High Yield Bond               | 10        | 0.4918     |
| Preferred Stock               | 7         | 0.2893     |
| Emerging Markets Fixed Income | 4         | 0.1695     |
| Unknown                       | 1         | 0.0493     |

**Replication basket** – JNK+PFF+PCY (cor=0.99)   basket lead at lag 1: 0.034

| ticker | weight_ols | name                                     | category                 | etf_signal |
|--------|------------|------------------------------------------|--------------------------|------------|
| JNK    | 0.487      | State Street SPDR Bloomberg High Yiel... | High Yield Bond          | FLAT       |
| PFF    | 0.257      | iShares Preferred and Income Securit...  | Preferred Stock          | LONG       |
| PCY    | 0.115      | Invesco Emerging Markets Sovereign De... | Emerging Markets Fixe... | FLAT       |

**Members**

|    | ticker | weight | name                                        | category                    | etf_signal |
|----|--------|--------|---------------------------------------------|-----------------------------|------------|
| 14 | JNK    | 0.0502 | State Street SPDR Bloomberg High Yield ...  | High Yield Bond             | FLAT       |
| 1  | HYG    | 0.0501 | iShares iBoxx \$ High Yield Corporate Bo... | High Yield Bond             | FLAT       |
| 20 | HYLB   | 0.0500 | Xtrackers USD High Yield Corporate Bond...  | High Yield Bond             | FLAT       |
| 19 | BBHY   | 0.0499 | JPMorgan BetaBuilders USD High Yield Co...  | High Yield Bond             | FLAT       |
| 17 | SHYG   | 0.0497 | iShares 0-5 Year High Yield Corporate B...  | High Yield Bond             | LONG       |
| 15 | SJNK   | 0.0496 | State Street SPDR Bloomberg Short Term ...  | High Yield Bond             | LONG       |
| 8  | PHB    | 0.0493 |                                             |                             | FLAT       |
| 16 | ANGL   | 0.0492 | VanEck Fallen Angel High Yield Bond ETF     | High Yield Bond             | FLAT       |
| 18 | FALN   | 0.0492 | iShares Fallen Angels USD Bond ETF          | High Yield Bond             | FLAT       |
| 6  | HYS    | 0.0488 | PIMCO 0-5 Year High Yield Corporat Bond...  | High Yield Bond             | FLAT       |
| 10 | PFF    | 0.0453 | iShares Preferred and Income Securities...  | Preferred Stock             | LONG       |
| 3  | HYLS   | 0.0450 | First Trust Tactical High Yield ETF         | High Yield Bond             | LONG       |
| 2  | EMB    | 0.0444 | iShares J.P. Morgan USD Emerging Market...  | Emerging Markets Fixed I... | FLAT       |
| 22 | VWOB   | 0.0443 | Vanguard Emerging Markets Government Bo...  | Emerging Markets Fixed I... | FLAT       |
| 12 | PFXF   | 0.0438 | VanEck Preferred Securities ex Financia...  | Preferred Stock             | LONG       |
| 21 | PCY    | 0.0435 | Invesco Emerging Markets Sovereign Debt...  | Emerging Markets Fixed I... | FLAT       |
| 7  | PGX    | 0.0421 | Invesco Preferred ETF                       | Preferred Stock             | FLAT       |
| 11 | PSK    | 0.0416 | State Street SPDR ICE Preferred Securit...  | Preferred Stock             | FLAT       |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 4  | FPE  | 0.0410 | First Trust Preferred Securities and In... | Preferred Stock             | LONG |
| 5  | PGF  | 0.0407 | Invesco Financial Preferred ETF            | Preferred Stock             | FLAT |
| 13 | HYEM | 0.0373 | VanEck Emerging Markets High Yield Bond... | Emerging Markets Fixed I... | LONG |
| 9  | VRP  | 0.0348 | Invesco Variable Rate Preferred ETF        | Preferred Stock             | LONG |

**Cluster 9 – Leveraged / Inverse (16/22)****Members:** 22 **Var explained:** 0.722 **Signal:** FLAT**Category breakdown**

| category                         | n_members | abs_weight |
|----------------------------------|-----------|------------|
| Leveraged / Inverse              | 16        | 0.7654     |
| Volatility                       | 4         | 0.1820     |
| Leveraged / Inverse Fixed Income | 1         | 0.0424     |
| Leveraged / Inverse Commodity    | 1         | 0.0102     |

**Replication basket** – SDS+UVXY+TZA (cor=1.00) basket lead at lag 1: -0.031

| ticker | weight_ols | name                                     | category            | etf_signal |
|--------|------------|------------------------------------------|---------------------|------------|
| SDS    | 0.632      | ProShares UltraShort S&P500              | Leveraged / Inverse | FLAT       |
| UVXY   | 0.118      | ProShares Ultra VIX Short Term Future... | Volatility          | FLAT       |
| TZA    | 0.133      | Direxion Small Cap Bear 3X ETF           | Leveraged / Inverse | FLAT       |

**Members**

|    | ticker | weight | name                               | category            | etf_signal |
|----|--------|--------|------------------------------------|---------------------|------------|
| 16 | SDS    | 0.0534 | ProShares UltraShort S&P500        | Leveraged / Inverse | FLAT       |
| 15 | SH     | 0.0533 | ProShares Short S&P500             | Leveraged / Inverse | FLAT       |
| 20 | SPXU   | 0.0533 | ProShares UltraPro Short S&P500    | Leveraged / Inverse | FLAT       |
| 21 | SPXS   | 0.0533 | Direxion Daily S&P 500 Bear 3X ETF | Leveraged / Inverse | FLAT       |
| 18 | DXD    | 0.0505 | ProShares UltraShort Dow30         | Leveraged / Inverse | FLAT       |
| 22 | SDOW   | 0.0505 | UltraPro Short Dow30               | Leveraged / Inverse | FLAT       |
| 2  | SQQQ   | 0.0502 | ProShares UltraPro Short QQQ       | Leveraged / Inverse | FLAT       |
| 10 | QID    | 0.0502 | ProShares UltraShort QQQ           | Leveraged / Inverse | FLAT       |
| 19 | PSQ    | 0.0502 | ProShares Short QQQ                | Leveraged / Inverse | FLAT       |

|    |      |        |                                            |                             |      |
|----|------|--------|--------------------------------------------|-----------------------------|------|
| 3  | TZA  | 0.0495 | Direxion Small Cap Bear 3X ETF             | Leveraged / Inverse         | FLAT |
| 11 | RWM  | 0.0495 | ProShares Short Russell2000                | Leveraged / Inverse         | FLAT |
| 14 | TWM  | 0.0495 | ProShares UltraShort Russell2000           | Leveraged / Inverse         | FLAT |
| 9  | UVXY | 0.0464 | ProShares Ultra VIX Short Term Futures ... | Volatility                  | FLAT |
| 8  | VIXY | 0.0462 | VIX Short-Term Futures ETF                 | Volatility                  | FLAT |
| 12 | FAZ  | 0.0460 | Direxion Financial Bear 3X ETF             | Leveraged / Inverse         | FLAT |
| 5  | VXZ  | 0.0448 | iPath Series B S&P 500 VIX Mid-Term Fut... | Volatility                  | FLAT |
| 7  | VXX  | 0.0446 | iPath Series B S&P 500 VIX Short-Term F... | Volatility                  | FLAT |
| 17 | SJB  | 0.0424 | ProShares Short High Yield                 | Leveraged / Inverse Fixe... | FLAT |
| 4  | EUM  | 0.0403 | ProShares Short MSCI Emerging Markets      | Leveraged / Inverse         | FLAT |
| 6  | EDZ  | 0.0403 | Direxion Emerging Markets Bear 3X ETF      | Leveraged / Inverse         | FLAT |
| 13 | ERY  | 0.0255 | Direxion Daily Energy Bear 2X ETF          | Leveraged / Inverse         | LONG |
| 1  | SCO  | 0.0102 | ProShares UltraShort Bloomberg Crude Oil   | Leveraged / Inverse Comm... | FLAT |

### Cluster 10 – Healthcare (16/17)

**Members:** 17 **Var explained:** 0.764 **Signal:** FLAT

#### Category breakdown

| category                  | n_members | abs_weight |
|---------------------------|-----------|------------|
| Healthcare                | 16        | 0.9419     |
| U.S. Small Cap Healthcare | 1         | 0.0581     |

**Replication basket** – FXH+XBI+VHT (cor=0.99) basket lead at lag 1: -0.011

| ticker | weight_ols | name                              | category   | etf_signal |
|--------|------------|-----------------------------------|------------|------------|
| FXH    | 0.363      | First Trust Health Care AlphaDEX  | Healthcare | FLAT       |
| XBI    | 0.185      | State Street SPDR S&P Biotech ETF | Healthcare | FLAT       |
| VHT    | 0.472      | Vanguard Health Care ETF          | Healthcare | FLAT       |

#### Members

|    | ticker | weight | name                                | category   | etf_signal |
|----|--------|--------|-------------------------------------|------------|------------|
| 15 | VHT    | 0.0653 | Vanguard Health Care ETF            | Healthcare | FLAT       |
| 16 | FHLC   | 0.0652 | Fidelity MSCI Health Care Index ETF | Healthcare | FLAT       |

|    |      |        |                                            |                           |      |
|----|------|--------|--------------------------------------------|---------------------------|------|
| 9  | FXH  | 0.0645 | First Trust Health Care AlphaDEX           | Healthcare                | FLAT |
| 14 | RSPH | 0.0639 | Invesco S&P 500 Equal Weight Health Car... | Healthcare                | FLAT |
| 13 | IYH  | 0.0637 | iShares U.S. Healthcare ETF                | Healthcare                | FLAT |
| 10 | IXJ  | 0.0622 | iShares Global Healthcare ETF              | Healthcare                | FLAT |
| 1  | XLV  | 0.0620 | State Street Health Care Select Sector ... | Healthcare                | FLAT |
| 3  | IBB  | 0.0604 | iShares Biotechnology ETF                  | Healthcare                | LONG |
| 6  | FBT  | 0.0595 | First Trust Amex Biotech Index Fund        | Healthcare                | FLAT |
| 8  | PSCH | 0.0581 | Invesco S&P SmallCap Health Care ETF       | U.S. Small Cap Healthcare | LONG |
| 7  | XHE  | 0.0574 | State Street SPDR S&P Health Care Equip... | Healthcare                | LONG |
| 12 | IHI  | 0.0574 | iShares U.S. Medical Devices ETF           | Healthcare                | FLAT |
| 5  | PTH  | 0.0545 | Invesco Dorsey Wright Healthcare Moment... | Healthcare                | FLAT |
| 17 | PPH  | 0.0542 | VanEck Pharmaceutical ETF                  | Healthcare                | LONG |
| 4  | XBI  | 0.0535 | State Street SPDR S&P Biotech ETF          | Healthcare                | FLAT |
| 2  | ARKG | 0.0511 | ARK Genomic Revolution ETF                 | Healthcare                | LONG |
| 11 | IHF  | 0.0471 | iShares U.S. Health Care Providers ETF     | Healthcare                | LONG |

### Cluster 11 – Materials (13/17)

**Members:** 17 **Var explained:** 0.756 **Signal:** FLAT

#### Category breakdown

| category          | n_members | abs_weight |
|-------------------|-----------|------------|
| Materials         | 13        | 0.7907     |
| Consumer Staples  | 1         | 0.0589     |
| Energy            | 1         | 0.0571     |
| Norway Equity     | 1         | 0.0519     |
| Industrial Metals | 1         | 0.0414     |

**Replication basket** – GNR+IYM+PICK (cor=0.99) basket lead at lag 1: 0.041

| ticker | weight_ols | name                                     | category  | etf_signal |
|--------|------------|------------------------------------------|-----------|------------|
| GNR    | 0.410      | State Street SPDR S&P Global Natural ... | Materials | FLAT       |
| IYM    | 0.369      | iShares U.S. Basic Materials ETF         | Materials | FLAT       |
| PICK   | 0.207      | iShares MSCI Global Select Metals & M... | Materials | FLAT       |

#### Members

|    | ticker | weight | name                                       | category          | etf_signal |
|----|--------|--------|--------------------------------------------|-------------------|------------|
| 11 | GNR    | 0.0653 | State Street SPDR S&P Global Natural Re... | Materials         | FLAT       |
| 6  | MXI    | 0.0648 | iShares Global Materials ETF               | Materials         | FLAT       |
| 10 | GUNR   | 0.0643 | FlexShares Global Upstream Natural Reso... | Materials         | LONG       |
| 7  | IYM    | 0.0641 | iShares U.S. Basic Materials ETF           | Materials         | FLAT       |
| 12 | PICK   | 0.0627 | iShares MSCI Global Select Metals & Min... | Materials         | FLAT       |
| 9  | VAW    | 0.0622 | Vanguard Materials ETF                     | Materials         | FLAT       |
| 17 | NANR   | 0.0613 | State Street SPDR S&P North American Na... | Materials         | FLAT       |
| 1  | XLB    | 0.0612 | State Street Materials Select Sector SP... | Materials         | FLAT       |
| 8  | RSPM   | 0.0611 | Invesco S&P 500 Equal Weight Materials ... | Materials         | FLAT       |
| 13 | SLX    | 0.0606 | VanEck Steel ETF                           | Materials         | LONG       |
| 4  | XME    | 0.0605 | State Street SPDR S&P Metals & Mining E... | Materials         | LONG       |
| 5  | MOO    | 0.0589 | VanEck Agribusiness ETF                    | Consumer Staples  | LONG       |
| 3  | COPX   | 0.0587 | Global X Copper Miners ETF                 | Materials         | LONG       |
| 16 | IGE    | 0.0571 | iShares North American Natural Resource... | Energy            | LONG       |
| 14 | NORW   | 0.0519 | Global X MSCI Norway ETF                   | Norway Equity     | FLAT       |
| 2  | URA    | 0.0438 | Global X Uranium ETF                       | Materials         | LONG       |
| 15 | DBB    | 0.0414 | Invesco DB Base Metals Fund                | Industrial Metals | FLAT       |

## Cluster 12 – Financials (13/15)

**Members:** 15   **Var explained:** 0.883   **Signal:** FLAT

### Category breakdown

| category            | n_members | abs_weight |
|---------------------|-----------|------------|
| Financials          | 13        | 0.8730     |
| Leveraged / Inverse | 1         | 0.0694     |
| Global Equity       | 1         | 0.0576     |

**Replication basket** – VFH+KBE+KCE (cor=1.00)   basket lead at lag 1: 0.031

| ticker | weight_ols | name                                     | category   | etf_signal |
|--------|------------|------------------------------------------|------------|------------|
| VFH    | 0.790      | Vanguard Financials ETF                  | Financials | FLAT       |
| KBE    | 0.223      | State Street SPDR S&P Bank ETF           | Financials | LONG       |
| KCE    | 0.126      | State Street SPDR S&P Capital Markets... | Financials | LONG       |

**Members**

|    | ticker | weight | name                                       | category            | etf_signal |
|----|--------|--------|--------------------------------------------|---------------------|------------|
| 10 | VFH    | 0.0703 | Vanguard Financials ETF                    | Financials          | FLAT       |
| 11 | FNCL   | 0.0703 | Fidelity MSCI Financials Index ETF         | Financials          | FLAT       |
| 7  | IYF    | 0.0701 | iShares U.S. Financial ETF                 | Financials          | FLAT       |
| 5  | FXO    | 0.0698 | First Trust Financials AlphaDEX            | Financials          | FLAT       |
| 8  | IYG    | 0.0696 | iShares U.S. Financial Services ETF        | Financials          | FLAT       |
| 13 | FAS    | 0.0694 | Direxion Financial Bull 3X ETF             | Leveraged / Inverse | FLAT       |
| 1  | XLFX   | 0.0691 | State Street Financial Select Sector SP... | Financials          | FLAT       |
| 9  | KBWB   | 0.0681 | Invesco KBW Bank ETF                       | Financials          | FLAT       |
| 3  | KBE    | 0.0659 | State Street SPDR S&P Bank ETF             | Financials          | LONG       |
| 15 | KCE    | 0.0659 | State Street SPDR S&P Capital Markets E... | Financials          | LONG       |
| 6  | IAT    | 0.0654 | iShares U.S. Regional Banks ETF            | Financials          | FLAT       |
| 14 | IAI    | 0.0652 | iShares U.S. Broker-Dealers & Securitie... | Financials          | FLAT       |
| 2  | KRE    | 0.0635 | State Street SPDR S&P Regional Banking ... | Financials          | FLAT       |
| 4  | KIE    | 0.0599 | State Street SPDR S&P Insurance ETF        | Financials          | FLAT       |
| 12 | PSP    | 0.0576 | Invesco Global Listed Private Equity ETF   | Global Equity       | LONG       |

**Cluster 13 – Mixed: DBP, GLTR, NUGT****Members:** 15   **Var explained:** 0.831   **Signal:** FLAT**Category breakdown**

| category                      | n_members | abs_weight |
|-------------------------------|-----------|------------|
| Materials                     | 5         | 0.3378     |
| Gold                          | 3         | 0.2011     |
| Precious Metals               | 2         | 0.1404     |
| Leveraged / Inverse Commodity | 2         | 0.1338     |
| Silver                        | 2         | 0.1332     |
| Platinum                      | 1         | 0.0537     |

**Replication basket** – GLTR+GDXJ+AGQ (cor=1.00)   basket lead at lag 1: 0.001

| ticker | weight_ols | name                                     | category        | etf_signal |
|--------|------------|------------------------------------------|-----------------|------------|
| GLTR   | 0.384      | abrdn Physical Precious Metals Basket... | Precious Metals | LONG       |

|      |       |                               |                          |      |
|------|-------|-------------------------------|--------------------------|------|
| GDXJ | 0.423 | VanEck Junior Gold Miners ETF | Materials                | FLAT |
| AGQ  | 0.125 | ProShares Ultra Silver        | Leveraged / Inverse C... | LONG |

**Members**

|    | ticker | weight | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 12 | DBP    | 0.0703 | Invesco DB Precious Metals Fund            | Precious Metals             | FLAT       |
| 7  | GLTR   | 0.0702 | abrdn Physical Precious Metals Basket S... | Precious Metals             | LONG       |
| 15 | NUGT   | 0.0683 | Direxion Daily Gold Miners Index Bull 2... | Leveraged / Inverse Comm... | FLAT       |
| 1  | GDX    | 0.0682 | VanEck Gold Miners ETF                     | Materials                   | FLAT       |
| 2  | GDXJ   | 0.0682 | VanEck Junior Gold Miners ETF              | Materials                   | FLAT       |
| 13 | GOAU   | 0.0675 | US Global GO Gold and Precious Metal Mi... | Materials                   | FLAT       |
| 6  | SIL    | 0.0674 | Global X Silver Miners ETF                 | Materials                   | FLAT       |
| 3  | GLD    | 0.0671 | SPDR Gold Shares                           | Gold                        | FLAT       |
| 10 | IAU    | 0.0671 | iShares Gold Trust Shares                  | Gold                        | FLAT       |
| 11 | SGOL   | 0.0670 | abrdn Physical Gold Shares ETF             | Gold                        | FLAT       |
| 4  | SLV    | 0.0667 | iShares Silver Trust                       | Silver                      | LONG       |
| 5  | SILJ   | 0.0665 | Amplify Junior Silver Miners ETF           | Materials                   | FLAT       |
| 9  | SIVR   | 0.0665 | abrdn Physical Silver Shares ETF           | Silver                      | LONG       |
| 14 | AGQ    | 0.0654 | ProShares Ultra Silver                     | Leveraged / Inverse Comm... | LONG       |
| 8  | PPLT   | 0.0537 | abrdn Physical Platinum Shares ETF         | Platinum                    | LONG       |

**Cluster 14 – Mixed: INDA, INDY, EPI****Members:** 15   **Var explained:** 0.386   **Signal:** FLAT**Category breakdown**

| category             | n_members | abs_weight |
|----------------------|-----------|------------|
| India Equity         | 3         | 0.3151     |
| Unknown              | 2         | 0.1252     |
| Latin America Equity | 1         | 0.0848     |
| Malaysia Equity      | 1         | 0.0765     |
| Brazil Equity        | 1         | 0.0748     |
| Indonesia Equity     | 1         | 0.0704     |
| Chile Equity         | 1         | 0.0698     |
| Philippines Equity   | 1         | 0.0660     |
| Vietnam Equity       | 1         | 0.0486     |

|                          |   |        |
|--------------------------|---|--------|
| Turkey Equity            | 1 | 0.0387 |
| Agricultural Commodities | 1 | 0.0251 |
| Natural Gas              | 1 | 0.0050 |

**Replication basket** – INDA+ILF+EIDO (cor=0.97) basket lead at lag 1: -0.021

| ticker | weight_ols | name                         | category             | etf_signal |
|--------|------------|------------------------------|----------------------|------------|
| INDA   | 0.481      | iShares MSCI India ETF       | India Equity         | FLAT       |
| ILF    | 0.258      | iShares Latin America 40 ETF | Latin America Equity | FLAT       |
| EIDO   | 0.123      | iShares MSCI Indonesia ETF   | Indonesia Equity     | FLAT       |

## Members

|    | ticker | weight | name                              | category                 | etf_signal |
|----|--------|--------|-----------------------------------|--------------------------|------------|
| 2  | INDA   | 0.1058 | iShares MSCI India ETF            | India Equity             | FLAT       |
| 4  | INDY   | 0.1049 | iShares India 50 ETF              | India Equity             | FLAT       |
| 5  | EPI    | 0.1044 | WisdomTree India Earnings Fund    | India Equity             | FLAT       |
| 14 | PIN    | 0.1040 |                                   |                          | FLAT       |
| 7  | ILF    | 0.0848 | iShares Latin America 40 ETF      | Latin America Equity     | FLAT       |
| 11 | EWM    | 0.0765 | iShares MSCI Malaysia Index Fund  | Malaysia Equity          | LONG       |
| 1  | EWZ    | 0.0748 | iShares MSCI Brazil ETF           | Brazil Equity            | FLAT       |
| 9  | EIDO   | 0.0704 | iShares MSCI Indonesia ETF        | Indonesia Equity         | FLAT       |
| 6  | ECH    | 0.0698 | iShares MSCI Chile ETF            | Chile Equity             | FLAT       |
| 10 | EPHE   | 0.0660 | iShares MSCI Philippines ETF      | Philippines Equity       | FLAT       |
| 15 | VNM    | 0.0486 | VanEck Vietnam ETF                | Vietnam Equity           | LONG       |
| 13 | TUR    | 0.0387 | iShares MSCI Turkey ETF           | Turkey Equity            | FLAT       |
| 3  | DBA    | 0.0251 | Invesco DB Agriculture Fund       | Agricultural Commodities | FLAT       |
| 12 | RSX    | 0.0212 |                                   |                          | LONG       |
| 8  | UNG    | 0.0050 | United States Natural Gas Fund LP | Natural Gas              | FLAT       |

## Cluster 15 – China Equity (13/15)

**Members:** 15 **Var explained:** 0.804 **Signal:** LONG since 2026-04-13 (18 days, 2.9%)

### Category breakdown

| category                | n_members | abs_weight |
|-------------------------|-----------|------------|
| China Equity            | 13        | 0.8766     |
| Emerging Markets Equity | 1         | 0.0643     |
| Hong Kong Equity        | 1         | 0.0591     |

**Replication basket** – CXSE+CNYY+MCHI (cor=0.99) basket lead at lag 1: -0.040

| ticker | weight_ols | name                                     | category     | etf_signal |
|--------|------------|------------------------------------------|--------------|------------|
| CXSE   | 0.454      | WisdomTree China ex-State-Owned Enter... | China Equity | LONG       |
| CNYY   | 0.207      | iShares MSCI China A ETF                 | China Equity | LONG       |
| MCHI   | 0.320      | iShares MSCI China ETF                   | China Equity | FLAT       |

## Members

|    | ticker | weight | name                                       | category                | etf_signal |
|----|--------|--------|--------------------------------------------|-------------------------|------------|
| 11 | CXSE   | 0.0736 | WisdomTree China ex-State-Owned Enterpr... | China Equity            | LONG       |
| 9  | GXC    | 0.0735 | State Street SPDR S&P China ETF            | China Equity            | FLAT       |
| 1  | MCHI   | 0.0729 | iShares MSCI China ETF                     | China Equity            | FLAT       |
| 5  | CQQQ   | 0.0712 | Invesco China Technology ETF               | China Equity            | LONG       |
| 2  | FXI    | 0.0711 | iShares China Large-Cap ETF                | China Equity            | FLAT       |
| 7  | CHIQ   | 0.0706 | Global X MSCI China Consumer Discretion... | China Equity            | FLAT       |
| 4  | KWEB   | 0.0679 | KraneShares CSI China Internet ETF         | China Equity            | LONG       |
| 6  | ASHR   | 0.0669 | Xtrackers Harvest CSI 300 China A-Share... | China Equity            | LONG       |
| 8  | CNYY   | 0.0661 | iShares MSCI China A ETF                   | China Equity            | LONG       |
| 13 | KGRN   | 0.0651 | KraneShares MSCI China Clean Technology... | China Equity            | FLAT       |
| 3  | EMQQ   | 0.0643 | EMQQ The Emerging Markets Internet ETF     | Emerging Markets Equity | FLAT       |
| 14 | CNXT   | 0.0595 | VanEck ChiNext Innovators ETF              | China Equity            | LONG       |
| 10 | KBA    | 0.0594 | KraneShares Boseri MSCI China A 50 Conn... | China Equity            | LONG       |
| 12 | EWK    | 0.0591 | iShares MSCI Hong Kong Index Fund          | Hong Kong Equity        | FLAT       |
| 15 | KURE   | 0.0587 | KraneShares MSCI All China Health Care ... | China Equity            | LONG       |

## Cluster 16 – Energy (10/14)

**Members:** 14 **Var explained:** 0.861 **Signal:** LONG since 2026-04-29 (6 days, -4.4%)

### Category breakdown

| category                    | n_members | abs_weight |
|-----------------------------|-----------|------------|
| Energy                      | 10        | 0.7398     |
| MLP / Energy Infrastructure | 4         | 0.2602     |

**Replication basket** – FENY+AMLP+XES (cor=1.00) basket lead at lag 1: 0.044

| ticker | weight_ols | name                                     | category                 | etf_signal |
|--------|------------|------------------------------------------|--------------------------|------------|
| FENY   | 0.657      | Fidelity MSCI Energy Index ETF           | Energy                   | LONG       |
| AMLP   | 0.199      | Alerian MLP ETF                          | MLP / Energy Infrastr... | LONG       |
| XES    | 0.134      | State Street SPDR S&P Oil & Gas Equip... | Energy                   | FLAT       |

## Members

|    | ticker | weight | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 11 | FENY   | 0.0759 | Fidelity MSCI Energy Index ETF             | Energy                      | LONG       |
| 9  | VDE    | 0.0758 | Vanguard Energy ETF                        | Energy                      | LONG       |
| 6  | IYE    | 0.0756 | iShares U.S. Energy ETF                    | Energy                      | LONG       |
| 1  | XLE    | 0.0752 | State Street Energy Select Sector SPDR ... | Energy                      | LONG       |
| 12 | IEO    | 0.0748 | iShares U.S. Oil & Gas Exploration & Pr... | Energy                      | LONG       |
| 10 | IXC    | 0.0747 | iShares Global Energy ETF                  | Energy                      | LONG       |
| 8  | XOP    | 0.0739 | State Street SPDR S&P Oil & Gas Explora... | Energy                      | LONG       |
| 14 | PXE    | 0.0738 | Invesco Energy Exploration & Production... | Energy                      | LONG       |
| 7  | OIH    | 0.0702 | VanEck Oil Services ETF                    | Energy                      | FLAT       |
| 13 | XES    | 0.0699 | State Street SPDR S&P Oil & Gas Equipme... | Energy                      | FLAT       |
| 5  | MLPX   | 0.0688 | Global X MLP & Energy Infrastructure ETF   | MLP / Energy Infrastruct... | LONG       |
| 3  | AMLP   | 0.0660 | Alerian MLP ETF                            | MLP / Energy Infrastruct... | LONG       |
| 4  | MLPA   | 0.0654 | Global X MLP ETF                           | MLP / Energy Infrastruct... | LONG       |
| 2  | EMLP   | 0.0600 | First Trust North American Energy Infra... | MLP / Energy Infrastruct... | LONG       |

## Cluster 17 – Real Estate (14/14)

**Members:** 14 **Var explained:** 0.962 **Signal:** LONG since 2026-04-10 (19 days, 3.8%)

### Category breakdown

| category    | n_members | abs_weight |
|-------------|-----------|------------|
| Real Estate | 14        | 1          |

**Replication basket** – SCHH+RWR+INDS (cor=1.00) basket lead at lag 1: -0.001

| ticker | weight_ols | name                                 | category    | etf_signal |
|--------|------------|--------------------------------------|-------------|------------|
| SCHH   | 0.541      | Schwab U.S. REIT ETF                 | Real Estate | LONG       |
| RWR    | 0.336      | State Street SPDR Dow Jones REIT ETF | Real Estate | LONG       |
| INDS   | 0.100      | Pacer Industrial Real Estate ETF     | Real Estate | FLAT       |

## Members

|    | ticker | weight | name                                       | category    | etf_signal |
|----|--------|--------|--------------------------------------------|-------------|------------|
| 8  | VNQ    | 0.0725 | Vanguard Real Estate ETF                   | Real Estate | LONG       |
| 9  | FREL   | 0.0725 | Fidelity MSCI Real Estate Index ETF        | Real Estate | LONG       |
| 11 | SCHH   | 0.0725 | Schwab U.S. REIT ETF                       | Real Estate | LONG       |
| 2  | IYR    | 0.0724 | iShares U.S. Real Estate ETF               | Real Estate | LONG       |
| 6  | RWR    | 0.0723 | State Street SPDR Dow Jones REIT ETF       | Real Estate | LONG       |
| 10 | USRT   | 0.0723 | iShares Core U.S. REIT ETF                 | Real Estate | LONG       |
| 13 | PSR    | 0.0721 | Invesco Active U.S. Real Estate Fund       | Real Estate | LONG       |
| 1  | XLRE   | 0.0719 | State Street Real Estate Select Sector ... | Real Estate | LONG       |
| 5  | ICF    | 0.0719 | iShares Select U.S. REIT ETF               | Real Estate | LONG       |
| 7  | REET   | 0.0715 | iShares Global REIT ETF                    | Real Estate | LONG       |
| 3  | RWO    | 0.0711 | State Street SPDR Dow Jones Global Real... | Real Estate | LONG       |
| 4  | REZ    | 0.0698 | iShares Residential and Multisector Rea... | Real Estate | LONG       |
| 14 | NURE   | 0.0689 | Nuveen Short-Term REIT ETF                 | Real Estate | LONG       |
| 12 | INDS   | 0.0683 | Pacer Industrial Real Estate ETF           | Real Estate | FLAT       |

## Cluster 18 – Municipal Bond (13/13)

**Members:** 13 **Var explained:** 0.819 **Signal:** FLAT

### Category breakdown

| category | n_members | abs_weight |
|----------|-----------|------------|
|----------|-----------|------------|

|                |    |   |
|----------------|----|---|
| Municipal Bond | 13 | 1 |
|----------------|----|---|

**Replication basket** – VTEB+HYD+CMF (cor=0.99) basket lead at lag 1: 0.188

| ticker | weight_ols | name                             | category       | etf_signal |
|--------|------------|----------------------------------|----------------|------------|
| VTEB   | 0.458      | Vanguard Tax-Exempt Bond ETF     | Municipal Bond | FLAT       |
| HYD    | 0.172      | VanEck High Yield Muni ETF       | Municipal Bond | LONG       |
| CMF    | 0.267      | iShares California Muni Bond ETF | Municipal Bond | FLAT       |

## Members

|    | ticker | weight | name                                       | category       | etf_signal |
|----|--------|--------|--------------------------------------------|----------------|------------|
| 13 | VTEB   | 0.0817 | Vanguard Tax-Exempt Bond ETF               | Municipal Bond | FLAT       |
| 11 | MUB    | 0.0813 | iShares National Muni Bond ETF             | Municipal Bond | FLAT       |
| 6  | CMF    | 0.0801 | iShares California Muni Bond ETF           | Municipal Bond | FLAT       |
| 9  | TFI    | 0.0801 | State Street SPDR Nuveen ICE Municipal ... | Municipal Bond | FLAT       |
| 8  | ITM    | 0.0800 | VanEck Intermediate Muni ETF               | Municipal Bond | FLAT       |
| 1  | FMB    | 0.0796 | First Trust Managed Municipal ETF          | Municipal Bond | FLAT       |
| 7  | NYF    | 0.0793 | iShares New York Muni Bond ETF             | Municipal Bond | FLAT       |
| 5  | PZA    | 0.0780 | Invesco National AMT-Free Municipal Bon... | Municipal Bond | FLAT       |
| 4  | MUNI   | 0.0769 | PIMCO Intermediate Municipal Bond Activ... | Municipal Bond | FLAT       |
| 2  | HYD    | 0.0748 | VanEck High Yield Muni ETF                 | Municipal Bond | LONG       |
| 3  | HYMB   | 0.0720 | State Street SPDR Nuveen ICE High Yield... | Municipal Bond | FLAT       |
| 12 | SUB    | 0.0694 | iShares Short-Term National Muni Bond E... | Municipal Bond | FLAT       |
| 10 | SHM    | 0.0668 | State Street SPDR Nuveen ICE Short Term... | Municipal Bond | LONG       |

## Cluster 19 – Consumer Staples (6/11)

**Members:** 11 **Var explained:** 0.748 **Signal:** LONG since 2026-04-27 (8 days, -0.3%)

### Category breakdown

| category                        | n_members | abs_weight |
|---------------------------------|-----------|------------|
| Consumer Staples                | 6         | 0.5724     |
| Utilities                       | 4         | 0.3530     |
| U.S. Small Cap Consumer Staples | 1         | 0.0746     |

**Replication basket** – VDC+FUTY+PSCC (cor=1.00) basket lead at lag 1: -0.009

| ticker | weight_ols | name                                     | category                 | etf_signal |
|--------|------------|------------------------------------------|--------------------------|------------|
| VDC    | 0.527      | Vanguard Consumer Staples ETF            | Consumer Staples         | LONG       |
| FUTY   | 0.361      | Fidelity MSCI Utilities Index ETF        | Utilities                | FLAT       |
| PSCC   | 0.085      | Invesco S&P SmallCap Consumer Staples... | U.S. Small Cap Consum... | LONG       |

## Members

|    | ticker | weight | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 5  | VDC    | 0.0983 | Vanguard Consumer Staples ETF              | Consumer Staples            | LONG       |
| 6  | FSTA   | 0.0982 | Fidelity MSCI Consumer Staples Index ETF   | Consumer Staples            | LONG       |
| 1  | XLP    | 0.0973 | State Street Consumer Staples Select Se... | Consumer Staples            | LONG       |
| 4  | IYK    | 0.0946 | iShares U.S. Consumer Staples ETF          | Consumer Staples            | LONG       |
| 3  | KXI    | 0.0943 | iShares Global Consumer Staples ETF        | Consumer Staples            | LONG       |
| 11 | PBJ    | 0.0897 | Invesco Food & Beverage ETF                | Consumer Staples            | FLAT       |
| 7  | IDU    | 0.0892 | iShares U.S. Utilities ETF                 | Utilities                   | FLAT       |
| 8  | VPU    | 0.0881 | Vanguard Utilities ETF                     | Utilities                   | FLAT       |
| 9  | FUTY   | 0.0881 | Fidelity MSCI Utilities Index ETF          | Utilities                   | FLAT       |
| 2  | XLU    | 0.0876 | State Street Utilities Select Sector SP... | Utilities                   | FLAT       |
| 10 | PSCC   | 0.0746 | Invesco S&P SmallCap Consumer Staples E... | U.S. Small Cap Consumer ... | LONG       |

## Cluster 20 – Broad Commodities (6/10)

**Members:** 10 **Var explained:** 0.912 **Signal:** FLAT

### Category breakdown

| category                      | n_members | abs_weight |
|-------------------------------|-----------|------------|
| Broad Commodities             | 6         | 0.6001     |
| Oil                           | 2         | 0.1983     |
| Energy Commodities            | 1         | 0.1013     |
| Leveraged / Inverse Commodity | 1         | 0.1003     |

**Replication basket** – GSG+UCO+DBC (cor=1.00) basket lead at lag 1: 0.055

| ticker | weight_ols | name                                     | category                 | etf_signal |
|--------|------------|------------------------------------------|--------------------------|------------|
| GSG    | 0.398      | iShares GSCI Commodity-Indexed Trust ... | Broad Commodities        | FLAT       |
| UCO    | 0.190      | ProShares Ultra Bloomberg Crude Oil      | Leveraged / Inverse C... | FLAT       |
| DBC    | 0.341      | Invesco DB Commodity Index Tracking F... | Broad Commodities        | FLAT       |

## Members

|    | ticker | weight | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 5  | GSG    | 0.1035 | iShares GSCI Commodity-Indexed Trust Fu... | Broad Commodities           | FLAT       |
| 2  | DBC    | 0.1033 | Invesco DB Commodity Index Tracking Fund   | Broad Commodities           | FLAT       |
| 8  | COMT   | 0.1033 | iShares GSCI Commodity Dynamic Roll Str... | Broad Commodities           | FLAT       |
| 7  | PDBC   | 0.1032 | Invesco Optimum Yield Diversified Commo... | Broad Commodities           | FLAT       |
| 9  | DBE    | 0.1013 | Invesco DB Energy Fund                     | Energy Commodities          | FLAT       |
| 10 | UCO    | 0.1003 | ProShares Ultra Bloomberg Crude Oil        | Leveraged / Inverse Comm... | FLAT       |
| 1  | USO    | 0.0997 | United States Oil Fund                     | Oil                         | FLAT       |
| 4  | DBO    | 0.0987 | Invesco DB Oil Fund                        | Oil                         | FLAT       |
| 3  | FTGC   | 0.0934 | First Trust Global Tactical Commodity S... | Broad Commodities           | FLAT       |
| 6  | DJP    | 0.0934 | iPath Bloomberg Commodity Index Total R... | Broad Commodities           | FLAT       |

## Cluster 21 – Inflation-Protected Fixed Income (10/10)

**Members:** 10   **Var explained:** 0.908   **Signal:** LONG since 2026-04-09 (20 days, 0.5%)

### Category breakdown

| category                         | n_members | abs_weight |
|----------------------------------|-----------|------------|
| Inflation-Protected Fixed Income | 10        | 1          |

**Replication basket** – TIP+STPZ+SPIP (cor=1.00)   basket lead at lag 1: -0.019

| ticker | weight_ols | name                                     | category                 | etf_signal |
|--------|------------|------------------------------------------|--------------------------|------------|
| TIP    | 0.627      | iShares TIPS Bond ETF                    | Inflation-Protected F... | LONG       |
| STPZ   | 0.194      | PIMCO 1-5 Year U.S. TIPS Index Exchan... | Inflation-Protected F... | LONG       |
| SPIP   | 0.136      | State Street SPDR Portfolio TIPS ETF     | Inflation-Protected F... | LONG       |

**Members**

|    | ticker | weight | name                                       | category                    | etf_signal |
|----|--------|--------|--------------------------------------------|-----------------------------|------------|
| 4  | TDTF   | 0.1033 | FlexShares iBoxx 5 Year Target Duration... | Inflation-Protected Fixe... | LONG       |
| 6  | TIPX   | 0.1032 | State Street SPDR Bloomberg 1a??10 Year... | Inflation-Protected Fixe... | LONG       |
| 3  | TIP    | 0.1028 | iShares TIPS Bond ETF                      | Inflation-Protected Fixe... | LONG       |
| 8  | SCHP   | 0.1023 | Schwab U.S. TIPS ETF                       | Inflation-Protected Fixe... | LONG       |
| 5  | TDDT   | 0.1007 | FlexShares iBoxx 3 Year Target Duration... | Inflation-Protected Fixe... | LONG       |
| 9  | STIP   | 0.0999 | iShares 0-5 Year TIPS Bond ETF             | Inflation-Protected Fixe... | LONG       |
| 10 | VTIP   | 0.0998 | Vanguard Short-Term Inflation-Protected... | Inflation-Protected Fixe... | LONG       |
| 7  | SPIP   | 0.0997 | State Street SPDR Portfolio TIPS ETF       | Inflation-Protected Fixe... | LONG       |
| 2  | STPZ   | 0.0995 | PIMCO 1-5 Year U.S. TIPS Index Exchange... | Inflation-Protected Fixe... | LONG       |
| 1  | LTPZ   | 0.0886 | Pimco 15 Year U.S. TIPS Index Exchange...  | Inflation-Protected Fixe... | LONG       |

**Cluster 22 – Japan Equity (7/9)****Members:** 9 **Var explained:** 0.888 **Signal:** FLAT**Category breakdown**

| category            | n_members | abs_weight |
|---------------------|-----------|------------|
| Japan Equity        | 7         | 0.7796     |
| Asia-Pacific Equity | 2         | 0.2204     |

**Replication basket** – EWJ+DXJ+VPL (cor=1.00) basket lead at lag 1: -0.032

| ticker | weight_ols | name                                | category            | etf_signal |
|--------|------------|-------------------------------------|---------------------|------------|
| EWJ    | 0.409      | iShares MSCI Japan Index Fund       | Japan Equity        | FLAT       |
| DXJ    | 0.405      | WisdomTree Japan Hedged Equity Fund | Japan Equity        | FLAT       |
| VPL    | 0.172      | Vanguard FTSE Pacific ETF           | Asia-Pacific Equity | LONG       |

**Members**

|   | ticker | weight | name                          | category     | etf_signal |
|---|--------|--------|-------------------------------|--------------|------------|
| 1 | EWJ    | 0.1141 | iShares MSCI Japan Index Fund | Japan Equity | FLAT       |

|   |      |        |                                        |                     |      |
|---|------|--------|----------------------------------------|---------------------|------|
| 8 | BBJP | 0.1140 | JPMorgan BetaBuilders Japan ETF        | Japan Equity        | FLAT |
| 9 | HEWJ | 0.1115 | iShares Currency Hedged MSCI Japan ETF | Japan Equity        | FLAT |
| 6 | DBJP | 0.1114 | Xtrackers MSCI Japan Hedged Equity ETF | Japan Equity        | FLAT |
| 3 | IPAC | 0.1113 | iShares Core MSCI Pacific ETF          | Asia-Pacific Equity | FLAT |
| 5 | FLJH | 0.1106 | Franklin FTSE Japan Hedged ETF         | Japan Equity        | FLAT |
| 2 | DXJ  | 0.1097 | WisdomTree Japan Hedged Equity Fund    | Japan Equity        | FLAT |
| 4 | VPL  | 0.1091 | Vanguard FTSE Pacific ETF              | Asia-Pacific Equity | LONG |
| 7 | EWJV | 0.1082 | iShares MSCI Japan Value ETF           | Japan Equity        | LONG |

**Cluster 23 – Mixed: SRLN, BKLN, FTSL**

**Members:** 9   **Var explained:** 0.560   **Signal:** LONG since 2026-04-02 (24 days, 1.2%)

**Category breakdown**

| category                          | n_members | abs_weight |
|-----------------------------------|-----------|------------|
| Bank Loan / Floating Rate         | 3         | 0.3632     |
| Interest Rate Hedged Fixed Income | 3         | 0.3245     |
| Floating Rate Fixed Income        | 3         | 0.3123     |

**Replication basket** – SRLN+LQDH+FLOT (cor=0.97)   basket lead at lag 1: -0.033

| ticker | weight_ols | name                                     | category                 | etf_signal |
|--------|------------|------------------------------------------|--------------------------|------------|
| SRLN   | 0.346      | State Street Blackstone Senior Loan E... | Bank Loan / Floating ... | LONG       |
| LQDH   | 0.319      | iShares Interest Rate Hedged Corporat... | Interest Rate Hedged ... | LONG       |
| FLOT   | 0.322      | iShares Floating Rate Bond ETF           | Floating Rate Fixed I... | FLAT       |

**Members**

|   | ticker | weight | name                                       | category                    | etf_signal |
|---|--------|--------|--------------------------------------------|-----------------------------|------------|
| 2 | SRLN   | 0.1249 | State Street Blackstone Senior Loan ETF    | Bank Loan / Floating Rate   | LONG       |
| 3 | BKLN   | 0.1214 | Invesco Senior Loan ETF                    | Bank Loan / Floating Rate   | FLAT       |
| 1 | FTSL   | 0.1170 | First Trust Senior Loan Fund               | Bank Loan / Floating Rate   | LONG       |
| 5 | LQDH   | 0.1145 | iShares Interest Rate Hedged Corporate ... | Interest Rate Hedged Fix... | LONG       |
| 7 | FLOT   | 0.1081 | iShares Floating Rate Bond ETF             | Floating Rate Fixed Inco... | FLAT       |
| 6 | IGBH   | 0.1057 | iShares Interest Rate Hedged Long-Term ... | Interest Rate Hedged Fix... | LONG       |

|   |      |        |                                            |                             |      |
|---|------|--------|--------------------------------------------|-----------------------------|------|
| 4 | IGHG | 0.1043 | ProShares Investment Grade-Interest Rat... | Interest Rate Hedged Fix... | LONG |
| 8 | FLRN | 0.1035 | State Street SPDR Bloomberg Investment ... | Floating Rate Fixed Inco... | FLAT |
| 9 | FLTR | 0.1007 | VanEck IG Floating Rate ETF                | Floating Rate Fixed Inco... | LONG |

**Cluster 24 – Cash / Ultra-Short Fixed Income (9/9)****Members:** 9 **Var explained:** 0.456 **Signal:** FLAT**Category breakdown**

| category                        | n_members | abs_weight |
|---------------------------------|-----------|------------|
| Cash / Ultra-Short Fixed Income | 9         | 1          |

**Replication basket** – GSY+NEAR+VNLA (cor=0.95) basket lead at lag 1: 0.028

| ticker | weight_ols | name                                     | category                 | etf_signal |
|--------|------------|------------------------------------------|--------------------------|------------|
| GSY    | 0.376      | Invesco Ultra Short Duration ETF         | Cash / Ultra-Short Fi... | FLAT       |
| NEAR   | 0.124      | iShares Short Duration Bond Active ETF   | Cash / Ultra-Short Fi... | FLAT       |
| VNLA   | 0.139      | Janus Henderson Short Duration Income... | Cash / Ultra-Short Fi... | LONG       |

**Members**

|   | ticker | weight | name                                       | category                    | etf_signal |
|---|--------|--------|--------------------------------------------|-----------------------------|------------|
| 5 | GSY    | 0.1359 | Invesco Ultra Short Duration ETF           | Cash / Ultra-Short Fixed... | FLAT       |
| 3 | FTSM   | 0.1285 | First Trust Enhanced Short Maturity ETF    | Cash / Ultra-Short Fixed... | FLAT       |
| 7 | SHV    | 0.1261 | iShares 0-1 Year Treasury Bond ETF         | Cash / Ultra-Short Fixed... | FLAT       |
| 6 | ICSH   | 0.1244 | iShares Ultra Short Duration Bond Activ... | Cash / Ultra-Short Fixed... | LONG       |
| 4 | NEAR   | 0.1090 | iShares Short Duration Bond Active ETF     | Cash / Ultra-Short Fixed... | FLAT       |
| 2 | MINT   | 0.1063 | PIMCO Enhanced Short Maturity Active Ex... | Cash / Ultra-Short Fixed... | FLAT       |
| 1 | VNLA   | 0.1031 | Janus Henderson Short Duration Income E... | Cash / Ultra-Short Fixed... | LONG       |
| 8 | BIL    | 0.0880 | State Street SPDR Bloomberg 1-3 Month T... | Cash / Ultra-Short Fixed... | LONG       |
| 9 | GBIL   | 0.0787 | Goldman Sachs Access Treasury 0-1 Year ... | Cash / Ultra-Short Fixed... | FLAT       |

**Cluster 25 – Energy (5/8)****Members:** 8 **Var explained:** 0.774 **Signal:** LONG since 2026-04-10 (19 days, 10.5%)**Category breakdown**

| category               | n_members | abs_weight |
|------------------------|-----------|------------|
| Energy                 | 5         | 0.6406     |
| Materials              | 2         | 0.2243     |
| Clean Energy / Climate | 1         | 0.1351     |

**Replication basket** – QCLN+REMX+ICLN (cor=0.99) basket lead at lag 1: 0.001

| ticker | weight_ols | name                                     | category  | etf_signal |
|--------|------------|------------------------------------------|-----------|------------|
| QCLN   | 0.342      | First Trust NASDAQ Clean Edge Green E... | Energy    | LONG       |
| REMX   | 0.205      | VanEck Rare Earth and Strategic Metal... | Materials | LONG       |
| ICLN   | 0.398      | iShares Global Clean Energy ETF          | Energy    | LONG       |

**Members**

|   | ticker | weight | name                                       | category               | etf_signal |
|---|--------|--------|--------------------------------------------|------------------------|------------|
| 7 | ERTH   | 0.1351 | Invesco MSCI Sustainable Future ETF        | Clean Energy / Climate | LONG       |
| 8 | QCLN   | 0.1339 | First Trust NASDAQ Clean Edge Green Ene... | Energy                 | LONG       |
| 6 | PBW    | 0.1312 | Invesco WilderHill Clean Energy ETF        | Energy                 | LONG       |
| 1 | ICLN   | 0.1308 | iShares Global Clean Energy ETF            | Energy                 | LONG       |
| 2 | TAN    | 0.1281 | Invesco Solar ETF                          | Energy                 | LONG       |
| 3 | FAN    | 0.1166 | First Trust Global Wind Energy ETF         | Energy                 | FLAT       |
| 4 | LIT    | 0.1153 | Global X Lithium & Battery Tech ETF        | Materials              | LONG       |
| 5 | REMX   | 0.1090 | VanEck Rare Earth and Strategic Metals ... | Materials              | LONG       |

**Cluster 26 – Currency (5/7)****Members:** 7 **Var explained:** 0.763 **Signal:** LONG since 2026-04-13 (18 days, 0.6%)**Category breakdown**

| category                   | n_members | abs_weight |
|----------------------------|-----------|------------|
| Currency                   | 5         | 0.7014     |
| International Fixed Income | 2         | 0.2986     |

**Replication basket** – UDN+BWX+FXF (cor=0.99) basket lead at lag 1: 0.026

| ticker | weight_ols | name                                     | category                 | etf_signal |
|--------|------------|------------------------------------------|--------------------------|------------|
| UDN    | 0.507      | Invesco DB USD Index Bearish ETF         | Currency                 | LONG       |
| BWX    | 0.337      | State Street SPDR Bloomberg Internati... | International Fixed I... | FLAT       |
| FXF    | 0.139      | Invesco CurrencyShares Swiss Franc Tr... | Currency                 | FLAT       |

## Members

|   | ticker | weight | name                                       | category                    | etf_signal |
|---|--------|--------|--------------------------------------------|-----------------------------|------------|
| 6 | UDN    | 0.1567 | Invesco DB USD Index Bearish ETF           | Currency                    | LONG       |
| 2 | IGOV   | 0.1502 | iShares International Treasury Bond ETF    | International Fixed Inco... | FLAT       |
| 1 | BWX    | 0.1484 | State Street SPDR Bloomberg Internation... | International Fixed Inco... | FLAT       |
| 4 | FXE    | 0.1477 | Invesco CurrencyShares Euro Currency Tr... | Currency                    | FLAT       |
| 5 | FXF    | 0.1392 | Invesco CurrencyShares Swiss Franc Trust   | Currency                    | FLAT       |
| 3 | FXB    | 0.1380 | Invesco CurrencyShares British Pound St... | Currency                    | LONG       |
| 7 | FXY    | 0.1198 | Invesco CurrencyShares Japanese Yen Tru... | Currency                    | LONG       |

**Cluster 27 – Mixed: UUP, EUO, TBT**

**Members:** 6 **Var explained:** 0.459 **Signal:** LONG since 2026-05-01 (4 days, -4.6%)

## Category breakdown

| category                         | n_members | abs_weight |
|----------------------------------|-----------|------------|
| Leveraged / Inverse Fixed Income | 2         | 0.3387     |
| Leveraged / Inverse Commodity    | 2         | 0.2501     |
| Currency                         | 1         | 0.2122     |
| Leveraged / Inverse Currency     | 1         | 0.1990     |

**Replication basket** – DUST+TBT+DZZ (cor=0.98) basket lead at lag 1: 0.004

| ticker | weight_ols | name                                     | category                 | etf_signal |
|--------|------------|------------------------------------------|--------------------------|------------|
| DUST   | 0.194      | Direxion Daily Gold Miners Index Bear... | Leveraged / Inverse C... | LONG       |
| TBT    | 0.275      | ProShares UltraShort Lehman 20 Year T... | Leveraged / Inverse F... | FLAT       |
| DZZ    | 0.081      | DB Gold Double Short ETN due February... | Leveraged / Inverse C... | FLAT       |

**Members**

|   | ticker | weight | name                                       | category                    | etf_signal |
|---|--------|--------|--------------------------------------------|-----------------------------|------------|
| 4 | UUP    | 0.2122 | Invesco DB USD Index Bullish Fund ETF      | Currency                    | FLAT       |
| 6 | EUO    | 0.1990 | ProShares UltraShort Euro                  | Leveraged / Inverse Curr... | FLAT       |
| 5 | TBT    | 0.1697 | ProShares UltraShort Lehman 20 Year Tre... | Leveraged / Inverse Fixe... | FLAT       |
| 3 | TBF    | 0.1690 | ProShares Short 20+ Year Treasury          | Leveraged / Inverse Fixe... | FLAT       |
| 2 | DUST   | 0.1644 | Direxion Daily Gold Miners Index Bear 2... | Leveraged / Inverse Comm... | LONG       |
| 1 | DZZ    | 0.0857 | DB Gold Double Short ETN due February 1... | Leveraged / Inverse Comm... | FLAT       |

## Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

| ticker | name                               | category                 | cluster_id | cluster_size | etf_signal | max_corr_to_others | median_corr_to_others | abs_pc1_loading | compo |
|--------|------------------------------------|--------------------------|------------|--------------|------------|--------------------|-----------------------|-----------------|-------|
| RSX    |                                    |                          | cluster_14 | 15           | LONG       | 0.371              | 0.053                 | 0.00019         |       |
| UNG    | United States Natural Gas Fund ... | Natural Gas              | cluster_14 | 15           | FLAT       | 0.492              | 0.082                 | 0.00016         |       |
| GBIL   | Goldman Sachs Access Treasury 0... | Cash / Ultra-Short Fi... | cluster_24 | 9            | FLAT       | 0.692              | 0.040                 | 0.00006         |       |
| DZZ    | DB Gold Double Short ETN due Fe... | Leveraged / Inverse C... | cluster_27 | 6            | FLAT       | 0.468              | 0.094                 | 0.00026         |       |
| FXV    | Invesco CurrencyShares Japanese... | Currency                 | cluster_26 | 7            | LONG       | 0.732              | 0.093                 | 0.00019         |       |
| VNLA   | Janus Henderson Short Duration ... | Cash / Ultra-Short Fi... | cluster_24 | 9            | LONG       | 0.635              | 0.215                 | 0.00031         |       |
| FTSM   | First Trust Enhanced Short Matu... | Cash / Ultra-Short Fi... | cluster_24 | 9            | FLAT       | 0.787              | 0.127                 | 0.00026         |       |
| BIL    | State Street SPDR Bloomberg 1-3... | Cash / Ultra-Short Fi... | cluster_24 | 9            | LONG       | 0.825              | 0.022                 | 0.00001         |       |
| DBA    | Invesco DB Agriculture Fund        | Agricultural Commodit... | cluster_14 | 15           | FLAT       | 0.591              | 0.178                 | 0.00036         |       |
| SHV    | iShares 0-1 Year Treasury Bond ... | Cash / Ultra-Short Fi... | cluster_24 | 9            | FLAT       | 0.825              | 0.088                 | 0.00019         |       |
| NEAR   | iShares Short Duration Bond Act... | Cash / Ultra-Short Fi... | cluster_24 | 9            | FLAT       | 0.720              | 0.180                 | 0.00036         |       |
| MINT   | PIMCO Enhanced Short Maturity A... | Cash / Ultra-Short Fi... | cluster_24 | 9            | FLAT       | 0.656              | 0.186                 | 0.00043         |       |
| ICSH   | iShares Ultra Short Duration Bo... | Cash / Ultra-Short Fi... | cluster_24 | 9            | LONG       | 0.793              | 0.158                 | 0.00034         |       |
| LDUR   | PIMCO Enhanced Low Duration Act... | Fixed Income             | cluster_4  | 59           | LONG       | 0.844              | 0.120                 | 0.00029         |       |
| TUR    | iShares MSCI Turkey ETF            | Turkey Equity            | cluster_14 | 15           | FLAT       | 0.487              | 0.219                 | 0.00051         |       |
| GSY    | Invesco Ultra Short Duration ETF   | Cash / Ultra-Short Fi... | cluster_24 | 9            | FLAT       | 0.793              | 0.174                 | 0.00039         |       |
| CMBS   | iShares CMBS Bond ETF              | Mortgage-Backed Fixed... | cluster_4  | 59           | FLAT       | 0.876              | 0.155                 | 0.00026         |       |
| FXF    | Invesco CurrencyShares Swiss Fr... | Currency                 | cluster_26 | 7            | FLAT       | 0.818              | 0.199                 | 0.00042         |       |
| LMBS   | First Trust Low Duration Opport... | Mortgage-Backed Fixed... | cluster_4  | 59           | FLAT       | 0.836              | 0.226                 | 0.00039         |       |
| SHM    | State Street SPDR Nuveen ICE Sh... | Municipal Bond           | cluster_18 | 13           | LONG       | 0.877              | 0.248                 | 0.00039         |       |
| VNM    | VanEck Vietnam ETF                 | Vietnam Equity           | cluster_14 | 15           | LONG       | 0.452              | 0.334                 | 0.00076         |       |
| SUB    | iShares Short-Term National Mun... | Municipal Bond           | cluster_18 | 13           | FLAT       | 0.877              | 0.261                 | 0.00046         |       |
| PPLT   | abrdn Physical Platinum Shares ... | Platinum                 | cluster_13 | 15           | LONG       | 0.750              | 0.269                 | 0.00066         |       |
| KURE   | KraneShares MSCI All China Heal... | China Equity             | cluster_15 | 15           | LONG       | 0.804              | 0.270                 | 0.00065         |       |
| DBB    | Invesco DB Base Metals Fund        | Industrial Metals        | cluster_11 | 17           | FLAT       | 0.751              | 0.357                 | 0.00076         |       |
| KBA    | KraneShares Bosera MSCI China A... | China Equity             | cluster_15 | 15           | LONG       | 0.855              | 0.254                 | 0.00065         |       |
| EIDO   | iShares MSCI Indonesia ETF         | Indonesia Equity         | cluster_14 | 15           | FLAT       | 0.514              | 0.371                 | 0.00097         |       |
| EPHE   | iShares MSCI Philippines ETF       | Philippines Equity       | cluster_14 | 15           | FLAT       | 0.578              | 0.407                 | 0.00097         |       |
| FLTR   | VanEck IG Floating Rate ETF        | Floating Rate Fixed I... | cluster_23 | 9            | LONG       | 0.864              | 0.331                 | 0.00065         |       |
| LTPZ   | Pimco 15 Year U.S. TIPS Index E... | Inflation-Protected F... | cluster_21 | 10           | LONG       | 0.920              | 0.223                 | 0.00049         |       |

## Long ETF Positions

| ticker | name                               | category               | cluster    | cluster_signal | entry_date | days | cum_return | prob_sell | nss   |
|--------|------------------------------------|------------------------|------------|----------------|------------|------|------------|-----------|-------|
| QTEC   | First Trust NASDAQ-100-Technolo... | Technology             | cluster_1  | LONG           | 2026-04-08 | 21   | 24.6%      | 0.00      | 1.000 |
| XSD    | State Street SPDR S&P Semicondu... | Technology             | cluster_1  | LONG           | 2026-04-08 | 21   | 45.8%      | 0.00      | 1.000 |
| VLUE   | iShares MSCI USA Value Factor E... | U.S. Value Factor E... | cluster_2  | LONG           | 2026-04-08 | 21   | 15.3%      | 0.00      | 1.000 |
| FLTR   | VanEck IG Floating Rate ETF        | Floating Rate Fixed... | cluster_23 | LONG           | 2026-04-07 | 22   | 0.6%       | 0.00      | 1.000 |
| ICSH   | iShares Ultra Short Duration Bo... | Cash / Ultra-Short ... | cluster_24 | FLAT           | 2022-10-24 | 886  | 19.5%      | 0.00      | 1.000 |
| BIL    | State Street SPDR Bloomberg 1-3... | Cash / Ultra-Short ... | cluster_24 | FLAT           | 2022-04-20 | 1015 | 18.0%      | 0.00      | 1.000 |
| PTNQ   | Pacer Trendpilot 100 ETF           | Trend-Following Equ... | cluster_1  | LONG           | 2026-04-15 | 16   | 9.0%       | 0.00      | 1.000 |
| FTXL   | First Trust Nasdaq Semiconducto... | Technology             | cluster_1  | LONG           | 2026-04-07 | 22   | 45.6%      | 0.00      | 1.000 |
| NXTG   | First Trust Indxx NextG ETF        | Technology             | cluster_1  | LONG           | 2026-04-08 | 21   | 16.3%      | 0.00      | 1.000 |
| EWT    | iShares MSCI Taiwan ETF            | Taiwan Equity          | cluster_7  | LONG           | 2026-04-08 | 21   | 23.5%      | 0.00      | 1.000 |
| BSCQ   | Invesco BulletShares 2026 Corpo... | Target Maturity Bond   | cluster_4  | LONG           | 2025-04-14 | 267  | 5.0%       | 0.00      | 1.000 |
| PSI    | Invesco Semiconductors ETF         | Technology             | cluster_1  | LONG           | 2026-04-07 | 22   | 43.8%      | 0.00      | 1.000 |
| IHF    | iShares U.S. Health Care Provid... | Healthcare             | cluster_10 | FLAT           | 2026-04-09 | 20   | 9.2%       | 0.00      | 1.000 |
| EWM    | iShares MSCI Malaysia Index Fund   | Malaysia Equity        | cluster_14 | FLAT           | 2026-04-17 | 14   | 4.3%       | 0.00      | 1.000 |
| SOXX   | iShares PHLX SOX Semiconductor ... | Technology             | cluster_1  | LONG           | 2026-04-08 | 21   | 32.9%      | 0.00      | 1.000 |
| SMH    | VanEck Semiconductor ETF           | Technology             | cluster_1  | LONG           | 2026-04-08 | 21   | 27.7%      | 0.00      | 1.000 |
| EWY    | iShares MSCI South Korea ETF       | South Korea Equity     | cluster_7  | LONG           | 2026-04-10 | 19   | 27.4%      | 0.00      | 1.000 |
| TQQQ   | ProShares UltraPro QQQ             | Leveraged / Inverse    | cluster_1  | LONG           | 2026-04-10 | 19   | 45.1%      | 0.00      | 1.000 |
| PSCT   | Invesco S&P SmallCap Informatio... | U.S. Small Cap Tech... | cluster_1  | LONG           | 2026-04-07 | 22   | 25.5%      | 0.00      | 1.000 |
| QYLD   | Global X NASDAQ 100 Covered Cal... | Options Income         | cluster_1  | LONG           | 2026-04-08 | 21   | 4.2%       | 0.00      | 1.000 |
| AIQ    | Global X Artificial Intelligenc... | Technology             | cluster_1  | LONG           | 2026-04-10 | 19   | 21.5%      | 0.00      | 1.000 |
| NULV   | Nuveen ESG Large-Cap Value ETF     | U.S. Large Cap Value   | cluster_5  | LONG           | 2026-04-09 | 20   | 5.2%       | 0.00      | 1.000 |
| IYW    | iShares U.S. Technology ETF        | Technology             | cluster_1  | LONG           | 2026-04-09 | 20   | 19.9%      | 0.00      | 1.000 |
| TAN    | Invesco Solar ETF                  | Energy                 | cluster_25 | LONG           | 2026-04-22 | 11   | 4.8%       | 0.00      | 1.000 |
| FV     | First Trust Dorsey Wright Focus... | Momentum Equity        | cluster_6  | LONG           | 2026-04-13 | 18   | 6.3%       | 0.00      | 1.000 |
| QLD    | ProShares Ultra QQQ                | Leveraged / Inverse    | cluster_1  | LONG           | 2026-04-10 | 19   | 28.6%      | 0.00      | 1.000 |
| ICLN   | iShares Global Clean Energy ETF    | Energy                 | cluster_25 | LONG           | 2026-04-10 | 19   | 9.5%       | 0.00      | 1.000 |
| XNTK   | State Street SPDR NYSE Technolo... | Technology             | cluster_1  | LONG           | 2026-04-09 | 20   | 19.7%      | 0.00      | 1.000 |
| QTUM   | Defiance Quantum ETF               | Technology             | cluster_1  | LONG           | 2026-04-09 | 20   | 21.5%      | 0.00      | 1.000 |
| XLK    | State Street Technology Select ... | Technology             | cluster_1  | LONG           | 2026-04-09 | 20   | 19.4%      | 0.00      | 1.000 |
| PBW    | Invesco WilderHill Clean Energy... | Energy                 | cluster_25 | LONG           | 2026-04-13 | 18   | 17.8%      | 0.00      | 1.000 |
| AIA    | iShares Asia 50 ETF                | Asia Equity            | cluster_7  | LONG           | 2026-04-10 | 19   | 16.8%      | 0.00      | 1.000 |
| QQQ    | Invesco QQQ Trust, Series 1        | U.S. Large Cap Grow... | cluster_1  | LONG           | 2026-04-10 | 19   | 13.7%      | 0.00      | 1.000 |
| FNDX   | Schwab Fundamental U.S. Large C... | U.S. Fundamental Eq... | cluster_5  | LONG           | 2026-04-09 | 20   | 4.7%       | 0.00      | 1.000 |
| LIT    | Global X Lithium & Battery Tech... | Materials              | cluster_25 | LONG           | 2026-03-31 | 26   | 20.2%      | 0.00      | 1.000 |
| IXN    | iShares Global Tech ETF            | Technology             | cluster_1  | LONG           | 2026-04-09 | 20   | 19.6%      | 0.00      | 1.000 |
| PTLC   | Pacer Trendpilot US Large Cap E... | U.S. Trend-Followin... | cluster_1  | LONG           | 2026-04-16 | 15   | 4.2%       | 0.00      | 1.000 |
| SKYY   | First Trust Cloud Computing ETF    | Technology             | cluster_6  | LONG           | 2026-04-17 | 14   | 10.5%      | 0.00      | 1.000 |
| SCHD   | Schwab US Dividend Equity ETF      | U.S. Dividend Equity   | cluster_5  | LONG           | 2026-04-17 | 14   | 1.6%       | 0.00      | 1.000 |
| QCLN   | First Trust NASDAQ Clean Edge G... | Energy                 | cluster_25 | LONG           | 2026-04-10 | 19   | 17.6%      | 0.00      | 1.000 |

|      |                                    |                        |            |      |            |    |       |      |       |
|------|------------------------------------|------------------------|------------|------|------------|----|-------|------|-------|
| PRF  | Invesco RAFI US 1000 ETF           | U.S. Fundamental Eq... | cluster_5  | LONG | 2026-04-09 | 20 | 5.3%  | 0.00 | 1.000 |
| IGM  | iShares Expanded Tech Sector ETF   | Technology             | cluster_1  | LONG | 2026-04-09 | 20 | 18.3% | 0.00 | 1.000 |
| FTEC | Fidelity MSCI Information Techn... | Technology             | cluster_1  | LONG | 2026-04-09 | 20 | 18.4% | 0.00 | 1.000 |
| VGT  | Vanguard Information Tech ETF      | Technology             | cluster_1  | LONG | 2026-04-09 | 20 | 18.4% | 0.00 | 1.000 |
| PTF  | Invesco Dorsey Wright Technolog... | Technology             | cluster_1  | LONG | 2026-04-07 | 22 | 25.1% | 0.00 | 1.000 |
| SRLN | State Street Blackstone Senior ... | Bank Loan / Floatin... | cluster_23 | LONG | 2026-04-02 | 24 | 1.8%  | 0.00 | 1.000 |
| MTUM | iShares MSCI USA Momentum Facto... | U.S. Momentum Equity   | cluster_6  | LONG | 2026-04-08 | 21 | 13.2% | 0.00 | 1.000 |
| TDIV | First Trust NASDAQ Technology D... | Technology             | cluster_1  | LONG | 2026-04-10 | 19 | 15.4% | 0.00 | 1.000 |
| IWX  | iShares Russell Top 200 Value E... | Value Equity           | cluster_5  | LONG | 2026-04-09 | 20 | 4.3%  | 0.00 | 1.000 |
| FXL  | First Trust Technology AlphaDEX    | Technology             | cluster_1  | LONG | 2026-04-08 | 21 | 14.7% | 0.00 | 1.000 |
| XLP  | State Street Consumer Staples S... | Consumer Staples       | cluster_19 | LONG | 2026-04-24 | 9  | 0.9%  | 0.00 | 1.000 |
| EIS  | iShares MSCI Israel ETF            | Israel Equity          | cluster_1  | LONG | 2026-04-10 | 19 | 7.4%  | 0.00 | 1.000 |
| XT   | iShares Future Exponential Tech... | Technology             | cluster_1  | LONG | 2026-04-10 | 19 | 10.1% | 0.00 | 1.000 |
| FXV  | Invesco CurrencyShares Japanese... | Currency               | cluster_26 | LONG | 2026-04-30 | 5  | -0.1% | 0.00 | 1.000 |
| AAXJ | iShares MSCI All Country Asia e... | Asia ex-Japan Equity   | cluster_7  | LONG | 2026-04-10 | 19 | 12.1% | 0.00 | 1.000 |
| CIBR | First Trust NASDAQ Cybersecurit... | Technology             | cluster_6  | LONG | 2026-04-17 | 14 | 10.0% | 0.00 | 1.000 |
| FPX  | First Trust US Equity Opportuni... | IPO Equity             | cluster_6  | LONG | 2026-04-08 | 21 | 8.8%  | 0.00 | 1.000 |
| EEMA | iShares MSCI Emerging Markets A... | Emerging Markets Eq... | cluster_7  | LONG | 2026-04-10 | 19 | 12.5% | 0.00 | 1.000 |
| ICVT | iShares Convertible Bond ETF       | Convertible Bond       | cluster_6  | LONG | 2026-04-08 | 21 | 8.6%  | 0.00 | 1.000 |
| BOTZ | Global X Robotics & Artificial ... | Technology             | cluster_1  | LONG | 2026-04-14 | 17 | 8.9%  | 0.00 | 1.000 |
| FTSL | First Trust Senior Loan Fund       | Bank Loan / Floatin... | cluster_23 | LONG | 2026-04-01 | 25 | 1.4%  | 0.00 | 1.000 |
| SUSA | iShares ESG Optimized MSCI USA ... | U.S. ESG Equity        | cluster_1  | LONG | 2026-04-10 | 19 | 8.5%  | 0.00 | 1.000 |
| DSI  | iShares ESG MSCI KLD 400 ETF       | Large Cap Blend Equ... | cluster_1  | LONG | 2026-04-10 | 19 | 10.5% | 0.00 | 1.000 |
| PRN  | Invesco Dorsey Wright Industria... | Industrials            | cluster_6  | LONG | 2026-04-08 | 21 | 11.7% | 0.00 | 1.000 |
| ONEQ | Fidelity Nasdaq Composite Index... | U.S. Equity            | cluster_1  | LONG | 2026-04-09 | 20 | 13.1% | 0.00 | 1.000 |
| ROBO | ROBO Global Robotics and Automa... | Technology             | cluster_1  | LONG | 2026-04-13 | 18 | 11.5% | 0.00 | 1.000 |
| VTIP | Vanguard Short-Term Inflation-P... | Inflation-Protected... | cluster_21 | LONG | 2026-04-01 | 25 | 0.8%  | 0.00 | 1.000 |
| RDIV | Invesco S&P Ultra Dividend Reve... | U.S. Dividend Equity   | cluster_5  | LONG | 2026-04-20 | 13 | 1.2%  | 0.00 | 1.000 |
| QQEW | First Trust Nasdaq-100 Select E... | U.S. Equal Weight E... | cluster_1  | LONG | 2026-04-16 | 15 | 6.5%  | 0.00 | 1.000 |
| JVAL | JPMorgan U.S. Value Factor ETF     | U.S. Value Factor E... | cluster_2  | LONG | 2026-04-09 | 20 | 7.2%  | 0.00 | 1.000 |
| STIP | iShares 0-5 Year TIPS Bond ETF     | Inflation-Protected... | cluster_21 | LONG | 2026-04-01 | 25 | 0.8%  | 0.00 | 1.000 |
| IWD  | iShares Russell 1000 Value ETF     | U.S. Large Cap Value   | cluster_5  | LONG | 2026-04-09 | 20 | 4.0%  | 0.00 | 0.999 |
| VDC  | Vanguard Consumer Staples ETF      | Consumer Staples       | cluster_19 | LONG | 2026-04-23 | 10 | 0.2%  | 0.00 | 0.999 |
| OEF  | iShares S&P 100 Fund               | U.S. Mega Cap Equity   | cluster_1  | LONG | 2026-04-10 | 19 | 9.2%  | 0.00 | 0.999 |
| VONV | Vanguard Russell 1000 Value ETF    | U.S. Large Cap Value   | cluster_5  | LONG | 2026-04-09 | 20 | 4.0%  | 0.00 | 0.999 |
| SPHB | Invesco S&P 500 High Beta ETF      | U.S. Equity            | cluster_1  | LONG | 2026-04-09 | 20 | 13.7% | 0.00 | 0.999 |
| FSTA | Fidelity MSCI Consumer Staples ... | Consumer Staples       | cluster_19 | LONG | 2026-04-23 | 10 | 0.2%  | 0.00 | 0.999 |
| VOOV | Vanguard S&P 500 Value ETF         | U.S. Large Cap Value   | cluster_5  | LONG | 2026-04-10 | 19 | 3.8%  | 0.00 | 0.999 |
| IOO  | iShares Global 100 ETF             | Global Equity          | cluster_1  | LONG | 2026-04-09 | 20 | 9.1%  | 0.00 | 0.999 |
| EES  | WisdomTree U.S. SmallCap Fund      | Small Cap Value Equ... | cluster_2  | LONG | 2026-04-07 | 22 | 8.0%  | 0.00 | 0.999 |
| SPYV | State Street SPDR Portfolio S&P... | U.S. Large Cap Value   | cluster_5  | LONG | 2026-04-10 | 19 | 3.8%  | 0.00 | 0.999 |
| ILCV | iShares Morningstar Large-Cap V... | U.S. Large Cap Value   | cluster_5  | LONG | 2026-04-10 | 19 | 4.3%  | 0.00 | 0.999 |
| CWB  | State Street SPDR Bloomberg Con... | Convertible Bond       | cluster_6  | LONG | 2026-04-08 | 21 | 8.3%  | 0.00 | 0.999 |
| IVE  | iShares S&P 500 Value ETF          | U.S. Large Cap Value   | cluster_5  | LONG | 2026-04-10 | 19 | 3.8%  | 0.00 | 0.999 |
| EMGF | iShares Emerging Markets Equity... | Emerging Markets Eq... | cluster_7  | LONG | 2026-04-10 | 19 | 10.4% | 0.00 | 0.999 |

|      |                                    |                        |            |      |            |    |       |      |       |
|------|------------------------------------|------------------------|------------|------|------------|----|-------|------|-------|
| HACK | Amplify Cybersecurity ETF          | Technology             | cluster_6  | LONG | 2026-04-20 | 13 | 7.2%  | 0.00 | 0.999 |
| ESGU | iShares ESG Aware MSCI USA ETF     | U.S. ESG Equity        | cluster_1  | LONG | 2026-04-10 | 19 | 8.0%  | 0.00 | 0.999 |
| SPYD | State Street SPDR Portfolio S&P... | U.S. Dividend Equity   | cluster_5  | LONG | 2026-04-15 | 16 | 2.2%  | 0.00 | 0.999 |
| CQQQ | Invesco China Technology ETF       | China Equity           | cluster_15 | LONG | 2026-04-15 | 16 | 8.7%  | 0.00 | 0.999 |
| QAI  | NYLI Hedge Multi-Strategy Track... | Alternatives           | cluster_1  | LONG | 2026-04-08 | 21 | 3.3%  | 0.00 | 0.999 |
| EDC  | Direxion Emerging Markets Bull ... | Leveraged / Inverse    | cluster_7  | LONG | 2026-04-13 | 18 | 27.4% | 0.00 | 0.999 |
| SPXL | Direxion Daily S&P 500 Bull 3X ... | Leveraged / Inverse    | cluster_1  | LONG | 2026-04-10 | 19 | 23.7% | 0.00 | 0.999 |
| UPRO | ProShares UltraPro S&P 500         | Leveraged / Inverse    | cluster_1  | LONG | 2026-04-10 | 19 | 23.6% | 0.00 | 0.999 |
| XLG  | Invesco S&P 500 Top 50 ETF         | U.S. Mega Cap Equity   | cluster_1  | LONG | 2026-04-10 | 19 | 9.5%  | 0.00 | 0.999 |
| PWB  | Invesco Large Cap Growth ETF       | Growth Equity          | cluster_1  | LONG | 2026-04-09 | 20 | 11.2% | 0.00 | 0.999 |
| PDP  | Invesco Dorsey Wright Momentum ... | Momentum Equity        | cluster_6  | LONG | 2026-04-08 | 21 | 7.7%  | 0.00 | 0.999 |
| IUSV | iShares Core S&P U.S. Value ETF    | U.S. Large Cap Value   | cluster_5  | LONG | 2026-04-10 | 19 | 3.7%  | 0.00 | 0.998 |
| XSOE | WisdomTree Emerging Markets Ex-... | Emerging Markets Eq... | cluster_7  | LONG | 2026-04-10 | 19 | 10.3% | 0.00 | 0.998 |
| NULG | Nuveen ESG Large-Cap Growth ETF    | U.S. Large Cap Grow... | cluster_1  | LONG | 2026-04-09 | 20 | 12.6% | 0.00 | 0.998 |
| IYK  | iShares U.S. Consumer Staples E... | Consumer Staples       | cluster_19 | LONG | 2026-04-28 | 7  | -0.1% | 0.00 | 0.998 |
| MGC  | Vanguard Mega Cap ETF              | U.S. Mega Cap Equity   | cluster_1  | LONG | 2026-04-10 | 19 | 8.5%  | 0.00 | 0.998 |
| RPG  | Invesco S&P 500 Pure Growth ETF    | Growth Equity          | cluster_6  | LONG | 2026-04-08 | 21 | 11.2% | 0.00 | 0.998 |
| SSO  | ProShares Ultra S&P500             | Leveraged / Inverse    | cluster_1  | LONG | 2026-04-10 | 19 | 15.4% | 0.00 | 0.998 |
| OMFL | Invesco Russell 1000 Dynamic Mu... | U.S. Multifactor Eq... | cluster_1  | LONG | 2026-04-09 | 20 | 5.8%  | 0.00 | 0.998 |
| ILCB | iShares Morningstar Large-Cap E... | U.S. Large Cap Equi... | cluster_1  | LONG | 2026-04-10 | 19 | 7.9%  | 0.00 | 0.998 |
| EEM  | iShares MSCI Emerging Index Fund   | Emerging Markets Eq... | cluster_7  | LONG | 2026-04-10 | 19 | 10.0% | 0.00 | 0.998 |
| VOO  | Vanguard S&P 500 ETF               | U.S. Large Cap Equi... | cluster_1  | LONG | 2026-04-10 | 19 | 7.7%  | 0.00 | 0.998 |
| IVV  | iShares Core S&P 500 ETF           | U.S. Large Cap Equi... | cluster_1  | LONG | 2026-04-10 | 19 | 7.7%  | 0.00 | 0.998 |
| SPY  | State Street SPDR S&P 500 ETF T... | U.S. Large Cap Equi... | cluster_1  | LONG | 2026-04-10 | 19 | 7.7%  | 0.00 | 0.998 |
| VV   | Vanguard Large-Cap ETF             | U.S. Large Cap Equi... | cluster_1  | LONG | 2026-04-10 | 19 | 7.7%  | 0.00 | 0.997 |
| EPS  | WisdomTree U.S. LargeCap Fund      | Large Cap Value Equ... | cluster_1  | LONG | 2026-04-09 | 20 | 7.1%  | 0.00 | 0.997 |
| BLCN | Siren NexGen Economy ETF           | Technology             | cluster_6  | LONG | 2026-04-14 | 17 | 7.8%  | 0.00 | 0.997 |
| IWL  | iShares Russell Top 200 ETF        | U.S. Equity            | cluster_1  | LONG | 2026-04-10 | 19 | 8.2%  | 0.00 | 0.997 |
| RWL  | Invesco S&P 500 Revenue ETF        | U.S. Revenue Weight... | cluster_5  | LONG | 2026-04-09 | 20 | 3.7%  | 0.00 | 0.997 |
| IEMG | iShares Core MSCI Emerging Mark... | Emerging Markets Eq... | cluster_7  | LONG | 2026-04-10 | 19 | 9.7%  | 0.00 | 0.997 |
| NURE | Nuveen Short-Term REIT ETF         | Real Estate            | cluster_17 | LONG | 2026-04-10 | 19 | 5.9%  | 0.00 | 0.997 |
| SCHB | Schwab U.S. Broad Market ETF       | U.S. Total Market E... | cluster_1  | LONG | 2026-04-10 | 19 | 7.5%  | 0.00 | 0.997 |
| SPTM | State Street SPDR Portfolio S&P... | U.S. Total Market E... | cluster_1  | LONG | 2026-04-10 | 19 | 7.5%  | 0.00 | 0.997 |
| IYY  | iShares Dow Jones U.S. ETF         | U.S. Total Market E... | cluster_1  | LONG | 2026-04-10 | 19 | 7.5%  | 0.00 | 0.997 |
| MLPA | Global X MLP ETF                   | MLP / Energy Infrac... | cluster_16 | LONG | 2026-04-30 | 5  | -1.3% | 0.00 | 0.997 |
| ITOT | iShares Core S&P Total U.S. Sto... | U.S. Total Market E... | cluster_1  | LONG | 2026-04-10 | 19 | 7.6%  | 0.00 | 0.997 |
| VUG  | Vanguard Growth ETF                | U.S. Large Cap Grow... | cluster_1  | LONG | 2026-04-10 | 19 | 12.0% | 0.00 | 0.997 |
| ESGE | iShares ESG Aware MSCI EM ETF      | Emerging Markets Eq... | cluster_7  | LONG | 2026-04-10 | 19 | 9.7%  | 0.00 | 0.996 |
| SCHX | Schwab U.S. Large-Cap ETF          | U.S. Large Cap Equi... | cluster_1  | LONG | 2026-04-10 | 19 | 7.6%  | 0.00 | 0.996 |
| GMF  | State Street SPDR S&P Emerging ... | Asia Equity            | cluster_7  | LONG | 2026-04-13 | 18 | 7.2%  | 0.00 | 0.996 |
| VTI  | Vanguard Total Stock Market ETF    | U.S. Total Market E... | cluster_1  | LONG | 2026-04-10 | 19 | 7.5%  | 0.00 | 0.996 |
| MGK  | Vanguard Mega Cap Growth ETF       | U.S. Mega Cap Growth   | cluster_1  | LONG | 2026-04-10 | 19 | 12.3% | 0.00 | 0.996 |
| IGV  | iShares Expanded Tech-Software ... | Technology             | cluster_6  | LONG | 2026-04-20 | 13 | 5.3%  | 0.00 | 0.996 |
| TDTT | FlexShares iBoxx 3 Year Target ... | Inflation-Protected... | cluster_21 | LONG | 2026-04-02 | 24 | 0.7%  | 0.00 | 0.996 |
| STPZ | PIMCO 1-5 Year U.S. TIPS Index ... | Inflation-Protected... | cluster_21 | LONG | 2026-04-06 | 23 | 0.7%  | 0.00 | 0.996 |

|      |                                    |                        |            |      |            |    |       |      |       |
|------|------------------------------------|------------------------|------------|------|------------|----|-------|------|-------|
| SPYX | State Street SPDR S&P 500 Fossi... | U.S. Equity            | cluster_1  | LONG | 2026-04-10 | 19 | 7.9%  | 0.00 | 0.995 |
| VTHR | Vanguard Russell 3000 ETF          | U.S. Total Market E... | cluster_1  | LONG | 2026-04-10 | 19 | 7.4%  | 0.00 | 0.995 |
| IWB  | iShares Russell 1000 ETF           | U.S. Large Cap Equi... | cluster_1  | LONG | 2026-04-10 | 19 | 7.4%  | 0.00 | 0.995 |
| IVW  | iShares S&P 500 Growth ETF         | U.S. Large Cap Grow... | cluster_1  | LONG | 2026-04-10 | 19 | 11.1% | 0.01 | 0.995 |
| ILCG | iShares Morningstar Large-Cap G... | U.S. Large Cap Grow... | cluster_1  | LONG | 2026-04-10 | 19 | 11.6% | 0.01 | 0.995 |
| AML  | Alerian MLP ETF                    | MLP / Energy Infras... | cluster_16 | LONG | 2026-04-30 | 5  | -1.4% | 0.00 | 0.995 |
| SPYG | State Street SPDR Portfolio S&P... | U.S. Large Cap Grow... | cluster_1  | LONG | 2026-04-10 | 19 | 11.2% | 0.01 | 0.995 |
| SCHG | Schwab U.S. Large-Cap Growth ETF   | U.S. Large Cap Grow... | cluster_1  | LONG | 2026-04-10 | 19 | 11.1% | 0.01 | 0.995 |
| VOOG | Vanguard S&P 500 Growth ETF        | U.S. Large Cap Grow... | cluster_1  | LONG | 2026-04-10 | 19 | 11.1% | 0.01 | 0.995 |
| SCHA | Schwab U.S. Small-Cap ETF          | U.S. Small Cap Equi... | cluster_2  | LONG | 2026-04-08 | 21 | 8.0%  | 0.01 | 0.995 |
| MGV  | Vanguard Mega Cap Value ETF        | U.S. Mega Cap Value    | cluster_5  | LONG | 2026-04-10 | 19 | 2.6%  | 0.01 | 0.995 |
| VONE | Vanguard Russell 1000 ETF          | U.S. Large Cap Equi... | cluster_1  | LONG | 2026-04-10 | 19 | 7.3%  | 0.01 | 0.994 |
| CATH | Global X S&P 500 Catholic Value... | Global Equity          | cluster_1  | LONG | 2026-04-10 | 19 | 7.6%  | 0.01 | 0.994 |
| IUSG | iShares Core S&P U.S. Growth ETF   | U.S. Large Cap Grow... | cluster_1  | LONG | 2026-04-10 | 19 | 10.9% | 0.01 | 0.994 |
| IWV  | iShares Russell 3000 Fund          | U.S. Total Market E... | cluster_1  | LONG | 2026-04-10 | 19 | 7.3%  | 0.01 | 0.994 |
| FXC  | Invesco CurrencyShares Canadian... | Currency               | cluster_3  | FLAT | 2026-04-17 | 14 | 0.3%  | 0.01 | 0.994 |
| ARKW | ARK Next Generation Internet ETF   | Technology             | cluster_6  | LONG | 2026-04-14 | 17 | 11.3% | 0.01 | 0.994 |
| IWO  | iShares Russell 2000 Growth Fund   | U.S. Small Cap Grow... | cluster_2  | LONG | 2026-04-09 | 20 | 10.2% | 0.01 | 0.994 |
| RFV  | Invesco S&P MidCap 400 Pure Val... | Leveraged / Inverse    | cluster_2  | LONG | 2026-04-07 | 22 | 5.1%  | 0.01 | 0.994 |
| TNA  | Direxion Small Cap Bull 3X ETF     | Leveraged / Inverse    | cluster_2  | LONG | 2026-04-09 | 20 | 23.3% | 0.01 | 0.994 |
| VTWG | Vanguard Russell 2000 Growth ETF   | U.S. Small Cap Grow... | cluster_2  | LONG | 2026-04-09 | 20 | 10.2% | 0.01 | 0.994 |
| GEM  | Goldman Sachs ActiveBeta Emergi... | Emerging Markets Eq... | cluster_7  | LONG | 2026-04-10 | 19 | 8.9%  | 0.01 | 0.994 |
| KXI  | iShares Global Consumer Staples... | Consumer Staples       | cluster_19 | LONG | 2026-04-28 | 7  | 1.0%  | 0.01 | 0.994 |
| IMCG | iShares Morningstar Mid-Cap Gro... | U.S. Mid Cap Growth    | cluster_6  | LONG | 2026-04-09 | 20 | 7.8%  | 0.01 | 0.994 |
| VFMO | Vanguard U.S. Momentum Factor E... | U.S. Momentum Equity   | cluster_6  | LONG | 2026-04-08 | 21 | 7.8%  | 0.01 | 0.993 |
| XSMO | Invesco S&P SmallCap Momentum E... | U.S. Small Cap Mome... | cluster_2  | LONG | 2026-04-07 | 22 | 10.3% | 0.01 | 0.993 |
| HDV  | iShares Core High Dividend ETF     | U.S. Dividend Equity   | cluster_5  | LONG | 2026-04-29 | 6  | -0.4% | 0.01 | 0.993 |
| PSC  | Principal U.S. Small-Cap ETF       | U.S. Small Cap Equi... | cluster_2  | LONG | 2026-04-09 | 20 | 6.5%  | 0.01 | 0.993 |
| PSCH | Invesco S&P SmallCap Health Car... | U.S. Small Cap Heal... | cluster_10 | FLAT | 2026-04-08 | 21 | 6.9%  | 0.01 | 0.993 |
| TILT | FlexShares Mornigstar US Market... | U.S. Multifactor Eq... | cluster_1  | LONG | 2026-04-09 | 20 | 6.1%  | 0.01 | 0.993 |
| DHS  | WisdomTree U.S. High Dividend F... | U.S. Dividend Equity   | cluster_5  | LONG | 2026-04-10 | 19 | 1.2%  | 0.01 | 0.992 |
| IWM  | iShares Russell 2000 Index Fund    | U.S. Small Cap Equi... | cluster_2  | LONG | 2026-04-09 | 20 | 7.7%  | 0.01 | 0.991 |
| UWM  | ProShares Ultra Russell2000        | Leveraged / Inverse    | cluster_2  | LONG | 2026-04-09 | 20 | 15.3% | 0.01 | 0.991 |
| VTWO | Vanguard Russell 2000 ETF          | U.S. Small Cap Equi... | cluster_2  | LONG | 2026-04-09 | 20 | 7.8%  | 0.01 | 0.991 |
| PFXF | VanEck Preferred Securities ex ... | Preferred Stock        | cluster_8  | LONG | 2026-04-13 | 18 | 4.1%  | 0.01 | 0.991 |
| VOT  | Vanguard Mid-Cap Growth ETF        | U.S. Mid Cap Growth    | cluster_6  | LONG | 2026-04-13 | 18 | 6.5%  | 0.01 | 0.990 |
| VIOG | Vanguard S&P Small-Cap 600 Grow... | U.S. Small Cap Grow... | cluster_2  | LONG | 2026-04-08 | 21 | 6.3%  | 0.01 | 0.990 |
| JHML | John Hancock Multifactor Large ... | U.S. Large Cap Equi... | cluster_1  | LONG | 2026-04-09 | 20 | 5.7%  | 0.01 | 0.990 |
| BLOK | Amplify Blockchain Technology E... | Technology             | cluster_6  | LONG | 2026-04-10 | 19 | 18.2% | 0.01 | 0.990 |
| QDF  | FlexShares Quality Dividend Ind... | Dividend Equity        | cluster_1  | LONG | 2026-04-10 | 19 | 5.7%  | 0.01 | 0.990 |
| SLX  | VanEck Steel ETF                   | Materials              | cluster_11 | FLAT | 2026-04-08 | 21 | 8.8%  | 0.01 | 0.990 |
| EEMV | iShares MSCI Emerging Markets M... | Emerging Markets Eq... | cluster_7  | LONG | 2026-04-09 | 20 | 6.7%  | 0.01 | 0.990 |
| SLYG | State Street SPDR S&P 600 Small... | U.S. Small Cap Grow... | cluster_2  | LONG | 2026-04-08 | 21 | 6.5%  | 0.01 | 0.989 |
| IWC  | iShares Microcap ETF               | U.S. Micro Cap Equi... | cluster_2  | LONG | 2026-04-08 | 21 | 9.2%  | 0.01 | 0.989 |
| SCHV | Schwab U.S. Large-Cap Value ETF    | U.S. Large Cap Value   | cluster_5  | LONG | 2026-04-09 | 20 | 2.5%  | 0.01 | 0.989 |

|      |                                    |                        |            |      |            |    |       |      |       |
|------|------------------------------------|------------------------|------------|------|------------|----|-------|------|-------|
| RPV  | Invesco S&P 500 Pure Value ETF     | Value Equity           | cluster_5  | LONG | 2026-04-09 | 20 | 1.3%  | 0.01 | 0.988 |
| IJT  | iShares S&P SmallCap 600 Growth... | U.S. Small Cap Grow... | cluster_2  | LONG | 2026-04-08 | 21 | 6.5%  | 0.01 | 0.988 |
| CNYA | iShares MSCI China A ETF           | China Equity           | cluster_15 | LONG | 2026-04-13 | 18 | 4.5%  | 0.01 | 0.988 |
| IPO  | Renaissance IPO ETF                | IPO Equity             | cluster_6  | LONG | 2026-04-10 | 19 | 14.1% | 0.01 | 0.988 |
| REZ  | iShares Residential and Multise... | Real Estate            | cluster_17 | LONG | 2026-04-14 | 17 | 3.2%  | 0.01 | 0.987 |
| KBA  | KraneShares Boseri MSCI China A... | China Equity           | cluster_15 | LONG | 2026-04-09 | 20 | 8.6%  | 0.01 | 0.987 |
| VPL  | Vanguard FTSE Pacific ETF          | Asia-Pacific Equity    | cluster_22 | FLAT | 2026-04-09 | 20 | 6.8%  | 0.01 | 0.987 |
| SCHM | Schwab U.S. Mid Cap ETF            | U.S. Mid Cap Equity    | cluster_2  | LONG | 2026-04-09 | 20 | 5.7%  | 0.01 | 0.986 |
| RWJ  | Invesco S&P SmallCap 600 Revenu... | U.S. Small Cap Equi... | cluster_2  | LONG | 2026-04-08 | 21 | 6.5%  | 0.01 | 0.986 |
| LRGF | iShares U.S. Equity Factor ETF     | U.S. Multifactor Eq... | cluster_1  | LONG | 2026-04-09 | 20 | 6.6%  | 0.02 | 0.985 |
| PRNT | 3D Printing (The) ETF              | Technology             | cluster_1  | LONG | 2026-04-14 | 17 | 8.4%  | 0.02 | 0.985 |
| GSLC | Goldman Sachs ActiveBeta U.S. L... | Commodity              | cluster_1  | LONG | 2026-04-10 | 19 | 6.7%  | 0.02 | 0.985 |
| DEM  | WisdomTree Emerging Markets Hig... | Emerging Markets Eq... | cluster_7  | LONG | 2026-04-08 | 21 | 5.1%  | 0.02 | 0.984 |
| VIOO | Vanguard S&P Small-Cap 600 ETF     | U.S. Small Cap Equi... | cluster_2  | LONG | 2026-04-08 | 21 | 6.1%  | 0.02 | 0.984 |
| SPSM | State Street SPDR Portfolio S&P... | U.S. Small Cap Equi... | cluster_2  | LONG | 2026-04-08 | 21 | 6.1%  | 0.02 | 0.982 |
| XMMO | Invesco S&P MidCap Momentum ETF    | U.S. Mid Cap Moment... | cluster_6  | LONG | 2026-04-07 | 22 | 11.1% | 0.02 | 0.982 |
| IJR  | iShares Core S&P Small-Cap ETF     | U.S. Small Cap Equi... | cluster_2  | LONG | 2026-04-08 | 21 | 6.0%  | 0.02 | 0.982 |
| ASHR | Xtrackers Harvest CSI 300 China... | China Equity           | cluster_15 | LONG | 2026-04-13 | 18 | 4.1%  | 0.02 | 0.981 |
| IWY  | iShares Russell Top 200 Growth ... | Growth Equity          | cluster_1  | LONG | 2026-04-13 | 18 | 8.9%  | 0.02 | 0.981 |
| DIV  | Global X Super Dividend ETF        | U.S. Dividend Equity   | cluster_5  | LONG | 2026-04-09 | 20 | 0.7%  | 0.02 | 0.980 |
| VTV  | Vanguard Value ETF                 | U.S. Large Cap Value   | cluster_5  | LONG | 2026-04-09 | 20 | 1.5%  | 0.02 | 0.980 |
| FTC  | First Trust Large Cap Growth Al... | Technology             | cluster_6  | LONG | 2026-04-09 | 20 | 6.4%  | 0.02 | 0.980 |
| PAVE | Global X U.S. Infrastructure De... | Industrials            | cluster_2  | LONG | 2026-04-09 | 20 | 4.1%  | 0.02 | 0.979 |
| TIPX | State Street SPDR Bloomberg 1a?... | Inflation-Protected... | cluster_21 | LONG | 2026-04-09 | 20 | 0.6%  | 0.02 | 0.979 |
| SPHQ | Invesco S&P 500 Quality ETF        | U.S. Quality / Fact... | cluster_1  | LONG | 2026-04-10 | 19 | 4.0%  | 0.02 | 0.979 |
| IWF  | iShares Russell 1000 Growth Fund   | U.S. Large Cap Grow... | cluster_1  | LONG | 2026-04-13 | 18 | 8.5%  | 0.02 | 0.978 |
| VONG | Vanguard Russell 1000 Growth ETF   | U.S. Large Cap Grow... | cluster_1  | LONG | 2026-04-13 | 18 | 8.5%  | 0.02 | 0.978 |
| PRFZ | Invesco RAFI US 1500 Small-Mid ... | U.S. Small/Mid Cap ... | cluster_2  | LONG | 2026-04-09 | 20 | 6.2%  | 0.02 | 0.978 |
| ISMD | Inspire Small/Mid Cap ETF          | U.S. Small/Mid Cap ... | cluster_2  | LONG | 2026-04-08 | 21 | 7.9%  | 0.02 | 0.976 |
| ACWI | iShares MSCI ACWI ETF              | Global Equity          | cluster_1  | LONG | 2026-04-10 | 19 | 6.1%  | 0.02 | 0.976 |
| SPGM | State Street SPDR Portfolio MSC... | Global Equity          | cluster_1  | LONG | 2026-04-09 | 20 | 5.9%  | 0.02 | 0.976 |
| FDN  | First Trust DJ Internet Index F... | Technology             | cluster_6  | LONG | 2026-04-08 | 21 | 9.3%  | 0.02 | 0.976 |
| REMX | VanEck Rare Earth and Strategic... | Materials              | cluster_25 | LONG | 2026-04-10 | 19 | 12.8% | 0.03 | 0.975 |
| QUS  | State Street SPDR MSCI USA Stra... | U.S. Quality / Fact... | cluster_1  | LONG | 2026-04-13 | 18 | 2.8%  | 0.03 | 0.975 |
| VBK  | Vanguard Small-Cap Growth ETF      | U.S. Small Cap Grow... | cluster_2  | LONG | 2026-04-09 | 20 | 8.0%  | 0.03 | 0.974 |
| CNXT | VanEck ChiNext Innovators ETF      | China Equity           | cluster_15 | LONG | 2026-04-10 | 19 | 11.1% | 0.03 | 0.973 |
| IMCB | iShares Morningstar Mid-Cap ETF    | U.S. Mid Cap Equity    | cluster_2  | LONG | 2026-04-09 | 20 | 4.2%  | 0.03 | 0.973 |
| IWN  | iShares Russell 2000 Value ETF     | U.S. Small Cap Value   | cluster_2  | LONG | 2026-04-08 | 21 | 5.9%  | 0.03 | 0.972 |
| VT   | Vanguard Total World Stock Inde... | Global Equity          | cluster_1  | LONG | 2026-04-10 | 19 | 6.0%  | 0.03 | 0.970 |
| PSCC | Invesco S&P SmallCap Consumer S... | U.S. Small Cap Cons... | cluster_19 | LONG | 2026-04-17 | 14 | -0.5% | 0.03 | 0.969 |
| CRBN | iShares Low Carbon Optimized MS... | Global Equity          | cluster_1  | LONG | 2026-04-10 | 19 | 6.6%  | 0.03 | 0.969 |
| VO   | Vanguard Mid-Cap ETF               | U.S. Mid Cap Equity    | cluster_2  | LONG | 2026-04-10 | 19 | 4.3%  | 0.03 | 0.968 |
| FYX  | First Trust Small Cap Core Alph... | U.S. Small Cap Equi... | cluster_2  | LONG | 2026-04-07 | 22 | 9.3%  | 0.03 | 0.968 |
| VTWV | Vanguard Russell 2000 Value ETF    | U.S. Small Cap Value   | cluster_2  | LONG | 2026-04-08 | 21 | 5.8%  | 0.03 | 0.967 |
| URTH | iShares MSCI World ETF             | Global Equity          | cluster_1  | LONG | 2026-04-10 | 19 | 5.7%  | 0.04 | 0.964 |

|      |                                    |                        |            |      |            |    |       |      |       |
|------|------------------------------------|------------------------|------------|------|------------|----|-------|------|-------|
| VNLA | Janus Henderson Short Duration ... | Cash / Ultra-Short ... | cluster_24 | FLAT | 2026-04-06 | 23 | 0.4%  | 0.04 | 0.964 |
| DWAS | Invesco Dorsey Wright SmallCap ... | U.S. Small Cap Mome... | cluster_6  | LONG | 2026-04-07 | 22 | 11.0% | 0.04 | 0.961 |
| VXF  | Vanguard Extended Market ETF       | U.S. Extended Marke... | cluster_2  | LONG | 2026-04-09 | 20 | 6.5%  | 0.04 | 0.960 |
| TDTF | FlexShares iBoxx 5 Year Target ... | Inflation-Protected... | cluster_21 | LONG | 2026-04-10 | 19 | 0.7%  | 0.04 | 0.958 |
| AOA  | iShares Core 80/20 Aggressive A... | Asset Allocation       | cluster_1  | LONG | 2026-04-10 | 19 | 4.7%  | 0.04 | 0.955 |
| OMFS | Invesco Russell 2000 Dynamic Mu... | U.S. Equity            | cluster_2  | LONG | 2026-04-08 | 21 | 6.3%  | 0.05 | 0.953 |
| SLYV | State Street SPDR S&P 600 Small... | U.S. Small Cap Value   | cluster_2  | LONG | 2026-04-08 | 21 | 5.6%  | 0.05 | 0.951 |
| DLN  | WisdomTree U.S. LargeCap Divide... | U.S. Dividend Equity   | cluster_5  | LONG | 2026-04-10 | 19 | 2.8%  | 0.05 | 0.949 |
| IJS  | iShares S&P SmallCap 600 Value ... | U.S. Small Cap Value   | cluster_2  | LONG | 2026-04-08 | 21 | 5.6%  | 0.05 | 0.947 |
| SPEM | State Street SPDR Portfolio Eme... | Emerging Markets Eq... | cluster_7  | LONG | 2026-04-10 | 19 | 6.2%  | 0.06 | 0.945 |
| MLPX | Global X MLP & Energy Infrastru... | MLP / Energy Infras... | cluster_16 | LONG | 2026-04-30 | 5  | -3.4% | 0.00 | 0.945 |
| DTD  | WisdomTree U.S. Total Dividend ... | U.S. Dividend Equity   | cluster_5  | LONG | 2026-04-10 | 19 | 2.9%  | 0.06 | 0.943 |
| VIOV | Vanguard S&P Small-Cap 600 Valu... | U.S. Small Cap Value   | cluster_2  | LONG | 2026-04-08 | 21 | 5.3%  | 0.06 | 0.943 |
| FNDA | Schwab Fundamental U.S. Small C... | U.S. Small Cap Fund... | cluster_2  | LONG | 2026-04-09 | 20 | 5.1%  | 0.06 | 0.938 |
| QMOM | Alpha Architect U.S. Quantitati... | U.S. Momentum Equity   | cluster_6  | LONG | 2026-04-08 | 21 | 5.7%  | 0.06 | 0.936 |
| VWO  | Vanguard FTSE Emerging Markets ... | Emerging Markets Eq... | cluster_7  | LONG | 2026-04-10 | 19 | 6.1%  | 0.06 | 0.936 |
| IJK  | iShares S&P Mid-Cap 400 Growth ... | U.S. Mid Cap Growth    | cluster_2  | LONG | 2026-04-09 | 20 | 6.1%  | 0.06 | 0.935 |
| MDYG | State Street SPDR S&P 400 Mid C... | U.S. Mid Cap Growth    | cluster_2  | LONG | 2026-04-09 | 20 | 6.0%  | 0.07 | 0.934 |
| ISCG | iShares Morningstar Small-Cap G... | Growth Equity          | cluster_2  | LONG | 2026-04-09 | 20 | 6.2%  | 0.07 | 0.933 |
| EWX  | State Street SPDR S&P Emerging ... | Emerging Markets Eq... | cluster_7  | LONG | 2026-04-09 | 20 | 7.7%  | 0.07 | 0.933 |
| AOR  | iShares Core 60/40 Balanced All... | Asset Allocation       | cluster_1  | LONG | 2026-04-10 | 19 | 3.6%  | 0.07 | 0.932 |
| PSR  | Invesco Active U.S. Real Estate... | Real Estate            | cluster_17 | LONG | 2026-04-10 | 19 | 4.2%  | 0.07 | 0.931 |
| ENZL | iShares MSCI New Zealand ETF       | New Zealand Equity     | cluster_3  | FLAT | 2026-04-30 | 5  | 2.2%  | 0.07 | 0.931 |
| SDOG | ALPS Sector Dividend Dogs ETF      | U.S. Dividend Equity   | cluster_5  | LONG | 2026-04-08 | 21 | 0.7%  | 0.07 | 0.927 |
| SMLF | iShares U.S. Small-Cap Equity F... | U.S. Small Cap Mult... | cluster_2  | LONG | 2026-04-08 | 21 | 6.1%  | 0.07 | 0.926 |
| SCHE | Schwab Emerging Markets Equity ... | Emerging Markets Eq... | cluster_7  | LONG | 2026-04-10 | 19 | 5.8%  | 0.07 | 0.926 |
| IVOG | Vanguard S&P Mid-Cap 400 Growth... | U.S. Mid Cap Growth    | cluster_2  | LONG | 2026-04-09 | 20 | 5.9%  | 0.08 | 0.925 |
| PXH  | Invesco RAFI Emerging Markets E... | Emerging Markets Eq... | cluster_7  | LONG | 2026-04-09 | 20 | 5.1%  | 0.08 | 0.924 |
| FEX  | First Trust Large Cap Core Alph... | Large Cap Blend Equ... | cluster_2  | LONG | 2026-04-09 | 20 | 2.9%  | 0.08 | 0.922 |
| FEM  | First Trust Emerging Markets Al... | Emerging Markets Eq... | cluster_7  | LONG | 2026-04-07 | 22 | 10.1% | 0.08 | 0.922 |
| PFF  | iShares Preferred and Income Se... | Preferred Stock        | cluster_8  | LONG | 2026-04-13 | 18 | 2.2%  | 0.08 | 0.920 |
| IWR  | iShares Russell Mid-Cap ETF        | U.S. Mid Cap Equity    | cluster_2  | LONG | 2026-04-09 | 20 | 3.7%  | 0.08 | 0.920 |
| IWS  | iShares Russell Mid-Cap Value E... | U.S. Mid Cap Value     | cluster_2  | LONG | 2026-04-08 | 21 | 3.4%  | 0.08 | 0.917 |
| VB   | Vanguard Small-Cap ETF             | U.S. Small Cap Equi... | cluster_2  | LONG | 2026-04-09 | 20 | 5.0%  | 0.09 | 0.913 |
| PWV  | Invesco Large Cap Value ETF        | Value Equity           | cluster_5  | LONG | 2026-04-06 | 23 | 3.0%  | 0.09 | 0.911 |
| SIZE | iShares MSCI USA Size Factor ETF   | U.S. Size Factor Eq... | cluster_2  | LONG | 2026-04-10 | 19 | 4.1%  | 0.09 | 0.909 |
| PSCI | Invesco S&P SmallCap Industrial... | U.S. Small Cap Indu... | cluster_2  | LONG | 2026-04-09 | 20 | 4.7%  | 0.09 | 0.906 |
| PFM  | Invesco Dividend Achievers ETF     | U.S. Dividend Equity   | cluster_5  | LONG | 2026-04-10 | 19 | 2.9%  | 0.10 | 0.902 |
| ETHO | Amplify Etho Climate Leadership... | U.S. ESG Equity        | cluster_2  | LONG | 2026-04-08 | 21 | 6.0%  | 0.10 | 0.900 |
| ICF  | iShares Select U.S. REIT ETF       | Real Estate            | cluster_17 | LONG | 2026-04-09 | 20 | 4.0%  | 0.10 | 0.900 |
| USRT | iShares Core U.S. REIT ETF         | Real Estate            | cluster_17 | LONG | 2026-04-10 | 19 | 4.2%  | 0.10 | 0.898 |
| IGHG | ProShares Investment Grade-Inte... | Interest Rate Hedge... | cluster_23 | LONG | 2026-04-06 | 23 | 1.0%  | 0.10 | 0.896 |
| NFRA | FlexShares STOXX Global Broad I... | Global Equity          | cluster_3  | FLAT | 2026-04-10 | 19 | 0.6%  | 0.10 | 0.896 |
| VIG  | Vanguard Div Appreciation ETF      | U.S. Dividend Equity   | cluster_5  | LONG | 2026-04-10 | 19 | 3.0%  | 0.10 | 0.895 |
| VYM  | Vanguard High Dividend Yield ETF   | U.S. Dividend Equity   | cluster_5  | LONG | 2026-04-09 | 20 | 1.6%  | 0.10 | 0.895 |

|      |                                    |                        |            |      |            |    |       |      |       |
|------|------------------------------------|------------------------|------------|------|------------|----|-------|------|-------|
| TIP  | iShares TIPS Bond ETF              | Inflation-Protected... | cluster_21 | LONG | 2026-04-13 | 18 | 0.3%  | 0.11 | 0.890 |
| IWP  | iShares Russell Midcap Growth E... | U.S. Mid Cap Growth    | cluster_6  | LONG | 2026-04-14 | 17 | 2.8%  | 0.11 | 0.888 |
| SCHP | Schwab U.S. TIPS ETF               | Inflation-Protected... | cluster_21 | LONG | 2026-04-13 | 18 | 0.3%  | 0.11 | 0.887 |
| QUAL | iShares MSCI USA Quality Factor... | U.S. Quality / Fact... | cluster_1  | LONG | 2026-04-13 | 18 | 4.1%  | 0.11 | 0.886 |
| GWX  | State Street SPDR S&P Internati... | International Equity   | cluster_3  | FLAT | 2026-04-10 | 19 | 4.8%  | 0.11 | 0.885 |
| DGRW | WisdomTree U.S. Quality Dividen... | U.S. Dividend Equity   | cluster_1  | LONG | 2026-04-13 | 18 | 4.1%  | 0.11 | 0.885 |
| FNX  | First Trust Mid Cap Core AlphaD... | U.S. Mid/Small Cap ... | cluster_2  | LONG | 2026-04-09 | 20 | 3.4%  | 0.12 | 0.882 |
| FXA  | Invesco CurrencyShares Australi... | Currency               | cluster_3  | FLAT | 2026-04-13 | 18 | 1.7%  | 0.12 | 0.881 |
| ONEY | State Street SPDR Russell 1000 ... | U.S. Dividend Equity   | cluster_5  | LONG | 2026-04-09 | 20 | 1.1%  | 0.12 | 0.881 |
| FNDE | Schwab Fundamental Emerging Mar... | Emerging Markets Eq... | cluster_7  | LONG | 2026-04-09 | 20 | 4.3%  | 0.12 | 0.879 |
| VIDI | Vident International Equity Str... | International Equity   | cluster_3  | FLAT | 2026-04-08 | 21 | 4.1%  | 0.12 | 0.875 |
| SPMD | State Street SPDR Portfolio S&P... | U.S. Mid Cap Equity    | cluster_2  | LONG | 2026-04-09 | 20 | 4.2%  | 0.13 | 0.873 |
| IJH  | iShares Core S&P Mid-Cap ETF       | U.S. Mid Cap Equity    | cluster_2  | LONG | 2026-04-09 | 20 | 4.2%  | 0.13 | 0.871 |
| IVOO | Vanguard S&P Mid-Cap 400 ETF       |                        | cluster_2  | LONG | 2026-04-09 | 20 | 4.2%  | 0.13 | 0.869 |
| MDY  | State Street SPDR S&P MIDCAP 40... | U.S. Mid Cap Equity    | cluster_2  | LONG | 2026-04-09 | 20 | 4.2%  | 0.13 | 0.869 |
| LQDH | iShares Interest Rate Hedged Co... | Interest Rate Hedge... | cluster_23 | LONG | 2026-04-02 | 24 | 1.1%  | 0.13 | 0.868 |
| IGBH | iShares Interest Rate Hedged Lo... | Interest Rate Hedge... | cluster_23 | LONG | 2026-04-06 | 23 | 1.4%  | 0.13 | 0.866 |
| XME  | State Street SPDR S&P Metals & ... | Materials              | cluster_11 | FLAT | 2026-04-10 | 19 | 6.1%  | 0.14 | 0.865 |
| AOM  | iShares Core 40/60 Moderate All... | Asset Allocation       | cluster_1  | LONG | 2026-04-10 | 19 | 2.4%  | 0.14 | 0.861 |
| SCHH | Schwab U.S. REIT ETF               | Real Estate            | cluster_17 | LONG | 2026-04-10 | 19 | 3.6%  | 0.14 | 0.860 |
| DUST | Direxion Daily Gold Miners Inde... | Leveraged / Inverse... | cluster_27 | LONG | 2026-04-30 | 5  | -9.0% | 0.00 | 0.856 |
| MVV  | ProShares Ultra MidCap400          | Leveraged / Inverse    | cluster_2  | LONG | 2026-04-09 | 20 | 8.0%  | 0.15 | 0.853 |
| RWR  | State Street SPDR Dow Jones REI... | Real Estate            | cluster_17 | LONG | 2026-04-10 | 19 | 3.8%  | 0.16 | 0.844 |
| PKB  | Invesco Building & Construction... | Industrials            | cluster_2  | LONG | 2026-04-10 | 19 | 1.8%  | 0.16 | 0.840 |
| ARKQ | ARK Autonomous Technology & Rob... | Technology             | cluster_6  | LONG | 2026-04-13 | 18 | 7.4%  | 0.16 | 0.839 |
| FXB  | Invesco CurrencyShares British ... | Currency               | cluster_26 | LONG | 2026-04-10 | 19 | 0.8%  | 0.16 | 0.837 |
| OUSA | ALPS O'Shares U.S. Quality Divi... | Dividend Equity        | cluster_5  | LONG | 2026-04-15 | 16 | 1.1%  | 0.17 | 0.834 |
| HYEM | VanEck Emerging Markets High Yi... | Emerging Markets Fi... | cluster_8  | LONG | 2026-04-13 | 18 | 1.1%  | 0.17 | 0.829 |
| XLRE | State Street Real Estate Select... | Real Estate            | cluster_17 | LONG | 2026-04-10 | 19 | 3.7%  | 0.17 | 0.826 |
| RSP  | Invesco S&P 500 Equal Weight ETF   | U.S. Equal Weight E... | cluster_2  | LONG | 2026-04-13 | 18 | 2.4%  | 0.17 | 0.826 |
| EMLP | First Trust North American Ener... | MLP / Energy Infrac... | cluster_16 | LONG | 2026-04-29 | 6  | -0.9% | 0.07 | 0.824 |
| SPIP | State Street SPDR Portfolio TIP... | Inflation-Protected... | cluster_21 | LONG | 2026-04-13 | 18 | 0.2%  | 0.18 | 0.819 |
| SVXY | ProShares Short VIX Short Term ... | Volatility             | cluster_1  | LONG | 2026-04-13 | 18 | 3.0%  | 0.18 | 0.817 |
| FPXI | First Trust International Equit... | International IPO E... | cluster_1  | LONG | 2026-04-09 | 20 | 8.1%  | 0.19 | 0.812 |
| VEU  | Vanguard FTSE All World Ex US E... | Global ex-US Equity    | cluster_3  | FLAT | 2026-04-09 | 20 | 4.1%  | 0.19 | 0.811 |
| CDC  | VictoryShares US EQ Income Enha... | U.S. Defensive Equi... | cluster_5  | LONG | 2026-04-09 | 20 | -0.2% | 0.19 | 0.810 |
| ARKK | ARK Innovation ETF                 | Technology             | cluster_6  | LONG | 2026-04-14 | 17 | 4.1%  | 0.19 | 0.810 |
| AIVL | WisdomTree U.S. AI Enhanced Val... | U.S. Value Factor E... | cluster_2  | LONG | 2026-04-09 | 20 | 1.4%  | 0.19 | 0.810 |
| VNM  | VanEck Vietnam ETF                 | Vietnam Equity         | cluster_14 | FLAT | 2026-04-09 | 20 | 6.4%  | 0.19 | 0.809 |
| AOK  | iShares Core 30/70 Conservative... | Asset Allocation       | cluster_1  | LONG | 2026-04-10 | 19 | 1.9%  | 0.19 | 0.807 |
| JHMM | John Hancock Multifactor Mid Ca... | U.S. Mid Cap Equity    | cluster_2  | LONG | 2026-04-09 | 20 | 3.1%  | 0.20 | 0.802 |
| RZV  | Invesco S&P SmallCap 600 Pure V... | Leveraged / Inverse    | cluster_2  | LONG | 2026-04-09 | 20 | 5.4%  | 0.20 | 0.797 |
| VXUS | Vanguard Total International St... | Global ex-US Equity    | cluster_3  | FLAT | 2026-04-09 | 20 | 3.9%  | 0.21 | 0.794 |
| PEY  | Invesco High Yield Equity Divid... | U.S. Dividend Equity   | cluster_5  | LONG | 2026-04-06 | 23 | 3.7%  | 0.21 | 0.792 |
| PXF  | Invesco RAFI Developed Markets ... | International Equity   | cluster_3  | FLAT | 2026-04-08 | 21 | 2.1%  | 0.21 | 0.787 |

|      |                                    |                        |            |      |            |    |       |      |       |
|------|------------------------------------|------------------------|------------|------|------------|----|-------|------|-------|
| FREL | Fidelity MSCI Real Estate Index... | Real Estate            | cluster_17 | LONG | 2026-04-10 | 19 | 3.8%  | 0.21 | 0.785 |
| IXUS | iShares Core MSCI Total Interna... | Global ex-US Equity    | cluster_3  | FLAT | 2026-04-09 | 20 | 4.0%  | 0.22 | 0.784 |
| DES  | WisdomTree U.S. SmallCap Divide... | U.S. Small Cap Divi... | cluster_2  | LONG | 2026-04-07 | 22 | 5.7%  | 0.22 | 0.780 |
| JPME | JPMorgan Diversified Return U.S... | U.S. Equity            | cluster_2  | LONG | 2026-04-08 | 21 | 1.7%  | 0.22 | 0.779 |
| DVY  | iShares Select Dividend ETF        | U.S. Dividend Equity   | cluster_5  | LONG | 2026-04-08 | 21 | -0.5% | 0.22 | 0.779 |
| NUSC | Nuveen ESG Small-Cap ETF           | U.S. Small Cap ESG ... | cluster_2  | LONG | 2026-04-09 | 20 | 4.0%  | 0.22 | 0.778 |
| EZM  | WisdomTree U.S. MidCap Fund        | Mid Cap Value Equit... | cluster_2  | LONG | 2026-04-09 | 20 | 3.6%  | 0.22 | 0.776 |
| VNQ  | Vanguard Real Estate ETF           | Real Estate            | cluster_17 | LONG | 2026-04-10 | 19 | 3.7%  | 0.22 | 0.775 |
| ARKG | ARK Genomic Revolution ETF         | Healthcare             | cluster_10 | FLAT | 2026-04-14 | 17 | 3.7%  | 0.22 | 0.775 |
| PNQI | Invesco Nasdaq Internet ETF        | Technology             | cluster_6  | LONG | 2026-04-14 | 17 | 4.2%  | 0.23 | 0.774 |
| DIA  | State Street SPDR Dow Jones Ind... | U.S. Large Cap Equi... | cluster_5  | LONG | 2026-04-10 | 19 | 3.5%  | 0.23 | 0.773 |
| CWI  | State Street SPDR MSCI ACWI ex-... | Global ex-US Equity    | cluster_3  | FLAT | 2026-04-09 | 20 | 3.6%  | 0.23 | 0.771 |
| DGRO | iShares Core Dividend Growth ETF   | U.S. Dividend Equity   | cluster_5  | LONG | 2026-04-10 | 19 | 1.8%  | 0.23 | 0.769 |
| IYR  | iShares U.S. Real Estate ETF       | Real Estate            | cluster_17 | LONG | 2026-04-10 | 19 | 3.6%  | 0.24 | 0.764 |
| CXSE | WisdomTree China ex-State-Owned... | China Equity           | cluster_15 | LONG | 2026-04-13 | 18 | 3.8%  | 0.24 | 0.759 |
| FDIS | Fidelity MSCI Consumer Discreti... | Consumer Discretion... | cluster_1  | LONG | 2026-04-13 | 18 | 5.3%  | 0.24 | 0.758 |
| VCR  | Vanguard Consumer Discretion ETF   | Consumer Discretion... | cluster_1  | LONG | 2026-04-13 | 18 | 5.4%  | 0.24 | 0.757 |
| ACWX | iShares MSCI ACWI ex U.S. ETF      | Global ex-US Equity    | cluster_3  | FLAT | 2026-04-09 | 20 | 3.7%  | 0.25 | 0.752 |
| XLY  | State Street Consumer Discretio... | Consumer Discretion... | cluster_1  | LONG | 2026-04-14 | 17 | 3.0%  | 0.25 | 0.751 |
| FINX | Global X FinTech ETF               | Financials             | cluster_6  | LONG | 2026-04-15 | 16 | 0.4%  | 0.25 | 0.749 |
| URA  | Global X Uranium ETF               | Materials              | cluster_11 | FLAT | 2026-04-13 | 18 | 7.8%  | 0.25 | 0.749 |
| FDVV | Fidelity High Dividend ETF         | U.S. Dividend Equity   | cluster_5  | LONG | 2026-04-10 | 19 | 3.5%  | 0.25 | 0.749 |
| VSS  | Vanguard FTSE All-Wld ex-US SmC... | Global ex-US Small ... | cluster_3  | FLAT | 2026-04-10 | 19 | 4.1%  | 0.25 | 0.747 |
| REET | iShares Global REIT ETF            | Real Estate            | cluster_17 | LONG | 2026-04-10 | 19 | 3.3%  | 0.26 | 0.745 |
| USMV | iShares MSCI USA Min Vol Factor... | U.S. Low Volatility... | cluster_5  | LONG | 2026-04-20 | 13 | -0.5% | 0.27 | 0.727 |
| VBR  | Vanguard Small-Cap Value ETF       | U.S. Small Cap Value   | cluster_2  | LONG | 2026-04-09 | 20 | 2.8%  | 0.29 | 0.705 |
| VIS  | Vanguard Industrials ETF           | Industrials            | cluster_2  | LONG | 2026-04-10 | 19 | 2.4%  | 0.30 | 0.697 |
| VOE  | Vanguard Mid-Cap Value ETF         | U.S. Mid Cap Value     | cluster_2  | LONG | 2026-04-09 | 20 | 0.7%  | 0.32 | 0.684 |
| FIDU | Fidelity MSCI Industrials Index... | Industrials            | cluster_2  | LONG | 2026-04-10 | 19 | 2.1%  | 0.33 | 0.675 |
| FYT  | First Trust Small Cap Value Alp... | Value Equity           | cluster_2  | LONG | 2026-04-07 | 22 | 6.3%  | 0.33 | 0.667 |
| VOX  | Vanguard Communication Services... | Communication Servi... | cluster_1  | LONG | 2026-04-10 | 19 | 5.1%  | 0.34 | 0.660 |
| RWO  | State Street SPDR Dow Jones Glo... | Real Estate            | cluster_17 | LONG | 2026-04-13 | 18 | 2.7%  | 0.34 | 0.657 |
| FNDC | Schwab Fundamental Internationa... | International Equity   | cluster_3  | FLAT | 2026-04-09 | 20 | 2.6%  | 0.35 | 0.648 |
| FCOM | Fidelity MSCI Communication Ser... | Communication Servi... | cluster_1  | LONG | 2026-04-10 | 19 | 5.0%  | 0.36 | 0.637 |
| PSP  | Invesco Global Listed Private E... | Global Equity          | cluster_12 | FLAT | 2026-04-13 | 18 | 3.7%  | 0.37 | 0.632 |
| IJJ  | iShares S&P Mid-Cap 400 Value E... | U.S. Mid Cap Value     | cluster_2  | LONG | 2026-04-09 | 20 | 2.3%  | 0.37 | 0.626 |
| UDN  | Invesco DB USD Index Bearish ETF   | Currency               | cluster_26 | LONG | 2026-04-10 | 19 | 0.6%  | 0.37 | 0.626 |
| MDYV | State Street SPDR S&P 400 Mid C... | U.S. Mid Cap Value     | cluster_2  | LONG | 2026-04-09 | 20 | 2.2%  | 0.38 | 0.619 |
| LDUR | PIMCO Enhanced Low Duration Act... | Fixed Income           | cluster_4  | LONG | 2026-04-09 | 20 | 0.1%  | 0.39 | 0.614 |
| SDY  | State Street SPDR S&P Dividend ... | U.S. Dividend Equity   | cluster_5  | LONG | 2026-04-20 | 13 | -0.5% | 0.39 | 0.611 |
| IVOV | Vanguard S&P Mid-Cap 400 Value ... | U.S. Mid Cap Value     | cluster_2  | LONG | 2026-04-09 | 20 | 2.1%  | 0.40 | 0.604 |
| XLI  | State Street Industrial Select ... | Industrials            | cluster_2  | LONG | 2026-04-10 | 19 | 1.4%  | 0.41 | 0.589 |
| FPE  | First Trust Preferred Securitie... | Preferred Stock        | cluster_8  | LONG | 2026-04-13 | 18 | 0.5%  | 0.42 | 0.577 |
| CFO  | VictoryShares US 500 Enhanced V... | U.S. Multifactor Eq... | cluster_2  | LONG | 2026-04-10 | 19 | 2.0%  | 0.42 | 0.577 |
| CFA  | VictoryShares US 500 Volatility... | U.S. Multifactor Eq... | cluster_2  | LONG | 2026-04-10 | 19 | 1.9%  | 0.43 | 0.574 |

|      |                                    |                        |            |      |            |    |       |      |        |
|------|------------------------------------|------------------------|------------|------|------------|----|-------|------|--------|
| RTH  | VanEck Retail ETF                  | Consumer Discretion... | cluster_1  | LONG | 2026-04-09 | 20 | 1.2%  | 0.43 | 0.566  |
| ERTH | Invesco MSCI Sustainable Future... | Clean Energy / Clim... | cluster_25 | LONG | 2026-04-09 | 20 | 3.6%  | 0.43 | 0.565  |
| SDIV | Global X SuperDividend ETF         | Global Dividend Equ... | cluster_7  | LONG | 2026-04-07 | 22 | 3.5%  | 0.44 | 0.561  |
| IPAY | Amplify Digital Payments ETF       | Financials             | cluster_6  | LONG | 2026-04-14 | 17 | 2.6%  | 0.44 | 0.553  |
| EQAL | Invesco Russell 1000 Equal Weig... | U.S. Equal Weight E... | cluster_2  | LONG | 2026-04-08 | 21 | 1.4%  | 0.45 | 0.545  |
| DON  | WisdomTree U.S. MidCap Dividend... | U.S. Mid Cap Divide... | cluster_2  | LONG | 2026-04-09 | 20 | 1.3%  | 0.47 | 0.531  |
| KCE  | State Street SPDR S&P Capital M... | Financials             | cluster_12 | FLAT | 2026-04-09 | 20 | 6.3%  | 0.47 | 0.527  |
| RWK  | Invesco S&P MidCap 400 Revenue ... | U.S. Mid Cap Equity    | cluster_2  | LONG | 2026-04-08 | 21 | 4.0%  | 0.48 | 0.524  |
| LVHD | Franklin U.S. Low Volatility Hi... | U.S. Low Volatility... | cluster_5  | LONG | 2026-04-28 | 7  | -1.4% | 0.49 | 0.492  |
| JPUS | JPMorgan Diversified Return U.S... | U.S. Multifactor Eq... | cluster_2  | LONG | 2026-04-08 | 21 | 0.8%  | 0.51 | 0.489  |
| JETS | U.S. Global Jets ETF               | Industrials            | cluster_2  | LONG | 2026-05-06 | 1  | 1.3%  | 0.48 | 0.477  |
| RDVY | First Trust Rising Dividend Ach... | U.S. Dividend Equity   | cluster_2  | LONG | 2026-04-09 | 20 | 2.4%  | 0.52 | 0.477  |
| KWEB | KraneShares CSI China Internet ... | China Equity           | cluster_15 | LONG | 2026-05-06 | 1  | -0.8% | 0.47 | 0.426  |
| EWC  | iShares MSCI Canada Index Fund     | Canada Equity          | cluster_3  | FLAT | 2026-04-09 | 20 | 2.4%  | 0.59 | 0.413  |
| IYJ  | iShares U.S. Industrials ETF       | Industrials            | cluster_2  | LONG | 2026-04-10 | 19 | 1.6%  | 0.59 | 0.410  |
| HYLS | First Trust Tactical High Yield... | High Yield Bond        | cluster_8  | LONG | 2026-04-08 | 21 | 0.4%  | 0.60 | 0.401  |
| VRP  | Invesco Variable Rate Preferred... | Preferred Stock        | cluster_8  | LONG | 2026-04-09 | 20 | 0.5%  | 0.61 | 0.394  |
| FTXR | First Trust Nasdaq Transportati... | Industrials            | cluster_2  | LONG | 2026-04-13 | 18 | 4.6%  | 0.61 | 0.388  |
| XOP  | State Street SPDR S&P Oil & Gas... | Energy                 | cluster_16 | LONG | 2026-04-30 | 5  | -6.8% | 0.06 | 0.385  |
| PXE  | Invesco Energy Exploration & Pr... | Energy                 | cluster_16 | LONG | 2026-04-30 | 5  | -6.5% | 0.08 | 0.373  |
| EWJV | iShares MSCI Japan Value ETF       | Japan Equity           | cluster_22 | FLAT | 2026-05-04 | 3  | 2.7%  | 0.53 | 0.372  |
| REM  | iShares Mortgage Real Estate ETF   | Real Estate            | cluster_2  | LONG | 2026-04-09 | 20 | 2.4%  | 0.65 | 0.349  |
| ACWV | iShares MSCI Global Min Vol Fac... | Global Equity          | cluster_5  | LONG | 2026-04-30 | 5  | -0.2% | 0.65 | 0.331  |
| BNDX | Vanguard Total International Bo... | International Fixed... | cluster_4  | LONG | 2026-05-06 | 1  | -0.2% | 0.64 | 0.331  |
| HYD  | VanEck High Yield Muni ETF         | Municipal Bond         | cluster_18 | FLAT | 2026-04-08 | 21 | 0.8%  | 0.68 | 0.317  |
| SDVY | First Trust SMID Cap Rising Div... | Dividend Equity        | cluster_2  | LONG | 2026-04-09 | 20 | 0.6%  | 0.69 | 0.307  |
| IEO  | iShares U.S. Oil & Gas Explorat... | Energy                 | cluster_16 | LONG | 2026-04-30 | 5  | -7.2% | 0.08 | 0.301  |
| SHYG | iShares 0-5 Year High Yield Cor... | High Yield Bond        | cluster_8  | LONG | 2026-04-07 | 22 | 0.8%  | 0.71 | 0.288  |
| VDE  | Vanguard Energy ETF                | Energy                 | cluster_16 | LONG | 2026-05-01 | 4  | -4.9% | 0.09 | 0.264  |
| XSLV | Invesco S&P SmallCap Low Volati... | U.S. Small Cap Low ... | cluster_2  | LONG | 2026-04-08 | 21 | 1.5%  | 0.74 | 0.261  |
| SIVR | abrdn Physical Silver Shares ETF   | Silver                 | cluster_13 | FLAT | 2026-05-06 | 1  | 2.1%  | 0.69 | 0.257  |
| SLV  | iShares Silver Trust               | Silver                 | cluster_13 | FLAT | 2026-05-06 | 1  | 2.1%  | 0.69 | 0.257  |
| IGF  | iShares Global Infrastructure E... | Global Equity          | cluster_3  | FLAT | 2026-04-24 | 9  | -1.0% | 0.58 | 0.251  |
| FENY | Fidelity MSCI Energy Index ETF     | Energy                 | cluster_16 | LONG | 2026-05-01 | 4  | -4.9% | 0.09 | 0.247  |
| IYE  | iShares U.S. Energy ETF            | Energy                 | cluster_16 | LONG | 2026-05-01 | 4  | -4.9% | 0.09 | 0.220  |
| LTPZ | Pimco 15 Year U.S. TIPS Index E... | Inflation-Protected... | cluster_21 | LONG | 2026-05-06 | 1  | -0.3% | 0.77 | 0.199  |
| IAGG | iShares International Aggregate... | International Fixed... | cluster_4  | LONG | 2026-05-06 | 1  | -0.2% | 0.74 | 0.193  |
| XLE  | State Street Energy Select Sect... | Energy                 | cluster_16 | LONG | 2026-05-01 | 4  | -4.9% | 0.10 | 0.183  |
| AGQ  | ProShares Ultra Silver             | Leveraged / Inverse... | cluster_13 | FLAT | 2026-05-07 | 0  | 0.0%  | 0.73 | 0.181  |
| SJNK | State Street SPDR Bloomberg Sho... | High Yield Bond        | cluster_8  | LONG | 2026-04-07 | 22 | 0.8%  | 0.83 | 0.167  |
| KBE  | State Street SPDR S&P Bank ETF     | Financials             | cluster_12 | FLAT | 2026-04-07 | 22 | 5.2%  | 0.83 | 0.165  |
| BAB  | Invesco Taxable Municipal Bond ... | Taxable Municipal B... | cluster_4  | LONG | 2026-05-06 | 1  | -0.3% | 0.82 | 0.121  |
| IXC  | iShares Global Energy ETF          | Energy                 | cluster_16 | LONG | 2026-05-01 | 4  | -4.8% | 0.14 | 0.111  |
| GNMA | iShares GNMA Bond ETF              | Mortgage-Backed Fix... | cluster_4  | LONG | 2026-05-06 | 1  | -0.1% | 0.93 | -0.025 |
| GLTR | abrdn Physical Precious Metals ... | Precious Metals        | cluster_13 | FLAT | 2026-05-07 | 0  | 0.0%  | 0.86 | -0.030 |

|      |                                     |                        |            |      |            |    |       |      |        |
|------|-------------------------------------|------------------------|------------|------|------------|----|-------|------|--------|
| PPH  | VanEck Pharmaceutical ETF           | Healthcare             | cluster_10 | FLAT | 2026-05-06 | 1  | -1.4% | 0.63 | -0.059 |
| SHM  | State Street SPDR Nuveen ICE Sh...  | Municipal Bond         | cluster_18 | FLAT | 2026-05-06 | 1  | -0.1% | 0.96 | -0.112 |
| IUSB | iShares Core Universal USD Bond...  | Aggregate Fixed Inc... | cluster_4  | LONG | 2026-05-06 | 1  | -0.2% | 0.97 | -0.130 |
| TOTL | State Street DoubleLine Total R...  | Active Fixed Income    | cluster_4  | LONG | 2026-05-06 | 1  | -0.2% | 0.98 | -0.139 |
| XHE  | State Street SPDR S&P Health Ca...  | Healthcare             | cluster_10 | FLAT | 2026-05-06 | 1  | 0.9%  | 0.89 | -0.158 |
| AGGY | WisdomTree Yield Enhanced U.S. ...  | Aggregate Fixed Inc... | cluster_4  | LONG | 2026-05-06 | 1  | -0.4% | 0.97 | -0.167 |
| FTCS | First Trust Capital Strength ETF    | U.S. Quality / Fact... | cluster_5  | LONG | 2026-05-04 | 3  | -0.5% | 0.94 | -0.176 |
| FDL  | First Trust Morningstar ETF         | U.S. Dividend Equity   | cluster_5  | LONG | 2026-05-01 | 4  | -2.3% | 0.47 | -0.179 |
| FBND | Fidelity Total Bond ETF             | Active Fixed Income    | cluster_4  | LONG | 2026-05-06 | 1  | -0.2% | 0.98 | -0.201 |
| GVI  | iShares Intermediate Government...  | Fixed Income           | cluster_4  | LONG | 2026-05-06 | 1  | -0.2% | 0.98 | -0.204 |
| IBB  | iShares Biotechnology ETF           | Healthcare             | cluster_10 | FLAT | 2026-05-06 | 1  | -2.2% | 0.89 | -0.206 |
| IGLB | iShares 10 Year Investment Grad...  | Investment Grade Co... | cluster_4  | LONG | 2026-05-06 | 1  | -0.5% | 0.98 | -0.209 |
| COPX | Global X Copper Miners ETF          | Materials              | cluster_11 | FLAT | 2026-05-06 | 1  | -1.1% | 0.96 | -0.216 |
| BOND | PIMCO Active Bond Exchange-Trad...  | Active Fixed Income    | cluster_4  | LONG | 2026-05-06 | 1  | -0.2% | 0.99 | -0.219 |
| PPLT | abrdn Physical Platinum Shares ...  | Platinum               | cluster_13 | FLAT | 2026-05-07 | 0  | 0.0%  | 0.87 | -0.233 |
| ERY  | Direxion Daily Energy Bear 2X E...  | Leveraged / Inverse    | cluster_9  | FLAT | 2026-05-07 | 0  | 0.0%  | 0.92 | -0.269 |
| VCLT | Vanguard Long-Term Corporate Bo...  | Investment Grade Co... | cluster_4  | LONG | 2026-05-06 | 1  | -0.5% | 0.99 | -0.272 |
| LQD  | iShares iBoxx \$ Investment Grad... | Investment Grade Co... | cluster_4  | LONG | 2026-05-06 | 1  | -0.4% | 0.99 | -0.287 |
| AGG  | iShares Core U.S. Aggregate Bon...  | Aggregate Fixed Inc... | cluster_4  | LONG | 2026-05-06 | 1  | -0.2% | 0.98 | -0.297 |
| SPLB | State Street SPDR Portfolio Lon...  | Investment Grade Co... | cluster_4  | LONG | 2026-05-06 | 1  | -0.6% | 0.99 | -0.300 |
| QLTA | iShares Aaa A Rated Corporate B...  | Investment Grade Co... | cluster_4  | LONG | 2026-05-06 | 1  | -0.3% | 0.99 | -0.306 |
| EZA  | iShares MSCI South Africa Index...  | South Africa Equity    | cluster_3  | FLAT | 2026-05-07 | 0  | 0.0%  | 0.95 | -0.307 |
| BND  | Vanguard Total Bond Market ETF      | Aggregate Fixed Inc... | cluster_4  | LONG | 2026-05-07 | 0  | 0.0%  | 0.99 | -0.335 |
| VMBS | Vanguard Mortgage-Backed Securi...  | Mortgage-Backed Fix... | cluster_4  | LONG | 2026-05-06 | 1  | -0.3% | 0.98 | -0.347 |
| SCHZ | Schwab US Aggregate Bond ETF        | Aggregate Fixed Inc... | cluster_4  | LONG | 2026-05-07 | 0  | 0.0%  | 0.99 | -0.377 |
| IGE  | iShares North American Natural ...  | Energy                 | cluster_11 | FLAT | 2026-05-01 | 4  | -2.3% | 0.52 | -0.392 |
| PHO  | Invesco Water Resources ETF         | Utilities              | cluster_2  | LONG | 2026-05-06 | 1  | -1.0% | 0.99 | -0.401 |
| MOO  | VanEck Agribusiness ETF             | Consumer Staples       | cluster_11 | FLAT | 2026-05-06 | 1  | -3.1% | 0.69 | -0.409 |
| AGZ  | iShares Agency Bond ETF             | Agency Fixed Income    | cluster_4  | LONG | 2026-05-06 | 1  | -0.2% | 0.98 | -0.442 |
| KURE | KraneShares MSCI All China Heal...  | China Equity           | cluster_15 | LONG | 2026-05-04 | 3  | 0.2%  | 0.91 | -0.508 |
| GUNR | FlexShares Global Upstream Natu...  | Materials              | cluster_11 | FLAT | 2026-05-05 | 2  | -2.0% | 0.96 | -0.881 |
| SPLG |                                     |                        | cluster_1  | LONG | 2026-04-10 | 19 |       |      |        |
| RSX  |                                     |                        | cluster_14 | FLAT | 2025-12-22 | 93 |       |      |        |
| FIVG |                                     |                        | cluster_1  | LONG | 2026-04-02 | 24 |       |      |        |
| PBS  |                                     |                        | cluster_6  | LONG | 2026-04-13 | 18 |       |      |        |
| IGN  |                                     |                        | cluster_1  | LONG | 2026-04-02 | 24 |       |      |        |

## Today's Crossovers

## Buy crossovers

| level   | id        | name                               | category               | cluster    | cluster_signal | spread_buy | prob_buy | nss    |
|---------|-----------|------------------------------------|------------------------|------------|----------------|------------|----------|--------|
| etf     | AGQ       | ProShares Ultra Silver             | Leveraged / Inverse... | cluster_13 | FLAT           | 1.613      | 0.91     | 0.181  |
| etf     | GLTR      | abrdn Physical Precious Metals ... | Precious Metals        | cluster_13 | FLAT           | 0.181      | 0.83     | -0.030 |
| etf     | PPLT      | abrdn Physical Platinum Shares ... | Platinum               | cluster_13 | FLAT           | 0.017      | 0.64     | -0.233 |
| etf     | ERY       | Direxion Daily Energy Bear 2X E... | Leveraged / Inverse    | cluster_9  | FLAT           | 0.009      | 0.65     | -0.269 |
| etf     | EZA       | iShares MSCI South Africa Index... | South Africa Equity    | cluster_3  | FLAT           | 0.263      | 0.64     | -0.307 |
| cluster | cluster_4 | Mixed: BND, AGG, SCHZ              | Mixed                  |            |                | 0.016      | 0.67     | -0.320 |
| etf     | BND       | Vanguard Total Bond Market ETF     | Aggregate Fixed Inc... | cluster_4  | LONG           | 0.013      | 0.65     | -0.335 |
| etf     | SCHZ      | Schwab US Aggregate Bond ETF       | Aggregate Fixed Inc... | cluster_4  | LONG           | 0.002      | 0.62     | -0.377 |

## Sell crossovers

| level   | id         | name                                | category               | cluster    | cluster_signal | spread_sell | prob_sell | nss    |
|---------|------------|-------------------------------------|------------------------|------------|----------------|-------------|-----------|--------|
| etf     | GNR        | State Street SPDR S&P Global Na...  | Materials              | cluster_11 | FLAT           | -0.175      | 0.92      | -0.424 |
| etf     | IDU        | iShares U.S. Utilities ETF          | Utilities              | cluster_19 | LONG           | -0.237      | 0.78      | -0.410 |
| etf     | PBJ        | Invesco Food & Beverage ETF         | Consumer Staples       | cluster_19 | LONG           | -0.054      | 0.90      | -0.361 |
| etf     | XLU        | State Street Utilities Select S...  | Utilities              | cluster_19 | LONG           | -0.040      | 0.71      | -0.194 |
| etf     | VPU        | Vanguard Utilities ETF              | Utilities              | cluster_19 | LONG           | -0.168      | 0.72      | -0.152 |
| etf     | FUTY       | Fidelity MSCI Utilities Index E...  | Utilities              | cluster_19 | LONG           | -0.051      | 0.72      | -0.146 |
| etf     | FTA        | First Trust Large Cap Value Alp...  | Value Equity           | cluster_5  | LONG           | -0.224      | 0.94      | 0.015  |
| etf     | IAT        | iShares U.S. Regional Banks ETF     | Financials             | cluster_12 | FLAT           | -0.040      | 0.94      | 0.061  |
| etf     | KBWB       | Invesco KBW Bank ETF                | Financials             | cluster_12 | FLAT           | -0.147      | 0.93      | 0.063  |
| etf     | HYG        | iShares iBoxx \$ High Yield Corp... | High Yield Bond        | cluster_8  | LONG           | -0.011      | 0.90      | 0.095  |
| etf     | FXO        | First Trust Financials AlphaDEX     | Financials             | cluster_12 | FLAT           | -0.115      | 0.90      | 0.100  |
| etf     | SPSB       | State Street SPDR Portfolio Sho...  | Fixed Income           | cluster_4  | LONG           | 0.000       | 0.88      | 0.117  |
| cluster | cluster_12 | Financials (13/15)                  | Financials             |            |                | -0.281      | 0.88      | 0.121  |
| etf     | IYG        | iShares U.S. Financial Services...  | Financials             | cluster_12 | FLAT           | -0.135      | 0.86      | 0.139  |
| etf     | ONEV       | State Street SPDR Russell 1000 ...  | U.S. Equity            | cluster_2  | LONG           | -0.070      | 0.83      | 0.169  |
| etf     | EFV        | iShares MSCI EAFE Value ETF         | Developed ex-US Equ... | cluster_3  | FLAT           | -0.011      | 0.78      | 0.199  |
| etf     | PKW        | Invesco BuyBack Achievers ETF       | U.S. Buyback Equity    | cluster_2  | LONG           | -0.072      | 0.76      | 0.232  |
| etf     | INTF       | iShares International Equity Fa...  | International Equity   | cluster_3  | FLAT           | -0.001      | 0.75      | 0.234  |
| etf     | EWL        | iShares MSCI Switzerland ETF        | Switzerland Equity     | cluster_3  | FLAT           | -0.012      | 0.69      | 0.302  |
| etf     | PGX        | Invesco Preferred ETF               | Preferred Stock        | cluster_8  | LONG           | -0.002      | 0.60      | 0.397  |
| etf     | DBB        | Invesco DB Base Metals Fund         | Industrial Metals      | cluster_11 | FLAT           | -0.005      | 0.60      | 0.397  |

**Net Signal Strength (NSS)**

NSS = P(buy crossover within 5 days) - P(sell crossover within 5 days). Range [-1, 1]; positive = bullish lean, negative = bearish lean.

**Top 20 most bullish (NSS desc)**

| level | id   | name                               | category               | cluster    | cluster_signal | prob_buy | prob_sell | nss |
|-------|------|------------------------------------|------------------------|------------|----------------|----------|-----------|-----|
| etf   | QTEC | First Trust NASDAQ-100-Technolo... | Technology             | cluster_1  | LONG           | 1        | 0         | 1   |
| etf   | XSD  | State Street SPDR S&P Semicondu... | Technology             | cluster_1  | LONG           | 1        | 0         | 1   |
| etf   | VLUE | iShares MSCI USA Value Factor E... | U.S. Value Factor E... | cluster_2  | LONG           | 1        | 0         | 1   |
| etf   | MINT | PIMCO Enhanced Short Maturity A... | Cash / Ultra-Short ... | cluster_24 | FLAT           | 1        | 0         | 1   |
| etf   | FLTR | VanEck IG Floating Rate ETF        | Floating Rate Fixed... | cluster_23 | LONG           | 1        | 0         | 1   |
| etf   | ICSH | iShares Ultra Short Duration Bo... | Cash / Ultra-Short ... | cluster_24 | FLAT           | 1        | 0         | 1   |
| etf   | SHV  | iShares 0-1 Year Treasury Bond ... | Cash / Ultra-Short ... | cluster_24 | FLAT           | 1        | 0         | 1   |
| etf   | BIL  | State Street SPDR Bloomberg 1-3... | Cash / Ultra-Short ... | cluster_24 | FLAT           | 1        | 0         | 1   |
| etf   | GBIL | Goldman Sachs Access Treasury 0... | Cash / Ultra-Short ... | cluster_24 | FLAT           | 1        | 0         | 1   |
| etf   | PTNQ | Pacer Trendpilot 100 ETF           | Trend-Following Equ... | cluster_1  | LONG           | 1        | 0         | 1   |
| etf   | FTXL | First Trust Nasdaq Semiconducto... | Technology             | cluster_1  | LONG           | 1        | 0         | 1   |
| etf   | IBDR | iShares iBonds Dec 2026 Term Co... | Target Maturity Bond   | cluster_4  | LONG           | 1        | 0         | 1   |
| etf   | NXTG | First Trust Indxx NextG ETF        | Technology             | cluster_1  | LONG           | 1        | 0         | 1   |
| etf   | EWI  | iShares MSCI Taiwan ETF            | Taiwan Equity          | cluster_7  | LONG           | 1        | 0         | 1   |
| etf   | BSCQ | Invesco BulletShares 2026 Corpo... | Target Maturity Bond   | cluster_4  | LONG           | 1        | 0         | 1   |
| etf   | GSY  | Invesco Ultra Short Duration ETF   | Cash / Ultra-Short ... | cluster_24 | FLAT           | 1        | 0         | 1   |
| etf   | PSI  | Invesco Semiconductors ETF         | Technology             | cluster_1  | LONG           | 1        | 0         | 1   |
| etf   | FLRN | State Street SPDR Bloomberg Inv... | Floating Rate Fixed... | cluster_23 | LONG           | 1        | 0         | 1   |
| etf   | IHF  | iShares U.S. Health Care Provid... | Healthcare             | cluster_10 | FLAT           | 1        | 0         | 1   |
| etf   | EWM  | iShares MSCI Malaysia Index Fund   | Malaysia Equity        | cluster_14 | FLAT           | 1        | 0         | 1   |

**Top 20 most bearish (NSS asc)**

| level | id   | name                               | category               | cluster    | cluster_signal | prob_buy | prob_sell | nss    |
|-------|------|------------------------------------|------------------------|------------|----------------|----------|-----------|--------|
| etf   | IHI  | iShares U.S. Medical Devices ETF   | Healthcare             | cluster_10 | FLAT           | 0.00     | 1.00      | -1.000 |
| etf   | DZZ  | DB Gold Double Short ETN due Fe... | Leveraged / Inverse... | cluster_27 | LONG           | 0.00     | 1.00      | -1.000 |
| etf   | PSQ  | ProShares Short QQQ                | Leveraged / Inverse    | cluster_9  | FLAT           | 0.00     | 1.00      | -1.000 |
| etf   | QID  | ProShares UltraShort QQQ           | Leveraged / Inverse    | cluster_9  | FLAT           | 0.00     | 1.00      | -1.000 |
| etf   | SQQQ | ProShares UltraPro Short QQQ       | Leveraged / Inverse    | cluster_9  | FLAT           | 0.00     | 1.00      | -0.999 |
| etf   | SCO  | ProShares UltraShort Bloomberg ... | Leveraged / Inverse... | cluster_9  | FLAT           | 0.00     | 1.00      | -0.996 |
| etf   | EUM  | ProShares Short MSCI Emerging M... | Leveraged / Inverse    | cluster_9  | FLAT           | 0.00     | 0.99      | -0.990 |

|         |           |                                    |                     |            |      |      |      |        |
|---------|-----------|------------------------------------|---------------------|------------|------|------|------|--------|
| etf     | SH        | ProShares Short S&P500             | Leveraged / Inverse | cluster_9  | FLAT | 0.00 | 0.99 | -0.989 |
| etf     | NANR      | State Street SPDR S&P North Ame... | Materials           | cluster_11 | FLAT | 0.01 | 0.99 | -0.988 |
| etf     | EDV       | Vanguard Extended Duration Trea... | Fixed Income        | cluster_4  | LONG | 0.01 | 1.00 | -0.986 |
| etf     | SDS       | ProShares UltraShort S&P500        | Leveraged / Inverse | cluster_9  | FLAT | 0.00 | 0.99 | -0.985 |
| etf     | GOAU      | US Global GO Gold and Precious ... | Materials           | cluster_13 | FLAT | 0.02 | 1.00 | -0.980 |
| etf     | SPXS      | Direxion Daily S&P 500 Bear 3X ... | Leveraged / Inverse | cluster_9  | FLAT | 0.00 | 0.98 | -0.979 |
| etf     | SPXU      | ProShares UltraPro Short S&P500    | Leveraged / Inverse | cluster_9  | FLAT | 0.00 | 0.98 | -0.978 |
| etf     | RWM       | ProShares Short Russell2000        | Leveraged / Inverse | cluster_9  | FLAT | 0.00 | 0.98 | -0.977 |
| etf     | PPA       | Invesco Aerospace & Defense ETF    | Industrials         | cluster_6  | LONG | 0.02 | 0.99 | -0.975 |
| etf     | EDZ       | Direxion Emerging Markets Bear ... | Leveraged / Inverse | cluster_9  | FLAT | 0.00 | 0.98 | -0.975 |
| cluster | cluster_9 | Leveraged / Inverse (16/22)        | Leveraged / Inverse |            |      | 0.00 | 0.97 | -0.973 |
| etf     | EIDO      | iShares MSCI Indonesia ETF         | Indonesia Equity    | cluster_14 | FLAT | 0.03 | 1.00 | -0.969 |
| etf     | TWM       | ProShares UltraShort Russell2000   | Leveraged / Inverse | cluster_9  | FLAT | 0.00 | 0.97 | -0.967 |

## Run Parameters

|                          | parameter                | value |
|--------------------------|--------------------------|-------|
| short_ma                 | short_ma                 | 5     |
| buy_ma                   | buy_ma                   | 35    |
| sell_ma                  | sell_ma                  | 20    |
| vol_window               | vol_window               | 20    |
| horizon                  | horizon                  | 5     |
| cluster_smoothing_window | cluster_smoothing_window | 3     |